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Transport of Sodium, Chloride, and Potassium

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INTRODUCTION TO NA, CL, AND K TRANSPORT

SODIUM AND CHLORIDE TRANSPORT

Daily sodium (Na^+) intake for adults in the United States is approximately 180 mmol (4.2 g) for men and 150 mmol (3.5 g) for women (<https://www.cdc.gov/nchs/nhanes/wwc.htm>). As Na^+ is the principal osmole in extracellular fluid, the total body content of Na^+ and chloride (Cl^-), its primary anion, determine the extracellular fluid volume. Renal excretion or retention of salt ($\text{Na}^+\text{-Cl}^-$) is thus the major determinant of the extracellular fluid volume, such that genetic loss- or gain-of-function in renal $\text{Na}^+\text{-Cl}^-$ transport can be associated with relative hypotension or hypertension, respectively. On a quantitative level, at a glomerular filtration rate (GFR) of 180 L/day and serum Na^+ of about 140 mmol/L, the kidney filters some 25,000 mmol/day of Na^+ ; this is equivalent to about 1.5 kg of salt, which would occupy roughly 10 times the extracellular space.¹ Minute changes in renal $\text{Na}^+\text{-Cl}^-$ excretion can thus have massive effects on the extracellular fluid volume. In addition, 99.6% of filtered $\text{Na}^+\text{-Cl}^-$ must be reabsorbed to excrete 140 mmol/L per day. Energetically, this renal absorption of Na^+ consumes 1 molecule of adenosine triphosphate (ATP) per five molecules of Na^+ .¹ This is gratifyingly economical, given that the absorption of $\text{Na}^+\text{-Cl}^-$ is primarily, but not exclusively, driven by basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$, which has a stoichiometry of three molecules of transported Na^+ per molecule of ATP.² This estimate reflects a net expenditure, however, because the cost of transepithelial $\text{Na}^+\text{-Cl}^-$ transport varies considerably along the nephron, from a predominance of passive transport by thin ascending limbs to the purely active transport mediated by the “aldosterone-sensitive distal nephron” (distal convoluted tubule [DCT], connecting tubule [CNT], and collecting duct).

As much as 60% to 70% of filtered $\text{Na}^+\text{-Cl}^-$ is reabsorbed along the proximal tubule (PT), and approximately 25% along the thick ascending limb (TAL; Fig. 6.1). Whereas the PT

can theoretically absorb as much as nine Na^+ molecules for each hydrolyzed ATP, paracellular Na^+ transport by the TAL doubles the efficiency of transepithelial $\text{Na}^+\text{-Cl}^-$ transport (six Na^+ molecules per ATP).^{1,3} By the time filtered fluid reaches the macula densa, more than 90% of filtered Na^+ has been reabsorbed,⁴ a percentage that varies only slightly, when dietary NaCl intake ranges from very low to very high.⁵ Thus the terminal segments of the nephron, while reabsorbing only 5% to 10% of filtered Na^+ , are a primary site of transport regulation. Here, renal $\text{Na}^+\text{-Cl}^-$ absorption occurs at full cost (3 Na^+ per ATP) in the aldosterone-sensitive distal nephron while affording the generation of considerable transepithelial gradients.¹

The nephron thus constitutes a serial arrangement of tubule segments with considerable heterogeneity in the physiologic consequences, mechanisms, and regulation of transepithelial $\text{Na}^+\text{-Cl}^-$ transport. These issues will be reviewed in this section in anatomic order.

PROXIMAL TUBULE

A primary function of the renal PT is the near-isosmotic reabsorption of two-thirds to three-quarters of the glomerular ultrafiltrate. This encompasses the reabsorption of at least 60% of filtered Na^+ with accompanying anions (Fig. 6.1), such that this nephron segment plays a critical role in the maintenance of extracellular fluid volume. Although all segments of the PT share the ability to transport a variety of inorganic and organic solutes, there are considerable differences in the transport characteristics and capacity of early, mid, and late segments of the PT. There is thus a gradual reduction in the volume of transported fluid and solutes as one proceeds along the proximal nephron. This corresponds to distinct ultrastructural characteristics in the tubular epithelium, moving from the S1 segment (early proximal convoluted tubule [PCT]) to the S2 segment (late PCT and beginning of the proximal straight tubule) and the S3 segment (remainder of the proximal straight tubule). Cells of the S1 segment are thus characterized by a tall brush

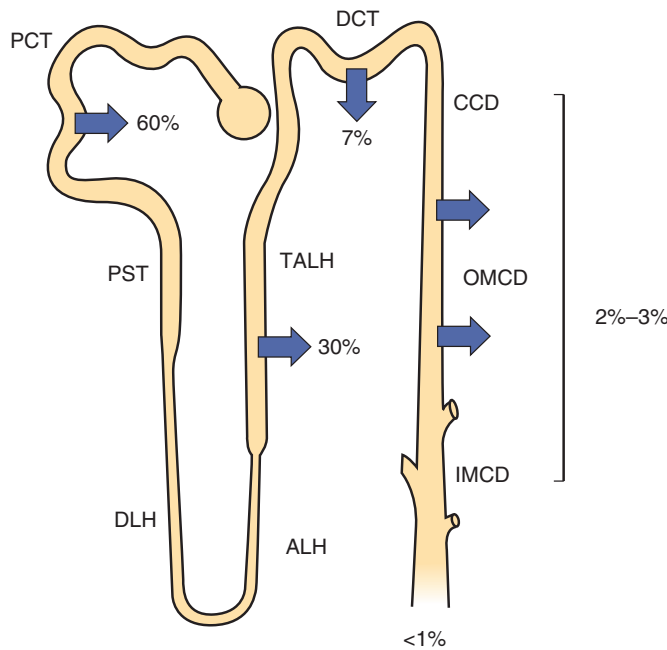


Fig. 6.1 Percentage reabsorption of filtered $\text{Na}^+\text{-Cl}^-$ along the euvoletic nephron. ALH, Thin ascending limb of the loop of Henle; CCD, cortical collecting duct; DCT, distal convoluted tubule; DLH, descending thin limb of the loop of Henle; IMCD, inner medullary collecting duct; OMCD, outer medullary collecting duct; PCT, proximal convoluted tubule; PST, proximal straight tubule; TALH, thick ascending limb of the loop of Henle. (From Moe OW, Baum M, Berry CA, Rector Jr FC. Renal transport of glucose, amino acids, sodium, chloride, and water. In: Brenner BM, ed. *Brenner and Rector's the Kidney*. Philadelphia: WB Saunders; 2004:413–452.)

border, with extensive lateral invaginations of the basolateral membrane.⁶ Numerous elongated mitochondria are located in lateral cell processes, with a proximity to the plasma membrane that is characteristic of epithelial cells involved in active transport. Ultrastructure of the S2 segment is similar, albeit with a shorter brush border, fewer lateral invaginations, and less prominent mitochondria. In epithelial cells of the S3 segment, lateral cell processes and invaginations are essentially absent, with small mitochondria that are randomly distributed within the cell.⁶ The extensive brush border of proximal tubular cells serves to amplify the apical cell surface that is available for reabsorption; again, this amplification is axially distributed, increasing the apical area 36-fold in S1 and 15-fold in S3.⁷ At the functional level, bicarbonate reabsorption rates decline by at least 80% between the first and last portions of the PT, whereas Cl^- reabsorption declines by approximately 50%.⁸

There is also considerable axial heterogeneity in the quantitative capacity of the proximal nephron for organic solutes such as glucose and amino acids, with predominant reabsorption of these substrates in S1 segments.⁹ The Na^+ -dependent reabsorption of glucose, amino acids, and other solutes in S1 segments results in a transepithelial potential difference (PD) that is initially lumen negative due to electrogenic removal of Na^+ from the lumen (Fig. 6.2).¹⁰ This is classically considered the first phase of volume reabsorption by the PT.¹¹ The lumen-negative PD serves to drive both

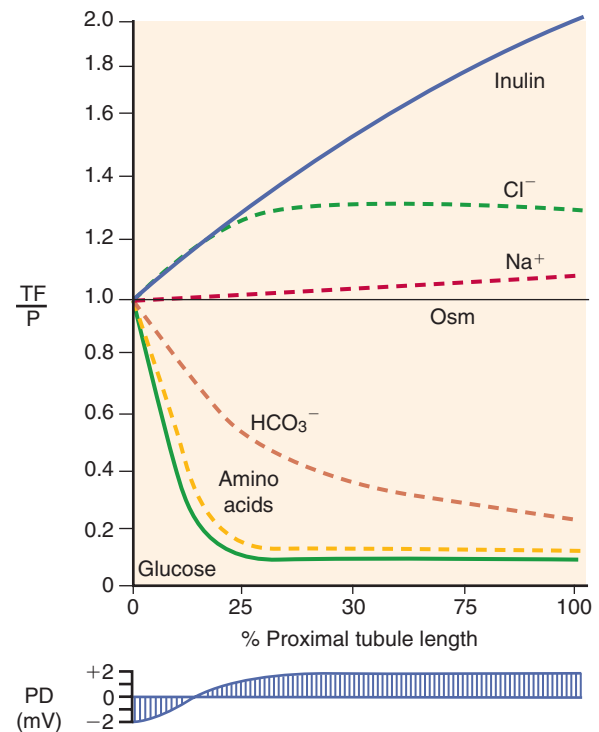


Fig. 6.2 Reabsorption of solutes along the proximal tubule in relation to the transepithelial potential difference (PD). Osm, Osmolality; TF/P, ratio of tubule fluid to plasma concentration. (From Rector Jr FC. Sodium, bicarbonate, and chloride absorption by the proximal tubule. *Am J Physiol*. 1983;244:F461–F471.)

paracellular Cl^- absorption and a backleak of Na^+ from the peritubular space to the lumen. Paracellular Cl^- absorption in this setting accomplishes the net transepithelial absorption of a solute such as glucose, along with equal amounts of Na^+ and Cl^- ; by contrast, backleak of Na^+ leads only to reabsorption of the organic solute, with no net transepithelial transport of Na^+ or Cl^- . The amount of Cl^- reabsorption that is driven by this lumen-negative PD thus depends on the relative permeability of the paracellular pathway to Na^+ and Cl^- . There appears to be considerable heterogeneity in the relative paracellular permeability to Na^+ and Cl^- ; for example, whereas superficial PCTs and proximal straight tubules in the rabbit are Cl^- selective, juxtamedullary PTs in this species are reportedly Na^+ selective.^{12,13} Regardless, the component of paracellular Cl^- transport that is driven by this lumen-negative PD is restricted to the very early PT.

The second phase of volume reabsorption by the PT is dominated by $\text{Na}^+\text{-Cl}^-$ reabsorption via paracellular and transcellular pathways.¹¹ In addition to the Na^+ -dependent reabsorption of organic solutes, the early PT has a much higher capacity for HCO_3^- absorption via the coupling of apical $\text{Na}^+\text{-H}^+$ exchange, carbonic anhydrase, and basolateral $\text{Na}^+\text{-HCO}_3^-$ cotransport.⁹ As the luminal concentrations of HCO_3^- and other solutes begin to drop, the concentration of $\text{Na}^+\text{-Cl}^-$ rises to a value greater than that of the peritubular space.¹⁴ This is accompanied by a reversal of the lumen-negative PD to a lumen-positive value generated by passive Cl^- diffusion (Fig. 6.2).¹⁵ This lumen-positive PD serves

to drive paracellular Na^+ transport, whereas the chemical gradient between the lumen and peritubular space provides the driving force for paracellular reabsorption of Cl^- . This passive paracellular pathway is thought to mediate about 40% of transepithelial $\text{Na}^+\text{-Cl}^-$ reabsorption by the mid to late PT.¹² Of note, however, there may be heterogeneity in the relative importance of this paracellular pathway, with evidence that active (i.e., transcellular) reabsorption predominates in PCTs from juxtamedullary versus superficial nephrons.¹⁶ Regardless, the combination of passive and active transport of $\text{Na}^+\text{-Cl}^-$ explains how the PT is able to reabsorb about 60% of filtered $\text{Na}^+\text{-Cl}^-$, despite $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity that is considerably lower than that of distal segments of the nephron (Fig. 6.3).¹⁷

The transcellular component of $\text{Na}^+\text{-Cl}^-$ reabsorption initially emerged from studies of the effect of cyanide, ouabain, luminal anion transport inhibitors, cooling, and luminal-peritubular K^+ removal.¹¹ For example, the luminal addition of SITS (4-acetamido-4'-isothiocyanostilbene-2,2'-disulfonic acid), an inhibitor of anion transporters, reduces volume reabsorption of PCTs perfused with a high Cl^- , low HCO_3^- solution that mimics the luminal composition of the late PT; this occurs in the absence of an effect on carbonic anhydrase.¹⁴ This transcellular component of $\text{Na}^+\text{-Cl}^-$ reabsorption is clearly electroneutral. For example, in the absence of anion gradients across the perfused PT, there is no change in transepithelial PD after the inhibition of active transport by ouabain, despite a marked reduction in volume reabsorption.¹⁸

Transcellular $\text{Na}^+\text{-Cl}^-$ reabsorption is accomplished by the coupling of luminal $\text{Na}^+\text{-H}^+$ exchange or $\text{Na}^+\text{-SO}_4^{2-}$ cotransport with a heterogeneous population of anion exchangers, as reviewed later.

PARACELLULAR $\text{Na}^+\text{-Cl}^-$ TRANSPORT

A number of factors serve to optimize the conditions for paracellular $\text{Na}^+\text{-Cl}^-$ transport by the mid to late PT. First, the PT is a low-resistance, so-called leaky epithelium, with tight junctions that are highly permeable to both Na^+ and Cl^- .^{12,13} Second, these tight junctions are preferentially permeable to Cl^- over HCO_3^- , a feature that helps generate the lumen-positive PD in the mid to late PT.¹⁴ Third, the increase in luminal $\text{Na}^+\text{-Cl}^-$ concentrations in the mid to late PT generates a chemical driving force for paracellular reabsorption of Cl^- .¹⁵ This increase in luminal $\text{Na}^+\text{-Cl}^-$ is the direct result of the robust reabsorption of HCO_3^- and other solutes by the early S1 segment, combined with the isosmotic reabsorption of filtered water.^{9,19}

A highly permeable paracellular pathway is a consistent feature of epithelia that function in the near-isosmolar reabsorption of $\text{Na}^+\text{-Cl}^-$, including the small intestine, PT, and gallbladder. Morphologically, the apical tight junction of proximal tubular cells and other leaky epithelia is considerably less complex than that of tight epithelia. Freeze-fracture microscopy thus reveals that the tight junction of proximal tubular cells is comparatively shallow, with as few as one junctional strand (Fig. 6.4); by contrast, high-resistance

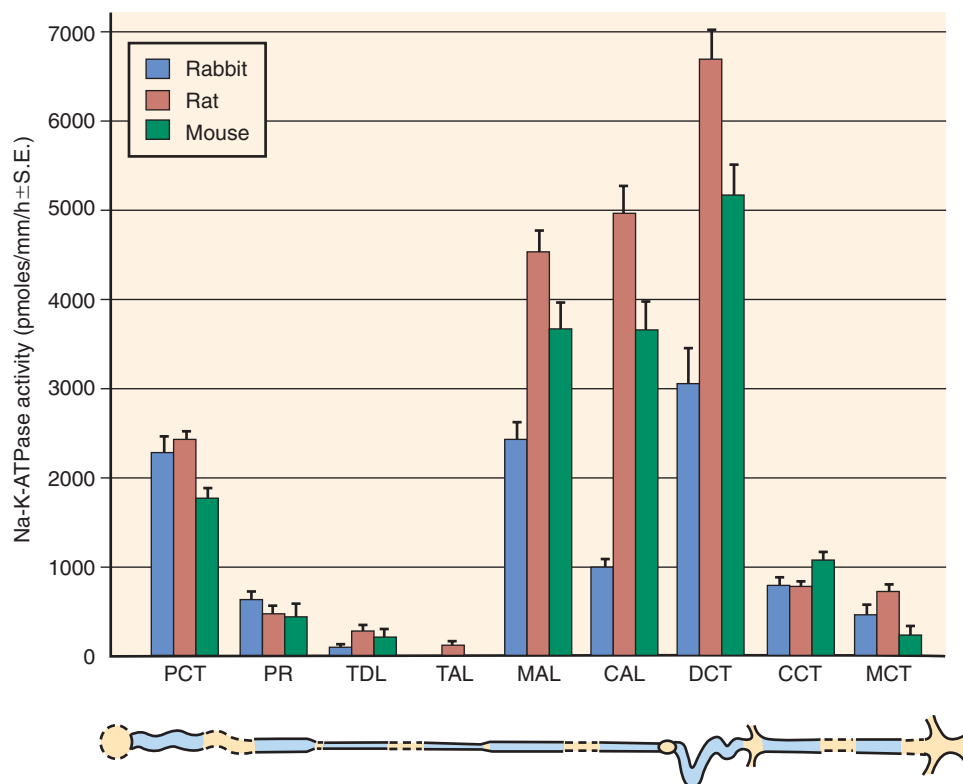


Fig. 6.3 Distribution of $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity along the nephron. *CCD*, Cortical collecting duct; *cTAL*, cortical thick ascending limb; *DCT*, distal convoluted tubule; *MCD*, medullary collecting duct; *mTAL*, medullary thick ascending limb; *PCT*, proximal convoluted tubule; *PST*, proximal straight tubule; *tAL*, thin ascending limb of the loop of Henle; *tDL*, descending thin limb of the loop of Henle. (From Katz AI, Doucet A, Morel F. Na-K-ATPase activity along the rabbit, rat, and mouse nephron. *Am J Physiol.* 1979;237:F114–F120.)

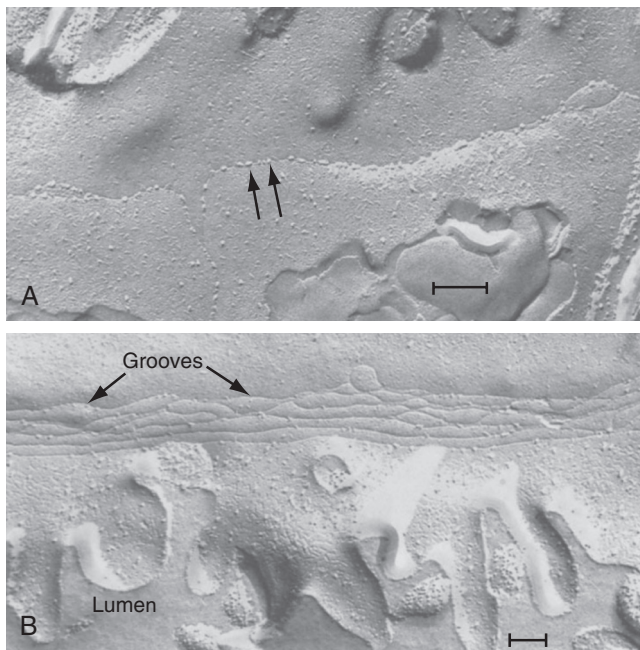


Fig. 6.4 Freeze-fracture electron microscopy images of tight junctions in mouse proximal and distal nephron. (A) Proximal convoluted tubule, a “leaky” epithelium; the tight junction contains only one junctional strand, seen as a groove in the fracture face (arrows). (B) Distal convoluted tubule, a “tight” epithelium. The tight junction is deeper and contains several anastomosing strands, seen as grooves in the fracture face. (From Claude P, Goodenough DA. Fracture faces of zonulae occludentes from “tight” and “leaky” epithelia. *J Cell Biol.* 1973;58: 390–400.)

epithelia have deeper tight junctions, with a complex and extensive network of junctional strands.²⁰ At the functional level, tight junctions of epithelia function as charge- and size-selective paracellular tight junction channels, physiologic characteristics that are thought to be conferred by integral membrane proteins that cluster together at the tight junction. Changes in the expression of these proteins can have marked effects on permeability without affecting the number of junctional strands.^{14,21,22} In particular, the charge and size selectivity of tight junctions appears to be conferred in large part by the claudins, a large (>20) gene family of tetraspan transmembrane proteins,^{23–25} one of which was recently crystallized.²⁶ The repertoire of claudins expressed by proximal tubular epithelial cells may thus determine the high paracellular permeability of this nephron segment. At a minimum, proximal tubular cells coexpress claudin-2, claudin-10, and claudin-17.^{14,27,28}

The robust expression of claudin-2 in the PT is of particular interest because this claudin can dramatically decrease the resistance of transfected epithelial cells.²² Overexpression of claudin-2, but not claudin-10, also increases Na^+ -dependent water flux in epithelial cell lines, suggesting that claudin-2 directly modulates paracellular water permeability.²⁹ Consistent with this cellular phenotype, targeted deletion of claudin-2 in mice generates a tight epithelium in the PT, with a reduction in Na^+ , Cl^- , and fluid absorption.³⁰ Loss of claudin-2 expression does not affect the ultrastructure of tight junctions, but leads to a reduction in paracellular cation permeability and secondary reduction in transepithelial Cl^- transport.³⁰

This action enhances the energy efficiency of proximal solute reabsorption. Although reabsorption is lower in claudin-2 knockout mice, overall sodium handling is normal because solute transport is increased along more distal segments. This requirement for more transcellular solute transport along the loop of Henle contributes to lower medullary oxygen tension, and increased susceptibility to renal ischemia.³¹ Terminal differentiation of proximal tubular claudin-2 expression requires the integrin β_1 -subunit, such that deletion of this protein in mice converts the PT to a tight epithelium expressing low levels of claudin-2.³²

The molecular identification of anion-selective claudins in the PT has lagged, but claudin-17 has been shown to generate a predominantly anion-selective paracellular conductance in Madin-Darby canine kidney (MDCK) C7 cells, whereas knockdown of the protein was able to reverse a predominantly cation-selective LLC-PK(1) epithelial cell line to an anion-selective cell line.²⁸ Claudin-17 is expressed along the PT, suggesting a significant role in paracellular chloride absorption by this nephron segment. Recently, it has been suggested that a form of claudin-10, claudin-10a, is another anion-selective paracellular pathway along the PT.³³

The reabsorption of HCO_3^- and other solutes from the glomerular ultrafiltrate would be expected to generate an osmotic gradient across the epithelium, resulting in a hypotonic lumen. This appears to be the case, although the absolute difference in osmolality between the lumen and peritubular space has been a source of considerable controversy.¹⁹ Another controversial issue has been the relative importance of paracellular versus transcellular water transport from this hypotonic lumen. These issues have been elegantly addressed through characterization of knockout mice with a targeted deletion of aquaporin-1, a water channel protein expressed at the apical and basolateral membranes of the PT. Mice deficient in aquaporin-1 have an 80% reduction in water permeability in perfused S2 segments, with a 50% reduction in transepithelial fluid transport.³⁴ Aquaporin-1 deficiency also results in a marked increase in luminal hypotonicity, providing definitive proof that near-isosmotic reabsorption by the PT requires transepithelial water transport via aquaporin-1.¹⁹ The residual water transport in the PTs of aquaporin-1 knockout mice is mediated in part by aquaporin-7 and/or by claudin-2-dependent paracellular water transport.^{30,35} Combined knockout of aquaporin-1 and claudin-2 in mice demonstrates sustained PT water reabsorption (25% of wild type), suggestive of compensation from other pathways.³⁶ Alternative pathways for water reabsorption may include cotransport of H_2O via the multiple Na^+ -dependent solute transporters in the early PT; this novel hypothesis is, however, a source of considerable controversy.^{37,38} A related issue is the relative importance of diffusional versus convective (solvent drag) transport of $\text{Na}^+\text{-Cl}^-$ across the paracellular tight junction; convective transport of $\text{Na}^+\text{-Cl}^-$ with water would seem to play a lesser role than diffusion, given the evidence that the transcellular pathway is the dominant transepithelial pathway for water in the PT.^{12,19,34,35}

TRANSCELLULAR $\text{Na}^+\text{-Cl}^-$ TRANSPORT

Apical Mechanisms

Apical $\text{Na}^+\text{-H}^+$ exchange plays a critical role in the transcellular and paracellular reabsorption of $\text{Na}^+\text{-Cl}^-$ by the PT. In addition

to providing an entry site in the transcellular transport of Na^+ , $\text{Na}^+\text{-H}^+$ exchange plays a dominant role in the functional “absorption” of HCO_3^- by the early PT (HCO_3^- does not actually move across the apical membrane, but rather is generated within cells together with H^+); as the movement of Na^+ and HCO_3^- drives osmotic water movement, it also acts to increase the luminal concentration of Cl^- , which in turn increases the driving forces for the passive paracellular transport of Cl^- .³⁹ Increases in luminal Cl^- also help drive the apical uptake of Cl^- during transcellular transport. Not surprisingly, there is a considerable reduction in fluid transport of perfused PTs exposed to concentrations of amiloride that are sufficient to inhibit proximal tubular $\text{Na}^+\text{-H}^+$ exchange.¹⁴

$\text{Na}^+\text{-H}^+$ exchange is predominantly mediated by the NHE proteins, encoded by the nine members of the *SLC9* gene family; NHE3 in particular plays an important role in proximal tubular physiology.⁴⁰ The NHE3 protein is expressed at the apical membrane of S1, S2, and S3 segments.⁴¹ The apical membrane of the PT also expresses alternative Na^+ -dependent H^+ transporters, including NHE8.^{40,42} NHE8 predominates over NHE3 in the neonatal PT, with subsequent induction of NHE3 and downregulation of NHE8 in mature, adult nephrons.⁴⁰ The primacy of NHE3 in mature PTs is illustrated

by the renal phenotype of NHE3 knockout mice, which have a 62% reduction in proximal fluid absorption and a 54% reduction in baseline chloride absorption.^{43,44} A recent study in which *Nhe3* was disrupted specifically in the kidney showed that it plays a role in maintaining blood pressure at baseline, and plasma Na^+ in response to increased or reduced dietary $\text{Na}^+\text{-Cl}^-$, with compensatory upregulation of $\text{Na}^+\text{-Cl}^-$ cotransporter (NCC) and epithelial Na^+ channel (ENaC) occurring.⁴⁵ The severe salt-wasting phenotype seen in global *Nhe3* knockout mice occurs due to its deletion in intestine.

Much as amiloride and other inhibitors of $\text{Na}^+\text{-H}^+$ exchange have revealed an important role for this transporter in transepithelial salt transport by the PT, evidence for the involvement of an apical anion exchanger first came from the use of anion transport inhibitors; DIDS (4,4'-diisothiocyanostilbene-2,2'-disulfonic acid), furosemide, and SITS all reduce fluid absorption from the lumen of PT segments perfused with solutions containing $\text{Na}^+\text{-Cl}^-$.¹⁴ In the simplest arrangement for the coupling of $\text{Na}^+\text{-H}^+$ exchange to Cl^- exchange, Cl^- would be exchanged with the OH^- ion during $\text{Na}^+\text{-Cl}^-$ transport (Fig. 6.5). Evidence for such a $\text{Cl}^-\text{-OH}^-$ exchanger was reported by a number of groups in the early 1980s that used membrane vesicles isolated from

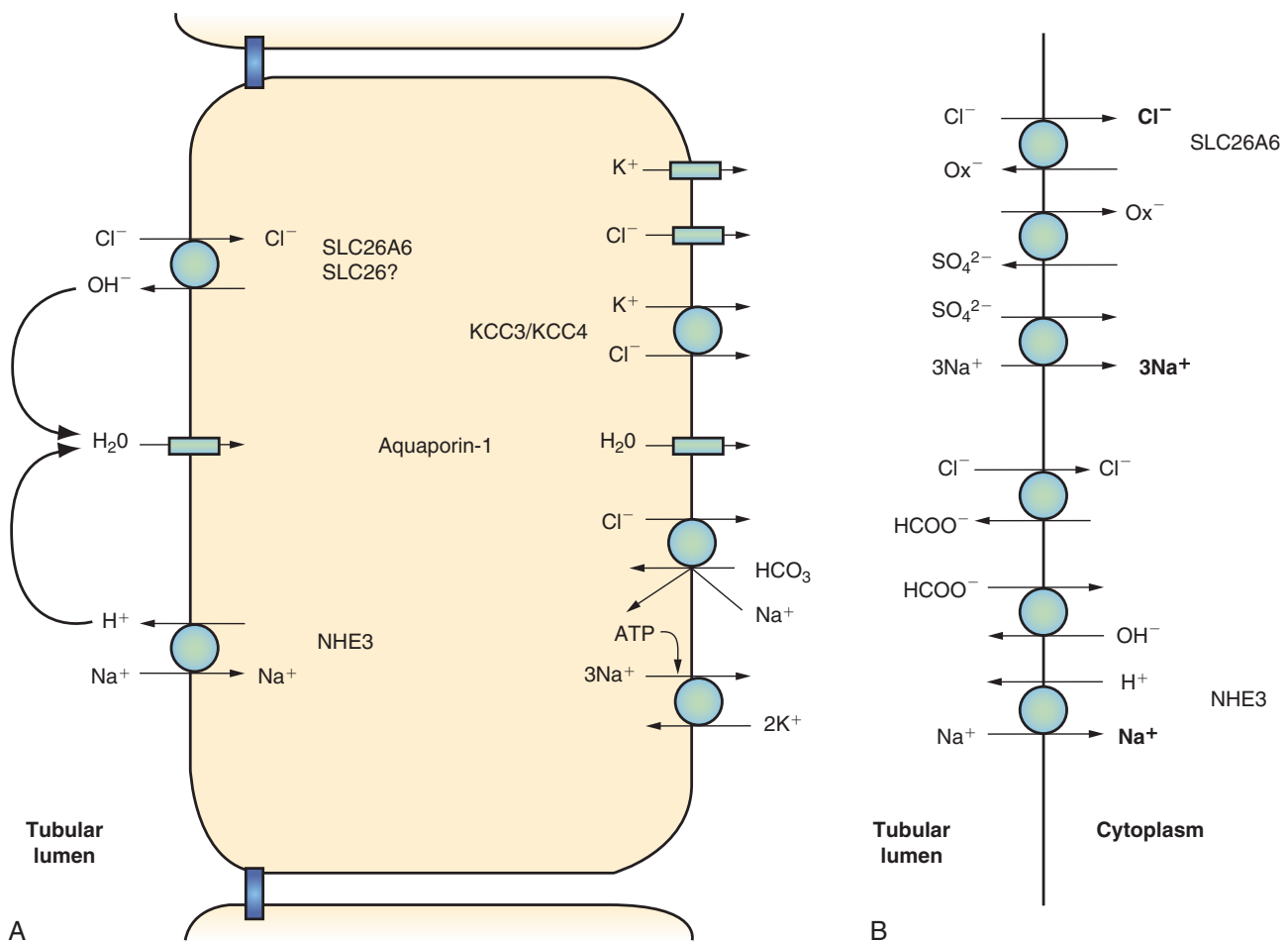


Fig. 6.5 Transepithelial $\text{Na}^+\text{-Cl}^-$ transport in the proximal tubule. (A) In the simplest scheme, Cl^- enters the apical membrane via a $\text{Cl}^-\text{-OH}^-$ exchanger, coupled to Na^+ entry via NHE3. (B) Alternative apical anion exchange activities that couple to $\text{Na}^+\text{-H}^+$ exchange and $\text{Na}^+\text{-SO}_4^{2-}$ cotransport. See text for details.

the PT.⁴⁶ These findings could not, however, be replicated in similar studies from other groups.^{46,47} Moreover, experimental evidence was provided for the existence of a dominant Cl⁻-formate exchange activity in brush border vesicles in the absence of significant Cl⁻-OH⁻ exchange.⁴⁷ It was postulated that recycling of formate by the back diffusion of formic acid would sustain the net transport of Na⁺-Cl⁻ across the apical membrane. Vesicle formate transport stimulated by a pH gradient (H⁺-formate cotransport or formate-OH⁻ exchange) is saturable, consistent with a carrier-mediated process rather than diffusion of formic acid across the apical membrane of the PT.⁴⁸ Transport studies using brush border vesicles have also detected the presence of Cl⁻-oxalate exchange mechanisms in the apical membrane of the PT, in addition to SO₄²⁻-oxalate exchange.^{39,49} Based on differences in the affinities and inhibitor sensitivity of the Cl⁻-oxalate and Cl⁻-formate exchange activities, it was suggested that there are two separate apical exchangers in the proximal nephron, a Cl⁻-formate exchanger and a Cl⁻-formate-oxalate exchanger capable of transporting both formate and oxalate (Fig. 6.5).

The physiological relevance of apical Cl⁻-formate and Cl⁻-oxalate exchange has been addressed by perfusing individual PT segments with solutions containing Na⁺-Cl⁻ and formate or oxalate. Both formate and oxalate significantly increased fluid transport under these conditions in rabbit, rat, and mouse PTs.⁴⁴ This increase in fluid transport was inhibited by DIDS, suggesting involvement of the DIDS-sensitive anion exchanger(s) detected in brush border vesicle studies. A similar mechanism for Na⁺-Cl⁻ transport in the DCT has also been detected, independent of thiazide-sensitive Na⁺-Cl⁻ cotransport.⁵⁰ Further experiments have indicated that the oxalate- and formate-dependent anion transporters in the PT are coupled to distinct Na⁺ entry pathways, to Na⁺-SO₄²⁻ cotransport and Na⁺-H⁺ exchange, respectively.⁵¹ The coupling of Cl⁻-oxalate transport to Na⁺-SO₄²⁻ cotransport requires the additional presence of SO₄²⁻-oxalate exchange, which has been demonstrated in brush border membrane vesicle studies.⁵² The obligatory role for NHE3 in formate-stimulated Cl⁻ transport was illustrated using *Nhe3* null mice, in which the formate effect is abolished; as expected, oxalate stimulation of Cl⁻ transport is preserved in the *Nhe3* null mice.⁴⁴ Finally, tubular perfusion data from superficial and juxtamedullary PCTs have suggested that there is heterogeneity in the dominant mode of anion exchange along the PT, such that Cl⁻-formate exchange is absent in juxtamedullary PCT, in which Cl⁻-OH⁻ exchange may instead be dominant.¹⁴

The molecular identity of the apical anion exchanger(s) involved in transepithelial Na⁺-Cl⁻ reabsorption by the PT has been the object of almost 3 decades of investigation. A key breakthrough was the observation that the SLC26A4 anion exchanger, also known as pendrin, is capable of Cl⁻-formate exchange when expressed in *Xenopus laevis* oocytes.⁵³ However, expression of SLC26A4 in the PT is minimal or absent in several species, and formate-stimulated Na⁺-Cl⁻ transport in this nephron segment is unimpaired in *Slc26a4* null mice.¹⁴ There is, however, robust expression of SLC26A4 in distal type B intercalated cells; the role of this exchanger in Cl⁻ transport by the distal nephron is reviewed elsewhere in this chapter (see the “Connecting Tubules and the Cortical Collecting Duct: Cl⁻ Transport” section).⁵⁴ Regardless, these data for SLC26A4 led to the identification and characterization of SLC26A6, a widely expressed member of the SLC26 family

that is expressed at the apical membrane of proximal tubular cells. Murine SLC26A6, when expressed in *Xenopus* oocytes, mediates the multiple modes of anion exchange that have been implicated in transepithelial Na⁺-Cl⁻ by the PT, including Cl⁻-formate, Cl⁻-OH⁻, Cl⁻-SO₄²⁻, and SO₄²⁻-oxalate exchange.⁵⁵ However, tubule perfusion experiments in mice deficient in SLC26A6 did not reveal a reduction in baseline Cl⁻ or fluid transport, indicative of considerable heterogeneity in apical Cl⁻ transport by the PT.⁵⁶ Candidates for the residual Cl⁻ transport in SLC26A6-deficient mice include SLC26A7 and SLC26A9, which are expressed at the apical membrane of PTs; however, these members of the SLC26 family appear to function as Cl⁻ channels rather than as exchangers.^{57–59} SLC26A2 may also contribute to apical anion exchange in the PT.⁶⁰ It does, however, appear that SLC26A6 is the dominant Cl⁻-oxalate exchanger of the proximal brush border; the usual increase in tubular fluid transport induced by oxalate is abolished in *Slc26a6* knockout mice, with an attendant loss of Cl⁻-oxalate exchange in brush border membrane vesicles.^{56,61}

Somewhat surprisingly, SLC26A6 mediates electrogenic Cl⁻-OH⁻ and Cl⁻-HCO₃⁻ exchange, and most if not all the members of this family are electrogenic in at least one mode of anion transport.^{14,55,58,62,63} This begs the question of how the electroneutrality of transcellular Na⁺-Cl⁻ transport is preserved. Notably, however, the stoichiometry and electrophysiology of Cl⁻-base exchange differ for individual members of the family; for example, SLC26A6 exchanges one Cl⁻ for two HCO₃⁻ anions, whereas SLC26A3 exchanges two Cl⁻ anions for one HCO₃⁻ anion.^{14,63} Coexpression of two or more electrogenic SLC26 exchangers in the same membrane may thus yield a net electroneutrality of apical Cl⁻ exchange. Alternatively, apical K⁺ channels in the PT may function to stabilize membrane potential during Na⁺-Cl⁻ absorption.⁶⁴

Another puzzle is why Cl⁻-formate exchange preferentially couples to Na⁺-H⁺ exchange mediated by NHE3 (Fig. 6.5), without evident coupling of Cl⁻-oxalate exchange to Na⁺-H⁺ exchange or Cl⁻-formate exchange to Na⁺-SO₄²⁻ cotransport; it is evident that SLC26A6 is capable of mediating SO₄²⁻-formate exchange, which would be necessary to support coupling between Na⁺-SO₄²⁻ cotransport and formate.^{44,55} Scaffolding proteins may serve to cluster these different transporters together in separate microdomains, leading to preferential coupling. Notably, whereas both SLC26A6 and NHE have been reported to bind to the scaffolding protein PDZK1, distribution of SLC26A6 is selectively impaired in *Pdzk1* knockout mice.⁶⁵ Petrovic and colleagues have also reported a novel activation of proximal Na⁺-H⁺ exchange by luminal formate, suggesting a direct effect of formate per se on NHE3; this may in part explain the preferential coupling of Cl⁻-formate exchange to NHE3.⁶⁶ Despite these intriguing observations, the relative importance of transcellular versus passive paracellular Cl⁻ reabsorption in the PT remains to be established with certainty.

Basolateral Mechanisms

As in other absorptive epithelia, basolateral Na⁺-K⁺-ATPase activity establishes the Na⁺ gradient for transcellular Na⁺-Cl⁻ transport by the PT and provides a major exit pathway for Na⁺. To preserve the electroneutrality of transcellular Na⁺-Cl⁻ transport, this exit of Na⁺ across the basolateral membrane must be balanced by an equal exit of Cl⁻.¹⁸ Several

exit pathways for Cl^- have been identified in proximal tubular cells, including $\text{K}^+\text{-Cl}^-$ cotransport, Cl^- channels, and various modalities of Cl^- - HCO_3^- exchange (Fig. 6.5).

Several lines of evidence support the existence of a swelling-activated basolateral $\text{K}^+\text{-Cl}^-$ cotransporter (KCC) in the PT.⁶⁷ The KCC proteins are encoded by four members of the cation-chloride cotransporter gene family; *Kcc1*, *Kcc3*, and *Kcc4* are all expressed in the kidney. In particular, there is very heavy coexpression of KCC3 and KCC4 at the basolateral membrane of the PT, from S1 to S3.⁶⁸ At the functional level, basolateral membrane vesicles from the renal cortex reportedly contain $\text{K}^+\text{-Cl}^-$ cotransport activity.⁶⁷ The use of ion-sensitive microelectrodes, combined with luminal charge injection and manipulation of bath K^+ and Cl^- , suggest the presence of an electroneutral KCC at the basolateral membrane of proximal straight tubules. Increases or decreases in basolateral K^+ increase or decrease intracellular Cl^- activity, respectively, with reciprocal effects of basolateral Cl^- on K^+ activity; these data are consistent with coupled $\text{K}^+\text{-Cl}^-$ transport.^{69,70} Notably, a 1-mmol/L concentration of furosemide, sufficient to inhibit all four of the KCCs, does not inhibit this $\text{K}^+\text{-Cl}^-$ cotransport under baseline conditions.⁶⁹ However, only 10% of baseline K^+ efflux in the PT is mediated by furosemide-sensitive $\text{K}^+\text{-Cl}^-$ cotransport, which is likely quiescent in the absence of cell swelling. Thus the activation of apical Na^+ -glucose transport in proximal tubular cells strongly activates a barium-resistant (Ba^{2+}) K^+ efflux pathway that is 75% inhibited by 1-mmol/L furosemide.⁷¹ In addition, a volume regulatory decrease (VRD) in Ba^{2+} -blocked PTs swollen by hypotonic conditions is blocked by 1-mmol/L furosemide.⁶⁷ Cell swelling in response to apical Na^+ absorption is postulated to activate a volume-sensitive basolateral KCC, which participates in transepithelial absorption of $\text{Na}^+\text{-Cl}^-$.¹⁴ Notably, targeted deletion of *Kcc3* and *Kcc4* in the respective knockout mice reduces VRD in the PT.⁷² Furthermore, perfused PTs from KCC3-deficient mice have a considerable reduction in transepithelial fluid transport, suggesting an important role for basolateral $\text{K}^+\text{-Cl}^-$ cotransport in transcellular $\text{Na}^+\text{-Cl}^-$ reabsorption.⁷³

The basolateral chloride conductance of mammalian proximal tubular cells is relatively low, suggesting a lesser role for Cl^- channels in transepithelial $\text{Na}^+\text{-Cl}^-$ transport. Basolateral anion substitutions have minimal effect on the membrane potential, despite considerable effects on intracellular Cl^- activity, nor for that matter do changes in basolateral membrane potential affect intracellular Cl^- .^{69,70,74} However, as with basolateral $\text{K}^+\text{-Cl}^-$ cotransport, basolateral Cl^- channels in the PT may be relatively inactive in the absence of cell swelling. Cell swelling thus activates both K^+ and Cl^- channels at the basolateral membranes of proximal tubular cells.^{14,75,76} Seki and associates have reported the presence of a basolateral Cl^- channel in S3 segments of the rabbit nephron, wherein they did not see an effect of the KCC inhibitor H74 on intracellular Cl^- activity.⁷⁷ The molecular identity of these and other basolateral Cl^- channels in the proximal nephron is not known with certainty, although S3 segments have been shown to express messenger RNA (mRNA) exclusively for the swelling-activated ClC-2 Cl^- channel; the role of this channel in transcellular $\text{Na}^+\text{-Cl}^-$ reabsorption is not as yet clear.⁷⁸

Finally, there is functional evidence for Na^+ -dependent and Na^+ -independent Cl^- - HCO_3^- exchange at the basolateral

membrane of proximal tubular cells.^{13,74,79} The impact of Na^+ -independent Cl^- - HCO_3^- exchange on basolateral exit is thought to be minimal.⁷⁴ First, this exchanger is expected to mediate Cl^- entry under physiologic conditions.⁷⁹ Second, there is only a modest difference between the rate of decrease in intracellular Cl^- activity and the combined removal of Na^+ and Cl^- versus Cl^- and HCO_3^- , suggesting that pure Cl^- - HCO_3^- exchange does not contribute significantly to Cl^- exit. By contrast, there is a 75% reduction in the rate of decrease in intracellular Cl^- activity after the removal of basolateral Na^+ .⁷⁴ The Na^+ -dependent Cl^- - HCO_3^- exchanger may thus play a considerable role in basolateral Cl^- exit, with recycled exit of Na^+ and HCO_3^- via the basolateral $\text{Na}^+\text{-HCO}_3^-$ cotransporter NBC1 (Fig. 6.5). The molecular identity of this proximal tubular Na^+ -dependent Cl^- - HCO_3^- exchanger is not as yet known.

REGULATION OF PROXIMAL TUBULAR $\text{Na}^+\text{-Cl}^-$ TRANSPORT

Glomerulotubular Balance

A fundamental property of the kidney is the phenomenon of glomerulotubular balance, wherein changes in the GFR are offset by changes in tubular reabsorption, thus maintaining a constant fractional reabsorption of fluid and $\text{Na}^+\text{-Cl}^-$ (Fig. 6.6). Although the distal nephron is capable of adjusting reabsorption in response to changes in tubular flow, the impact of GFR on $\text{Na}^+\text{-Cl}^-$ reabsorption by the PT is particularly pronounced (Fig. 6.7).⁸⁰ Glomerulotubular balance is independent of direct neuronal and systemic hormonal control, and is thought to be mediated by the additive effects of luminal and peritubular factors.⁸¹

Until recently, there was some controversy regarding the role of luminal factors in glomerulotubular balance because experiments performed using isolated rabbit PTs failed to demonstrate a significant effect of tubular flow on fluid absorption.⁸² This issue has largely been resolved, however,

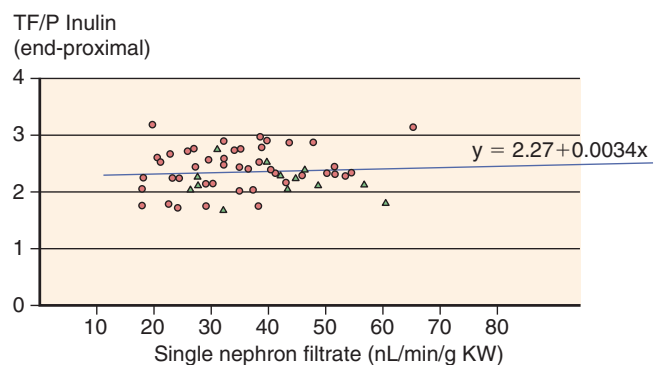


Fig. 6.6 Glomerulotubular balance. The tubular fluid-to-plasma ratio of the nonreabsorbable marker, inulin (TF/P inulin), at the end of the proximal tubule, which is used as a measure of fractional water absorption by the proximal tubule, does not change as a function of single-nephron glomerular filtration rate. Measurements were done during antidiuresis (triangles) and water diuresis (circles). (From Schnermann J, Wahl M, Liebau G, Fischbach H. Balance between tubular flow rate and net fluid reabsorption in the proximal convolution of the rat kidney. I. Dependency of reabsorptive net fluid flux upon proximal tubular surface area at spontaneous variations of filtration rate. *Pflügers Arch.* 1968;304:90–103.)

with clear evidence that fluid shear stress (FSS) increases solute and water absorption.⁸³ Du and coworkers reported linear flow dependence of fluid and HCO_3^- transport in isolated perfused murine PTs (Fig. 6.8),^{81,84} mediated by NHE3 and the H^+ -ATPase, as discussed later. These data were analyzed using a mathematical model that estimated microvillus torque as a function of tubular flow; accounting for increases in tubular diameter, which reduce torque, there is a linear relationship between calculated torque and fluid and HCO_3^- absorption.^{81,84} Consistent with an effect of torque rather than flow per se, increasing viscosity of the perfusate by the addition of dextran increases the effect on fluid transport; the extra viscosity increases the hydrodynamic effect of flow and thus increases torque. The mathematical analysis of Du and associates provides an excellent explanation of the discrepancy between their results and those of Burg and Orloff.⁸² Whereas Burg and Orloff performed their

experiments in rabbits, the more recent report used mice; other studies that had found an effect of flow used perfusion of rat PTs, presumably more similar to mouse than rabbit.^{80–82,84} Increased flow has a considerably greater effect on tubular diameter in the rabbit PT, thus reducing the increase in torque. Mathematical analysis of the rabbit data thus predicts a 43% increase in torque due to a 41% increase in tubule diameter at a threefold increase in flow; this corresponds to the statistically insignificant 36% increase in volume reabsorption reported by Burg and Orloff.⁸²

Pharmacologic inhibition reveals that tubular flow activates proximal HCO_3^- reabsorption mediated by NHE3 and apical H^+ -ATPase.⁸¹ The flow-dependent increase in proximal fluid and HCO_3^- reabsorption is also attenuated in NHE3-deficient knockout mice.^{81,84} Inhibition of the actin cytoskeleton with cytochalasin D reduces the effect of flow on fluid and HCO_3^- transport. This maneuver blocked the effects of flow on NHE3 and Na^+/K^+ -ATPase, but not on H^+ -ATPase. It also blocks the effects of FSS on movement of these transporters to the plasma membranes, suggesting that flow-dependent movement of microvilli activates these transport proteins via their linkage to the cytoskeleton (see Fig. 6.12 for NHE3).⁸⁵ FSS induces densely distributed peripheral actin bands and increases the formation of tight junctions and adherens junctions in cultured tubule cells; this junctional buttressing is hypothesized to maximize flow-activated transcellular salt and water absorption.⁸⁶

The roles of dopamine, angiotensin II (Ang II), and calcium on FSS-induced sodium and bicarbonate transport have been examined. Luminal dopamine completely inhibited the flow-induced increase in Na^+ transport,⁸³ with the major effects being mediated through the D_{1A} receptor. Deletion of AT_{1A} receptors in mice also abrogated flow-induced increments in Na^+ transport,⁸⁷ but these effects may be related to the profound basal reductions in NHE3 activity. When AT_1 receptor blockers are employed, flow-induced Na^+ transport remained.^{87,88} Intracellular activation of IP₃ through a local calcium signal may mediate the effects of flow on NHE3 activity, although increased calcium influx does not appear to play a role.⁸⁹ Flow and torque were not found to have any effects on chloride absorption, suggesting no convective flow of chloride through the paracellular pathway.

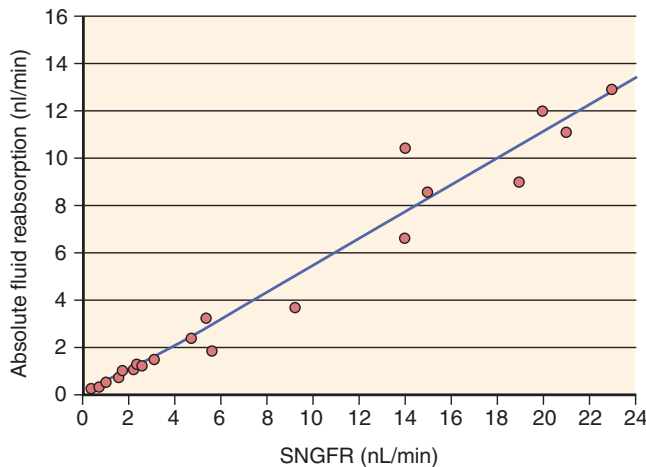


Fig. 6.7 Glomerulotubular balance. Shown is the linear increase in absolute fluid reabsorption by the late proximal tubule as a function of single-nephron glomerular filtration rate (SNGFR). (From Spitzer A, Brandis M. Functional and morphologic maturation of the superficial nephrons. Relationship to total kidney function. *J Clin Invest.* 1974;53: 279–287.)

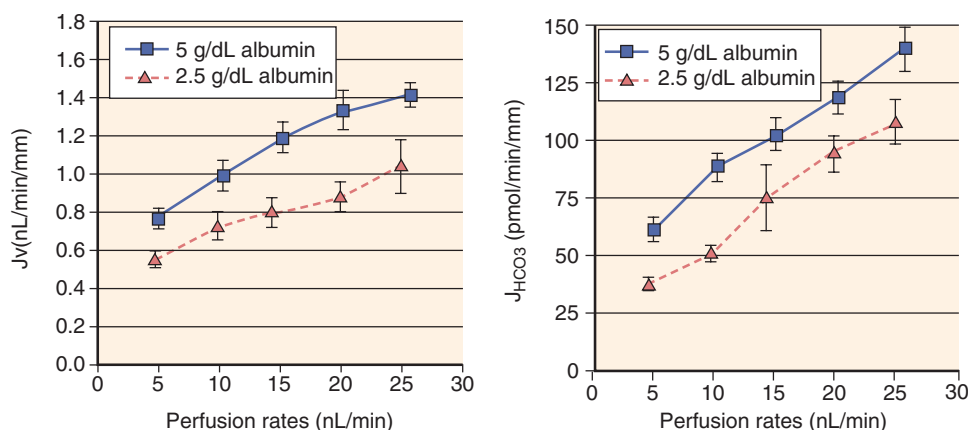


Fig. 6.8 Glomerulotubular balance; flow-dependent increases in fluid (J_v) and HCO_3^- (J_{HCO_3}) absorption by perfused mouse proximal tubules. Absorption also increases when bath albumin concentration increases from 2.5 to 5 g/dL. (From Du Z, Yan Q, Duan Y, et al. Axial flow modulates proximal tubule NHE3 and H-ATPase activities by changing microvillus bending moments. *Am J Physiol Renal Physiol.* 2006;290:F289–F296.)

Another mechanism for glomerulotubular balance operating from the luminal side involves limiting solute concentration. Solutes, such as bicarbonate, amino acids, and glucose, that are reabsorbed coupled to sodium will be depleted earlier along the PT when flow is low, thereby limiting reabsorption rates along the segment as a whole.⁹⁰

Peritubular factors also play an important additive role in glomerulotubular balance, perhaps accounting for the difficulties in documenting flow-induced alterations using isolated rabbit PTs. Specifically, increases in GFR result in an increase in filtration fraction and an attendant increase in postglomerular protein and peritubular oncotic pressure. It has long been appreciated that changes in peritubular protein concentration have important effects on proximal tubular $\text{Na}^+\text{-Cl}^-$ reabsorption; these effects are also seen in combined capillary and tubular perfusion experiments.^{81,91} Peritubular protein also has an effect in isolated perfused PT segments, where the effect of hydrostatic pressure is abolished.⁸¹ Increases in peritubular protein concentration have an additive effect on the flow-dependent activation of proximal fluid and HCO_3^- absorption (Fig. 6.8). The effect of peritubular protein on HCO_3^- absorption, which is a predominantly transcellular phenomenon, suggests that changes in peritubular oncotic pressure do not affect transport

via the paracellular pathway.¹⁴ However, the mechanism of the stimulatory effect of peritubular protein on transcellular transport is still not completely clear.⁸¹ There are also changes in absorption that correlate with changes in peritubular hydrostatic pressure, as occurs during expansion or contraction of the extracellular fluid volume.⁹¹

Neurohumoral Influences

Fluid and $\text{Na}^+\text{-Cl}^-$ reabsorption by the PT are affected by a number of hormones and neurotransmitters. The major hormonal influences on renal $\text{Na}^+\text{-Cl}^-$ transport are shown in Fig. 6.9. Renal sympathetic tone exerts a particularly important stimulatory influence, as does Ang II; dopamine is a major inhibitor of proximal tubular $\text{Na}^+\text{-Cl}^-$ reabsorption.

Unilateral denervation of the rat kidney causes a marked natriuresis and a 40% reduction in proximal $\text{Na}^+\text{-Cl}^-$ reabsorption, without effects on single-nephron GFR or on the contralateral innervated kidney.⁹² By contrast, low-frequency electrical stimulation of renal sympathetic nerves increases proximal tubular fluid absorption, with a 32% drop in natriuresis and no change in GFR.⁹³ Basolateral epinephrine and/or norepinephrine stimulate proximal $\text{Na}^+\text{-Cl}^-$ reabsorption via both α - and β -adrenergic receptors. Several lines of evidence suggest that α_1 -adrenergic receptors exert a

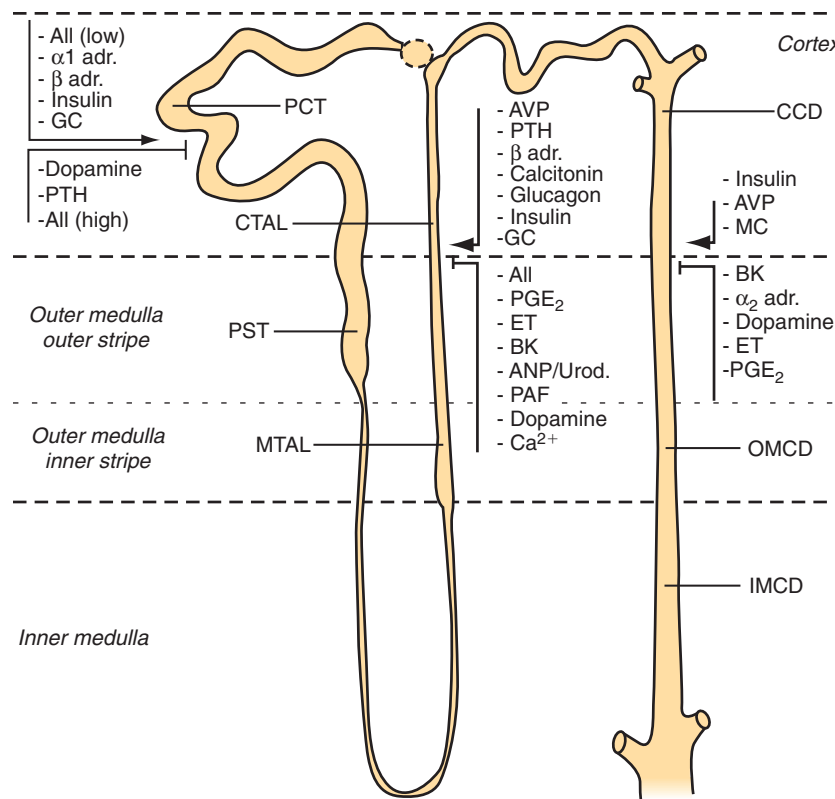


Fig. 6.9 Neurohumoral influences on $\text{Na}^+\text{-Cl}^-$ absorption by the proximal tubule, thick ascending limb, and collecting duct. Factors that stimulate ($\rightarrow$) and inhibit ($\leftarrow$) sodium reabsorption are as follows: *ANG II*, Angiotensin II (low and high referring to picomolar and micromolar concentrations, respectively); *ANP/Urod*, atrial natriuretic peptide and urodilatin; *AVP*, arginine vasopressin; *BK*, bradykinin; *CCD*, cortical collecting duct; *cTAL*, cortical thick ascending limb; *ET*, endothelin; *GC*, glucocorticoids; *IMCD*, inner medullary collecting duct; *MC*, mineralocorticoids; *mTAL*, medullary thick ascending limb of the loop of Henle; *OMCD*, outer medullary collecting duct; *PAF*, platelet-activating factor; *PCT*, proximal convoluted tubule; *PGE₂*, prostaglandin E₂; *PST*, proximal straight tubule; *PTH*, parathyroid hormone; α_1 adr, α_1 -adrenergic agonist; β adr, β -adrenergic agonist. (From Feraille E, Doucet A. Sodium-potassium-adenosine triphosphatase-dependent sodium transport in the kidney: hormonal control. *Physiol Rev.* 2001;81:345–418.)

stimulatory effect on proximal $\text{Na}^+\text{-Cl}^-$ transport via activation of basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$ and apical $\text{Na}^+\text{-H}^+$ exchange; the role of α_2 -adrenergic receptors is more controversial.⁹⁴ Ligand-dependent recruitment of the scaffolding protein NHE regulatory factor-1 (NHERF-1) by β_2 -adrenergic receptors results in direct activation of apical NHE3, bypassing the otherwise negative effect of downstream cyclic adenosine monophosphate (cAMP; see later).^{95,96} Ang II has potent effects on proximal $\text{Na}^+\text{-Cl}^-$ reabsorption and therefore on blood pressure. Genetic deletion of $\text{AT}_{1\text{A}}$ receptors from PT cells reduced proximal fluid reabsorption, lowered basal blood pressure, shifted the pressure natriuresis, and attenuated the hypertensive response to Ang II infused chronically.⁹⁶ Although basal abundances of NHE3 and NaPi2 were similar in PTs from control and knockout mice, the abundance of NHE3 and NaPi2 was lower following Ang II infusion in mice lacking $\text{AT}_{1\text{A}}$, suggesting that these effects are mediated, at least in part, through these two prominent Na^+ transport pathways.

Despite this clear stimulatory effect, it has also been appreciated for 3 decades that Ang II has a bimodal (sometimes called “biphasic”) effect on the PT Na^+ transport in rats, rabbits, and mice; stimulation of $\text{Na}^+\text{-Cl}^-$ reabsorption occurs at 10^{-12} to 10^{-10} M, whereas inhibition of $\text{Na}^+\text{-Cl}^-$ reabsorption occurs at concentrations greater than 10^{-7} M (Fig. 6.10).⁹⁷ Note, however, that plasma Ang II concentrations typically do not exceed 10^{-9} M, even during pathological states, such as 2 kidney/1 clip Goldblatt hypertension.⁹⁸ Furthermore, this biphasic role of Ang II may not hold true for all species, and in human PT samples, obtained during nephrectomy, concentrations up to 10^{-6} M Ang II stimulate $\text{Na}^+\text{-Cl}^-$ reabsorption, primarily owing to a stimulatory effect of the nitric oxide (NO)–cyclic guanosine monophosphate

(cGMP) pathway on extracellular signal-regulated kinase (ERK) phosphorylation.⁹⁹ Although the plasma Ang II concentration is typically below the inhibitory concentration, it should be noted, as discussed further later, that the concentration of Ang II in the lumen of the PT frequently exceeds that in plasma, and thus plasma concentrations may not be the sole determining factor. Given the substantial effects of proximal $\text{AT}_{1\text{A}}$ receptor deletion to reduce blood pressure and shift the pressure natriuresis,⁹⁶ it seems likely that the predominant effect of Ang II is stimulatory along the PT under most physiological conditions.

Further complexity of Ang II signaling arises from the presence of AT_1 receptors at both the luminal and basolateral membranes in the PT.¹⁰⁰ Ang II application to the luminal or peritubular side of perfused tubules has a similar bimodal effect on fluid transport, albeit with more potent effects at the luminal side.¹⁰¹ Traditionally, experiments using receptor antagonists and knockout mice have indicated that the stimulatory and inhibitory effects of Ang II are both mediated via AT_1 receptors due to signaling at the luminal and basolateral membranes.¹⁰² However, other work has identified that AT_2 receptors working through NO-cGMP pathway are able to downregulate NHE3 and $\text{Na}^+\text{-K}^+\text{-ATPase}$, leading to natriuresis and reduced blood pressure.¹⁰³ Finally, Ang II is also synthesized and secreted by the PT, exerting a potent autocrine effect on proximal tubular $\text{Na}^+\text{-Cl}^-$ reabsorption.¹⁰⁴ Proximal tubular cells thus express mRNA for angiotensinogen, renin, and angiotensin-converting enzyme (ACE),⁹⁴ allowing for the autocrine generation of Ang II. Indeed, luminal concentrations of Ang II can be 100- to 1000-fold higher than circulating levels of the hormone.⁹⁴ Proximal tubular and systemic synthesis of Ang II may be subject to different control. In fact, intrarenal Ang II appears to stimulate proximal $\text{Na}^+\text{-Cl}^-$ and fluid reabsorption even when dietary salt intake is high, thereby helping to prevent rises in glomerular filtration from increasing late proximal flow.¹⁰⁵ It should be recalled, however, that in this, as in many salt loading studies in rodents, animals received 1% saline as drinking solution. Thus the high salt load was accompanied by free water deprivation, a circumstance shown recently to lead to stress and inflammation.¹⁰⁶

The PT is also a target for natriuretic hormones; in particular, dopamine synthesized in the PT has negative autocrine effects on proximal $\text{Na}^+\text{-Cl}^-$ reabsorption.⁹⁴ Proximal tubular cells have the requisite enzymatic machinery for the synthesis of dopamine, using L-dopa reabsorbed from the glomerular ultrafiltrate. Dopamine synthesis by proximal tubular cells and release into the tubular lumen are increased after volume expansion or a high-salt diet, resulting in a considerable natriuresis.^{107,108} Luminal dopamine antagonizes the stimulatory effect of epinephrine on volume absorption in perfused PCTs, consistent with an autocrine effect of dopamine released into the tubular lumen.^{107,109} Dopamine primarily exerts its natriuretic effect via D_1 -like dopamine receptors (D_1 and D_5 in humans); as is the case for the AT_1 receptors for Ang II, D_1 receptors are expressed at the apical and luminal membranes of PTs.^{100,110} Targeted deletion of the $\text{D}_{1\text{A}}$ and D_5 receptors in mice leads to hypertension by mechanisms that include reduced proximal tubular natriuresis.^{111,112} The proximal tubular-specific deletion of aromatic amino acid decarboxylase, which produces dopamine, generates mice that are a vivid demonstration of the role of intrarenal

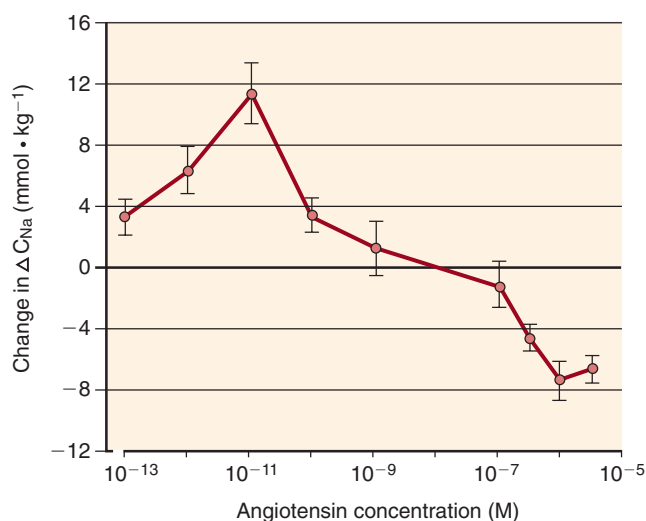


Fig. 6.10 Biphasic effect of angiotensin II (Ang II) on Na^+ reabsorption in microperfused proximal tubules. The steady-state transepithelial Na^+ concentration gradient (peritubular-luminal), ΔC_{Na} , that developed in a stationary split droplet is used as an indication of the rate of active Na^+ reabsorption. This is plotted as a function of peritubular Ang II concentration; low concentrations activate Na^+ absorption by the proximal tubule, whereas higher concentrations inhibit it. (From Harris PJ, Navar LG. Tubular transport responses to angiotensin. *Am J Physiol.* 1985;248:F621–F630)

dopamine. This intrarenal dopamine deficiency leads to upregulation of sodium transporters along the nephron, upregulation of the intrarenal renin–angiotensin axis, decreased natriuresis in response to L-dopa, and reduced medullary cyclooxygenase-2 (COX-2) expression, with reduced urinary prostaglandin levels. These mice also exhibit salt-sensitive hypertension and ultimately a significantly shorter life span compared with wild type mice.¹¹³

The natriuretic effect of dopamine in the PT is modulated by atrial natriuretic peptide (ANP), which inhibits apical $\text{Na}^+\text{-H}^+$ exchange via a dopamine-dependent mechanism.¹⁴ ANP appears to induce recruitment of the D_1 dopamine receptor to the plasma membrane of proximal tubular cells, thus sensitizing the tubule to the effect of dopamine.¹¹⁴ The inhibitory effect of ANP on basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$ occurs via a D_1 -dependent mechanism, with a synergistic inhibition of $\text{Na}^+\text{-K}^+\text{-ATPase}$ by the two hormones.¹¹⁴ Furthermore, dopamine and D_1 receptors appear to play critical permissive roles in the *in vivo* natriuretic effect of ANP.¹⁴

Finally, there is considerable crosstalk between the major antinatriuretic and natriuretic influences on the PT. For example, ANP inhibits Ang II–dependent stimulation of proximal tubular fluid absorption, presumably via the dopamine-dependent mechanisms discussed earlier.^{14,115} Dopamine also decreases the expression of AT_1 receptors for Ang II in cultured proximal tubular cells.¹¹⁶ Furthermore, the provision of L-dopa in the drinking water of rats decreases AT_1 receptor expression in the PT, suggesting that dopamine synthesis in the PT resets the sensitivity to Ang II.¹¹⁶ Ang II signaling through AT_1 receptors decreases expression of the D_5 dopamine receptor, whereas renal cortical expression of AT_1 receptors is in turn increased in knockout mice deficient in the D_5 receptor.¹¹⁷ Similar interactions have been found between proximal tubular AT_1 receptors and the D_2 -like D_3 receptor.¹¹⁸

Regulation of Proximal Tubular Transporters

The apical $\text{Na}^+\text{-H}^+$ exchanger NHE3 and the basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$ are primary targets for signaling pathways elicited by the various antinatriuretic and natriuretic stimuli discussed earlier; NHE3 mediates the rate-limiting step in transepithelial $\text{Na}^+\text{-Cl}^-$ absorption and, as such, is the dominant target for regulatory pathways.⁸⁴ NHE3 is regulated by the combined effects of direct phosphorylation and dynamic, carboxyl-terminal interaction with scaffolding proteins and signal transduction proteins, which primarily regulate transport via changes in trafficking of the exchanger protein to and from the brush border membrane (Fig. 6.11).^{40,119} Basal activity of the exchanger is also dependent on carboxyl-terminal binding of casein kinase 2 (CK2); phosphorylation of serine 719 by CK2 contributes significantly to the transport activity of NHE3 by modulating membrane trafficking of the transport protein.¹²⁰

Increases in cAMP have a profound inhibitory effect on apical $\text{Na}^+\text{-H}^+$ exchange in the PT. Intracellular cAMP is increased in response to dopamine signaling via D_1 -like receptors and/or parathyroid hormone (PTH)–dependent signaling via the PTH receptor; whereas Ang II–dependent activation of NHE3 is associated with a reduction in cAMP.¹²¹ PTH is a potent inhibitor of NHE3, presumably so as to promote the distal delivery of $\text{Na}^+\text{-HCO}_3^-$ and an attendant stimulation of distal calcium reabsorption.¹²² The activation of protein kinase A (PKA) by increased cAMP results in direct phosphorylation of NHE3; although several sites in NHE3 are phosphorylated by PKA, the phosphorylation of serine 552 (S552) and 605 (S605) has been specifically implicated in the inhibitory effect of cAMP on $\text{Na}^+\text{-H}^+$ exchange.¹²³ So-called phospho-specific antibodies, which specifically recognize the phosphorylated forms of S552 and S605, have demonstrated dopamine-dependent increases in the phosphorylation of both these serines.¹²⁴ Moreover,

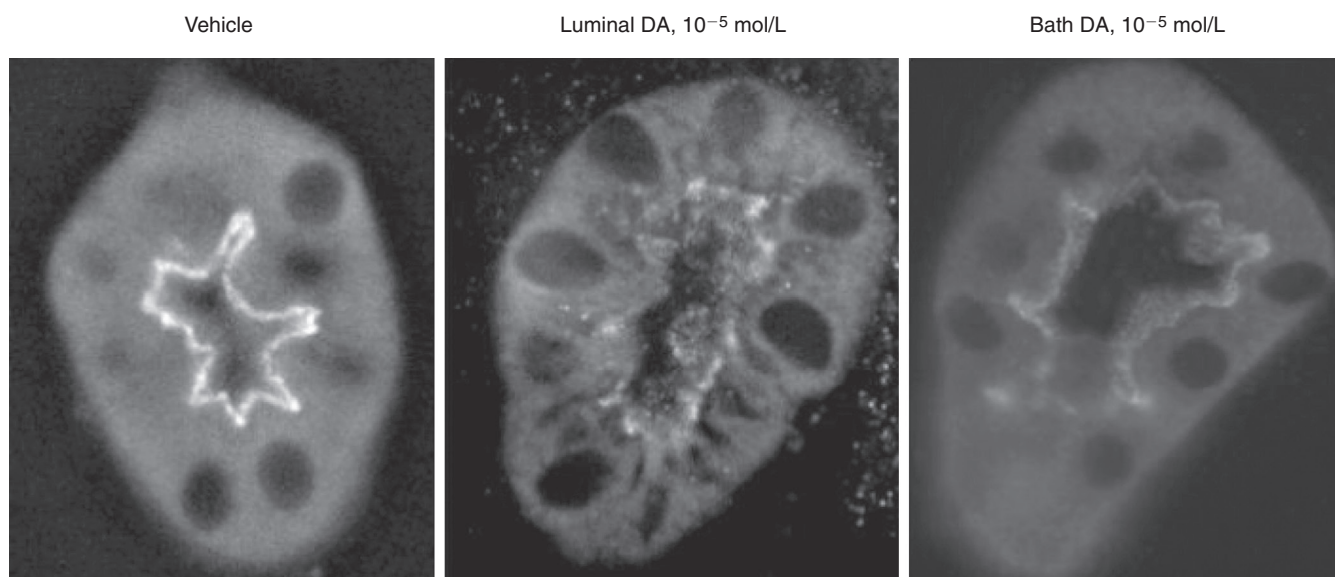


Fig. 6.11 Effect of dopamine on trafficking of the $\text{Na}^+\text{-H}^+$ exchanger NHE3 in the proximal tubule. Microdissected proximal convoluted tubules were perfused for 30 minutes with 10^{-5} mol/L dopamine (DA), in the lumen or the bath, inducing a retraction of immunoreactive NHE3 protein from the apical membrane. (From Bacic D, Kaissling B, McLeroy P, et al. Dopamine acutely decreases apical membrane Na/H exchanger NHE3 protein in mouse renal proximal tubule. *Kidney Int.* 2003;64:2133–2141.)

immunostaining of rat kidney has revealed that S552-phosphorylated NHE3 localizes at the coated pit region of the brush border membrane, where the oligomerized inactive form of NHE3 predominates.^{124,125} The cAMP-stimulated phosphorylation of NHE3 by PKA thus results in a redistribution of the transporter from the microvillar membrane to an inactive submicrovillar population (Fig. 6.11). Notably, however, phosphorylation of these residues appears to be necessary but not sufficient for regulation of NHE3.⁴⁰ A number of regulators of NHE3, including gastrin and uroguanylin, have been found to exert a functional effect through phosphorylation of S552 and/or S605.^{126,127}

The regulation of NHE3 by cAMP also requires the participation of a family of homologous scaffolding proteins that contain protein–protein interaction motifs known as PDZ domains (named for the PSD95, *Drosophila* disc large, and ZO-1 proteins in which these domains were first discovered; Fig. 6.12). The first of these proteins, NHERF-1, was purified as a cellular factor required for the inhibition of NHE3 by PKA.¹²⁸ NHERF-2 was in turn cloned by yeast two-hybrid screens as a protein that interacts with the carboxyl-terminus of NHE3; NHERF-1 and NHERF-2 have very similar effects on the regulation of NHE3 in cultured cells. The related protein PDZK1 interacts with NHE3 and a number of other epithelial transporters and is required for expression of the anion exchanger SLC26A6 at brush border membranes of the PT.⁶⁵

NHERF-1 and NHERF-2 are both expressed in human and mouse PT cells; NHERF-1 colocalizes with NHE3 in microvilli of the brush border, whereas NHERF-2 is predominantly expressed at the base of microvilli in the vesicle-rich

domain.¹²⁸ The NHERFs assemble a multiprotein, dynamically regulated signaling complex that includes NHE3 and several other transport proteins. In addition to NHE3, they bind to the actin-associated protein, ezrin, thus linking NHE3 to the cytoskeleton; this linkage to the cytoskeleton may be particularly important for the mechanical activation of NHE3 by microvillar bending, as has been implicated in glomerulotubular balance (see earlier discussion).^{81,84,128} Ezrin also interacts directly with NHE3, binding to a separate binding site within the carboxyl-terminus of the transport protein.¹¹⁹ Ezrin functions as an anchoring protein for PKA, bringing PKA into close proximity with NHE3 and facilitating its phosphorylation (Fig. 6.12).¹²⁸ Analysis of knockout mice for *Nherf-1* has revealed that it is not required for baseline activity of NHE3; as expected, however, it is required for cAMP-dependent regulation of the exchanger by PTH.¹²⁸ One long-standing paradox has been that β -adrenergic receptors, which increase cAMP in the PT, cause an activation of apical $\text{Na}^+\text{-H}^+$ exchange.⁹⁴ This was resolved by the observation that the first PDZ domain of NHERF-1 interacts with the β_2 -adrenergic receptor in an agonist-dependent fashion; this interaction serves to disrupt the interaction between the second PDZ domain and NHE3, resulting in a stimulation of the exchanger, despite the catecholamine-dependent increase in cAMP.¹²⁸

As discussed earlier, at concentrations higher than 10^{-7} M (Fig. 6.10), Ang II has an inhibitory effect on proximal tubular $\text{Na}^+\text{-Cl}^-$ absorption.⁹⁷ This inhibition is dependent on the activation of brush border phospholipase A_2 (PLA_2), which results in the liberation of arachidonic acid.¹⁰¹ Metabolism of arachidonic acid by cytochrome P450 monooxygenases, in turn, generates 20-hydroxyeicosatetraenoic acid (20-HETE) and epoxyeicosatrienoic acids (EETs), compounds that inhibit NHE3 and the basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$.^{94,129} EETs and 20-HETE have also been implicated in the reduction in proximal $\text{Na}^+\text{-Cl}^-$ absorption that occurs during pressure natriuresis, inhibiting $\text{Na}^+\text{-K}^+\text{-ATPase}$ and retracting NHE3 from the brush border membrane.¹³⁰

Antinatriuretic stimuli such as Ang II acutely increase the expression of NHE3 at the apical membrane, at least in part by inhibiting the generation of cAMP.¹²¹ Low-dose Ang II 0.1 nmol/L (10^{-10} M) also increases exocytic insertion of NHE3 into the plasma membrane via a mechanism that is dependent on phosphatidylinositol-3-kinase (PI3K).¹³¹ Treatment of rats with captopril thus results in a retraction of NHE3 and associated proteins from the brush border of PT cells.¹³² Glucocorticoids also increase NHE3 activity due to transcriptional induction of the *Nhe3* gene and an acute stimulation of exocytosis of the exchanger to the plasma membrane.⁴⁰ Glucocorticoid-dependent exocytosis of NHE3 appears to require NHERF-2, which acts in this context as a scaffolding protein for the glucocorticoid-induced serine-threonine kinase SGK1 (see the “Regulation of $\text{Na}^+\text{-Cl}^-$ Transport in the Connecting Tubule and Cortical Collecting Duct: Aldosterone” section).¹³³ The acute effect of dexamethasone has thus been shown to require direct phosphorylation of serine 663 in the NHE3 protein by SGK1.¹³⁴

Finally, many of the natriuretic and antinatriuretic pathways that influence NHE3 have parallel effects on the basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$ (see Feraille and Doucet⁹⁴ for a detailed review). The molecular mechanisms underlying inhibition of $\text{Na}^+\text{-K}^+\text{-ATPase}$ by dopamine have been particularly well characterized.

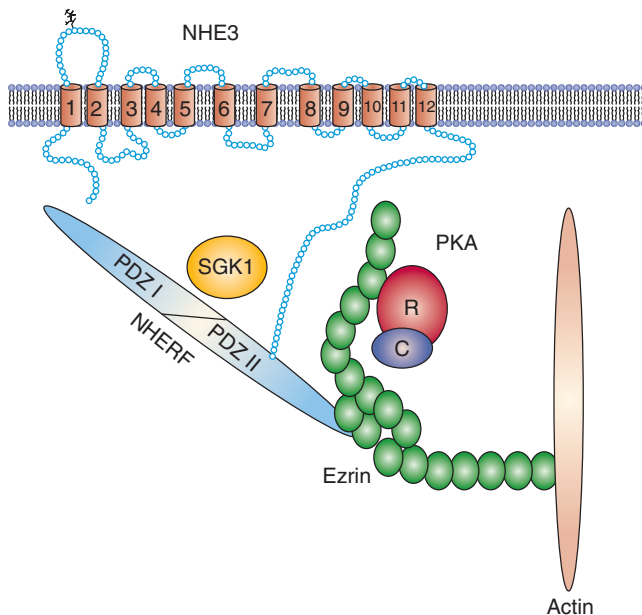


Fig. 6.12 Scaffolding protein NHERF ($\text{Na}^+\text{-H}^+$ exchanger regulatory factor) links the $\text{Na}^+\text{-H}^+$ exchanger NHE3 to the cytoskeleton and signaling proteins. NHERF binds to ezrin, which in turn links to protein kinase A (PKA) and the actin cytoskeleton. NHERF also binds to SGK1 (serum- and glucocorticoid-regulated kinase 1), which activates NHE3. C, Catalytic; PDZ, domain named for the PSD95, *Drosophila* disc large (*Drosophila*), and ZO-1 proteins; R, regulatory.

Inhibition by dopamine is associated with removal of active $\text{Na}^+\text{-K}^+\text{-ATPase}$ units from the basolateral membrane, somewhat analogous to the effect on NHE3 expression at the apical membrane.¹³⁵ This inhibitory effect is primarily mediated by protein kinase C (PKC), which directly phosphorylates the α_1 -subunit of $\text{Na}^+\text{-K}^+\text{-ATPase}$, the predominant α -subunit in the kidney.⁹⁴ The effect of dopamine requires phosphorylation of serine 18 of the α_1 -subunit by PKC; this phosphorylation does not affect enzymatic activity of the $\text{Na}^+\text{-K}^+\text{-ATPase}$, but rather induces a conformational change that enhances the binding of PI3K to an adjacent, proline-rich domain. The PI3K recruited by this phosphorylated α_1 -subunit then stimulates the dynamin-dependent endocytosis of the $\text{Na}^+\text{-K}^+\text{-ATPase}$ complex via clathrin-coated pits.¹³⁵

LOOP OF HENLE

The loop of Henle encompasses the thin descending limb, thin ascending limb, and TAL. The descending and ascending thin limbs function in passive absorption of water and $\text{Na}^+\text{-Cl}^-$, respectively, whereas the TAL reabsorbs about 30% of filtered $\text{Na}^+\text{-Cl}^-$ via active transport.^{136,137} There is considerable cellular and functional heterogeneity along the entire length of the loop of Henle, with consequences for the transport of water, $\text{Na}^+\text{-Cl}^-$, and other solutes. The thin descending limb begins in the outer medulla after an abrupt transition from S3 segments of the PT, marking the boundary between the outer and inner stripes of the outer medulla. Thin descending limbs end at a hairpin turn at the end of the loop of Henle. Short-looped nephrons that originate from superficial and midcortical nephrons have a short descending limb within the inner stripe of the outer medulla; these tubules merge abruptly into the TAL close to the hairpin turn of the loop (see also discussion later). Long-looped nephrons originating from juxtamedullary glomeruli have a long ascending thin limb that then merges with the TAL. The TALs of long-looped nephrons begin at the boundary between the inner and outer medulla, whereas the TALs of short-looped nephrons may be entirely cortical. The ratio of medullary to cortical TAL for a given nephron is a function of the depth of its origin, such that superficial nephrons are primarily composed of cortical TALs, whereas juxtamedullary nephrons primarily possess medullary TALs.

The TAL begins abruptly after the thin ascending limb of long-looped nephrons and after the aquaporin-negative segment of short-limbed nephrons.¹³⁸ The TAL extends into the renal cortex, where it meets its parent glomerulus at the vascular pole; the plaque of cells at this junction form the macula densa, which function as the tubular sensor for tubuloglomerular feedback (TGF) and tubular regulation of renin release by the juxtaglomerular apparatus. Cells in the medullary TAL are 7 to 8 μm in height, with extensive invaginations of the basolateral plasma membrane and interdigitations between adjacent cells.⁶ As in the PT, these lateral cell processes contain numerous elongated mitochondria, perpendicular to the basement membrane. Cells in the cortical TAL are considerably shorter, 2 μm in height at the end of the cortical TAL in rabbits, with fewer mitochondria and a simpler basolateral membrane.⁶ Macula densa cells also lack the lateral cell processes and interdigitations characteristic of medullary TAL cells.⁶ However, scanning electron microscopy has revealed that the TAL of rat and

hamster contains two morphologic subtypes, a rough-surfaced cell type (R cells) with prominent apical microvilli and a smooth-surfaced cell type (S cells) with an abundance of subapical vesicles.^{6,139–141} In the hamster TAL, cells can also be separated into those with high apical and low basolateral K^+ conductance and weak basolateral Cl^- conductance (LBC cells) versus a second population with low apical and high basolateral K^+ conductance combined with high basolateral Cl^- conductance (HBC).^{140,142} The relative frequency of the morphologic and functional subtypes in the cortical and medullary TAL suggests that HBC cells correspond to S cells and LBC cells to R cells.¹⁴⁰

TRANSPORT CHARACTERISTICS OF THE DESCENDING THIN LIMB

It has long been appreciated that the osmolality of tubular fluid increases progressively between the corticomedullary junction and papillary tip due to active secretion of solutes or passive absorption of water along the descending thin limb.¹⁴³ Subsequent reports have revealed a very high water permeability of perfused outer medullary thin descending limbs in the absence of significant permeability to $\text{Na}^+\text{-Cl}^-$.¹⁴⁴ Notably, however, the permeability properties of descending thin limbs vary as a function of depth in the inner medulla and inclusion in short- versus long-looped nephrons.^{145,146} Descending thin limbs from short-looped nephrons contain type I cells, very flat, endothelial-like cells, with intermediate-depth tight junctions suggesting a relative tight epithelium.^{145,146} The epithelium of descending limbs from long-looped nephrons is initially more complex, with taller type II cells possessing more elaborate apical microvilli and more prominent mitochondria. In the lower medullary portion of long-looped nephrons, these cells change into a type III morphology, endothelial-like cells similar to the type I cells from short-looped nephrons.¹⁴⁵ The permeability properties appear to change as a function of cell type, with a progressive axial drop in water permeability of long-looped descending limbs; the water permeability of descending thin limbs in the middle part of the inner medulla is thus about 42% that of outer medullary thin descending limbs.¹⁴⁷ Furthermore, the distal 20% of descending thin limbs have a very low water permeability.¹⁴⁷ These changes in water permeability along the descending thin limb are accompanied by a progressive increase in $\text{Na}^+\text{-Cl}^-$ permeability, although the ionic permeability remains considerably less than that of the ascending thin limb.¹⁴⁶

Consistent with a primary role in passive water and solute absorption, $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity in the descending thin limb is almost undetectable,¹⁷ suggesting that these cells do not actively transport $\text{Na}^+\text{-Cl}^-$; those ion transport pathways that have been identified in descending thin limb cells are thought to contribute primarily to cellular volume regulation.¹⁴⁸ In contrast to the relative lack of $\text{Na}^+\text{-Cl}^-$ transport, transcellular water reabsorption by the thin descending limb is a critical component of the renal countercurrent concentrating mechanism (see Chapter 10).^{136,144}

$\text{Na}^+\text{-Cl}^-$ TRANSPORT BY THE THIN ASCENDING LIMB

Fluid entering the thin ascending limb has a very high concentration of $\text{Na}^+\text{-Cl}^-$ due to osmotic equilibration by the water-permeable descending limbs. The passive reabsorption of this delivered $\text{Na}^+\text{-Cl}^-$ by the thin ascending limb is a

critical component of the passive equilibration model of the renal countercurrent multiplication system (see Chapter 10). Consistent with this role, the permeability properties of the thin ascending limb are dramatically different from those of the descending thin limb, with a much higher permeability to $\text{Na}^+\text{-Cl}^-$ and vanishingly low water permeability.^{146,149} Passive $\text{Na}^+\text{-Cl}^-$ reabsorption by thin ascending limbs occurs via a combination of paracellular Na^+ transport and transcellular Cl^- transport.^{137,142,150–153} The inhibition of paracellular conductance by protamine thus selectively inhibits Na^+ transport across perfused thin ascending limbs, consistent with paracellular transport of Na^+ .¹⁵⁰ As in the descending limb, thin ascending limbs have a modest $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity (Fig. 6.3); however, the active transport of Na^+ across thin ascending limbs accounts for only an estimated 2% of Na^+ reabsorption by this nephron segment.¹⁵⁴ Chloride channel blockers reduce Cl^- permeability of the thin ascending limb, consistent with passive transcellular Cl^- transport.¹⁵³ Direct measurement of the membrane potential of impaled hamster thin ascending limbs has also yielded evidence for apical and basolateral Cl^- channel activity.¹⁴² This transepithelial transport of Cl^- , but not Na^+ , is activated by vasopressin, with a pharmacologic profile that is consistent with direct activation of thin ascending limb Cl^- channels.¹⁵⁵

Both apical and basolateral Cl^- transport in the thin ascending limb appear to be mediated by the CLC-K1 Cl^- channel in cooperation with the Barttin subunit (see also the “ $\text{Na}^+\text{-Cl}^-$ Transport by the Thick Ascending Limb: Basolateral Mechanisms” section). Immunofluorescence and in situ hybridization indicate a selective expression of CLC-K1 (homologous to the human chloride channel Kb, CLC-NKB) in thin ascending limbs, although single-tubule, reverse transcriptase-polymerase chain reaction (RT-PCR) studies

have suggested additional expression in the TAL, DCT, and cortical collecting duct (CCD).^{156–158} Notably, immunofluorescence and immunogold labeling indicate that CLC-K1 is expressed exclusively at both the apical and basolateral membranes of thin ascending limbs, such that both the luminal and basolateral Cl^- channels of this nephron segment are encoded by the same gene.^{142,156} Homozygous knockout mice with a targeted deletion of *Clc-k1* have a vasopressin-resistant nephrogenic diabetes insipidus, reminiscent of the phenotype of aquaporin-1 knockout mice.^{136,159} Given that CLC-K1 is potentially expressed in the TAL, dysfunction of this nephron segment might also contribute to the renal phenotype of *Clc-k1* knockout mice; however, the closely homologous channel CLC-K2 (CLC-NKB) is clearly expressed in the TAL, where it can likely substitute for CLC-K1 .¹⁵⁸ Furthermore, loss-of-function mutations in *CLC-NKB* cause Bartter syndrome, which is phenocopied in *Clc-k2* knockout mice, indicating that CLC-K2 , rather than CLC-K1 , is critical for transport function of the TAL.^{160–162}

Detailed characterization of *Clc-k1* knockout mice has revealed a selective impairment in Cl^- transport by the thin ascending limb.¹³⁷ Whereas Cl^- absorption is profoundly reduced, Na^+ absorption by thin ascending limbs is not significantly impaired (Fig. 6.13). The diffusion voltage induced by a transepithelial $\text{Na}^+\text{-Cl}^-$ gradient is reversed by the absence of CLC-K1 , from +15.5 mV in homozygous wild type controls (+/+) to –7.6 mV in homozygous knockout mice (–/–). This change in diffusion voltage is due to the dominance of paracellular Na^+ transport in the CLC-K1 -deficient –/– mice, leading to a lumen-negative potential; this corresponds to a marked reduction in the relative permeability of Cl^- to that of Na^+ ($P_{\text{Cl}}/P_{\text{Na}}$), from 4.02 to 0.63 (Fig. 6.13). Protamine, an inhibitor of paracellular Na^+ transport, has a comparable effect on

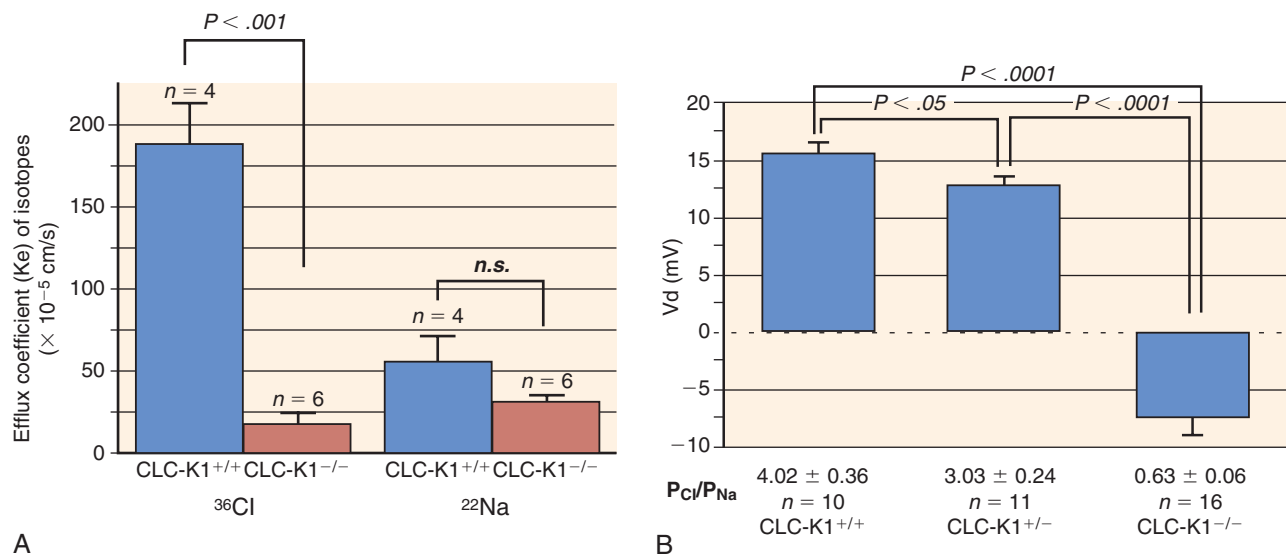


Fig. 6.13 Role of the CLC-K1 chloride channel in Na^+ and Cl^- transport by the thin ascending limbs. Homozygous knockout mice ($\text{CLC-K1}^{-/-}$) are compared with their littermate controls ($\text{CLC-K1}^{+/+}$). (A) Efflux coefficients for $^{36}\text{Cl}^-$ and $^{22}\text{Na}^+$ in the thin ascending limbs. Cl^- absorption is essentially abolished in the knockout mice, whereas there is no significant effect of CLC-K1 deficiency on Na^+ transport. (B) The diffusion voltage (VD), induced by a transepithelial $\text{Na}^+\text{-Cl}^-$ gradient, is reversed by the absence of CLC-K1 , from +15.5 mV in controls to –7.6 mV in homozygous knockout mice. This change in diffusion voltage is due to the dominance of paracellular Na^+ transport in the CLC-K1 -deficient –/– mice, leading to a lumen-negative potential; this corresponds to a marked reduction in the relative permeability of Cl^- to that of Na^+ ($P_{\text{Cl}}/P_{\text{Na}}$), from 4.02 to 0.63. (From Liu W, Morimoto T, Kondo Y, et al. Analysis of NaCl transport in thin ascending limb of the loop of Henle in CLC-K1 null mice. *Am J Physiol Renal Physiol*. 2002;282:F451–F457.)

the diffusion voltage in $-/-$ mice versus $+/-$ and $+/+$ mice that have been treated with 5-nitro-2-(3-phenylpropylamino)-benzoate (NPPB) to inhibit CLC-K1; the respective diffusion voltages are 7.9 mV ($-/-$ plus protamine), 8.6 mV ($+/-$ plus protamine and NPPB), and 9.8 mV ($+/+$ plus protamine and NPPB). Therefore the paracellular Na^+ conductance is unimpaired and essentially the same in *Clc-k1* knockout mice when compared with littermate controls. This study thus provided elegant proof for the relative independence of paracellular and transcellular conductances for Na^+ and Cl^- , respectively, in thin ascending limbs.¹³⁷

CLC-K1 associates with Barttin, an accessory subunit identified via positional cloning of the gene for Bartter syndrome with sensorineural deafness (see the “[Na⁺-Cl⁻ Transport by the Thick Ascending Limb: Basolateral Mechanisms](#)” section).¹⁶³ Barttin is expressed with CLC-K1 in thin ascending limbs, in addition to the TAL, DCT, and α -intercalated cells.^{158,163} Rat CLC-K1 is unique among the CLC-K orthologs and paralogs (CLC-K1/2 in rodents, CLC-NKB/NKA in humans) in that it can generate Cl^- channel activity in the absence of coexpression with Barttin; however, its human ortholog CLC-NKA is nonfunctional in the absence of Barttin.^{156,158,164} Regardless, Barttin coimmunoprecipitates with CLC-K1 and increases expression of the channel protein at the cell membrane.^{158,164} This so-called chaperone function seems to involve the transmembrane core of Barttin, whereas domains within the cytoplasmic carboxy terminus modulate channel properties (open probability and unitary conductance).¹⁶⁴

With respect to regulation in this nephron segment, vasopressin has stimulatory effects on Cl^- transport by the thin ascending limb, acting as in principal cells and TAL through V2 receptors and cAMP.¹⁵⁵ Water deprivation induces a fourfold increase in CLC-K1 mRNA, indicating transcriptional effects of vasopressin or medullary tonicity.¹⁶⁵ Basolateral calcium in turn inhibits Cl^- and Na^+ transport in the thin ascending limb via activation of the calcium-sensing receptor (CaSR).¹⁶⁶

NA⁺-CL⁻ TRANSPORT BY THE THICK ASCENDING LIMB

Apical Na⁺-Cl⁻ Transport

The TAL reabsorbs about 30% of filtered $\text{Na}^+\text{-Cl}^-$ (Fig. 6.1). In addition to an important role in the defense of the extracellular fluid volume, $\text{Na}^+\text{-Cl}^-$ reabsorption by the water-impermeable TAL is a critical component of the renal countercurrent multiplication system. The separation of $\text{Na}^+\text{-Cl}^-$ and water by the TAL is thus responsible for the capability of the kidney to dilute or concentrate the urine. In concert with the countercurrent mechanism, $\text{Na}^+\text{-Cl}^-$ reabsorption by the thin ascending limb and TAL increases medullary tonicity, facilitating water absorption by the collecting duct.

Notwithstanding the morphological heterogeneity described earlier, the cells of the medullary TAL, cortical TAL, and macula densa share the same basic transport mechanisms (Fig. 6.14). $\text{Na}^+\text{-Cl}^-$ reabsorption by the TAL is thus a secondarily active process, driven by the favorable electrochemical gradient for Na^+ established by the basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$.^{14,167} Na^+ , K^+ , and Cl^- are cotransported across the apical membrane by an electroneutral $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter, which generally requires the simultaneous presence of all three ions.¹⁴ Of note, under certain circumstances,

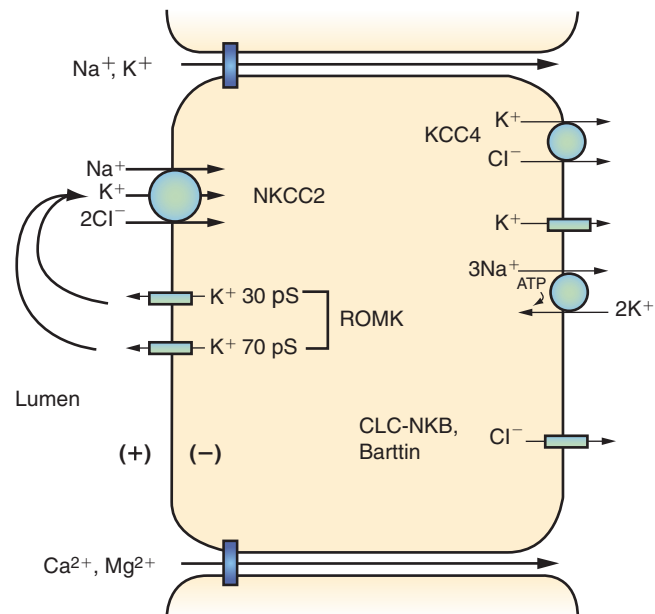


Fig. 6.14 Transepithelial $\text{Na}^+\text{-Cl}^-$ transport pathways in the thick ascending limb. CLC-K2, Cl^- channel, Barttin; KCC4, $\text{K}^+\text{-Cl}^-$ cotransporter-4; NKCC2, $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter-2; ROMK, renal outer medullary K^+ channel.

apical $\text{Na}^+\text{-Cl}^-$ transport in the TAL appears to be K^+ independent; this issue is reviewed later (see the “[Regulation of Na⁺-Cl⁻ Transport by the Thick Ascending Limb](#)” section). Apical $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransport is mediated by the cation-chloride cotransporter NKCC2, encoded by *SLC12A1*.¹⁶⁸ This is a member of the cation chloride cotransporter family of proteins that includes the thiazide-sensitive transporter NCC, and the potassium chloride cotransporters. Functional expression of NKCC2 in *Xenopus* oocytes yields Cl^- - and Na^+ -dependent uptake of Rb^+ (a radioactive substitute for K^+) and Cl^- - and K^+ -dependent uptake of $^{22}\text{Na}^+$.^{97,168–170} As expected, NKCC2 is sensitive to micromolar concentrations of furosemide, bumetanide, and other loop diuretics.¹⁶⁸

Immunofluorescence indicates expression of NKCC2 protein along the entire length of the TAL.¹⁶⁸ In particular, immunoelectron microscopy reveals expression in both rough (R) and smooth (S) cells of the TAL (see earlier discussion).¹⁴¹ NKCC2 expression in subapical vesicles is particularly prominent in smooth cells, suggesting a role for vesicular trafficking in the regulation of NKCC2 (see the “[Regulation of Na⁺-Cl⁻ Transport by the Thick Ascending Limb](#)” section).¹⁴¹ NKCC2 is also expressed in macula densa cells, which have been shown to possess apical $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransport activity.^{141,171} This latter observation is of considerable significance, given the role of the macula densa in TGF and renal renin secretion; luminal loop diuretics block TGF and the suppression of renin release by luminal Cl^- .¹⁴

Alternative splicing of exon 4 of the *SLC12A1* gene yields NKCC2 proteins that differ within transmembrane domain 2 and the adjacent intracellular loop; the functional significance of these variants appears primarily related to differences in binding to chloride.¹⁷² There are thus three different variants of exon 4, denoted “A,” “B,” and “F”; the variable inclusion of these cassette exons yields NKCC2-A, NKCC2-B,

and NKCC2-F proteins.^{168,170} Kinetic characterization reveals that these isoforms differ dramatically in ion affinities.^{168,170} In particular, NKCC2-F has a very low affinity for Cl^- ($K_m = 113 \text{ mmol/L}$) and NKCC2-B has a very high affinity ($K_m = 8.9 \text{ mmol/L}$); NKCC2-A has an intermediate affinity for Cl^- ($K_m = 44.7 \text{ mmol/L}$).¹⁷⁰ These isoforms differ in axial distribution along the tubule, with the F cassette expressed in the inner stripe of the outer medulla, the A cassette in the outer stripe, and the B cassette in cortical TAL.¹⁴ There is thus an axial distribution of the anion affinity of NKCC2 along the TAL, from a low-affinity, high-capacity transporter (NKCC2-F) to a high-affinity, low-capacity transporter (NKCC2-B). Although technically compromised by the considerable homology between the 3' end of these 96-base pair exons, *in situ* hybridization has suggested that rabbit macula densa exclusively expresses the NKCC2-B isoform.¹⁴ Notably, however, selective knockout of the B cassette exon 4 does not eliminate NKCC2 expression in the murine macula densa, which also seems to express NKCC2-A by *in situ* hybridization.¹⁷³ The comparative phenotypes of NKCC2-A and NKCC2-B knockout mice are consistent with the relative Cl^- affinity of each isoform, with NKCC2-B functioning as a high-affinity, low-capacity isoform and NKCC2-A functioning as a low-affinity, high-capacity isoform. Thus targeted deletion of NKCC2-A selectively reduces TGF responses at the higher range of tubular flow rates (a low-affinity, high-capacity situation), whereas NKCC2-B deletion reduces responses at low flow rates.¹⁷⁴ Loss of NKCC2-A almost abolishes the suppression of plasma renin activity by isotonic saline infusion, which is, if anything, more robust in NKCC2-B knockout mice than wild type littermates.¹⁷⁴

It should be mentioned in this context that the $\text{Na}^+\text{-H}^+$ exchanger NHE3 functions as an alternative mechanism for apical Na^+ absorption by the TAL. There is also evidence in mouse cortical TAL for $\text{Na}^+\text{-Cl}^-$ transport via parallel $\text{Na}^+\text{-H}^+$ and $\text{Cl}^-\text{-HCO}_3^-$ exchange, although the role of this mechanism in transepithelial $\text{Na}^+\text{-Cl}^-$ transport seems less prominent than in the PT.¹⁴ Indeed, apical $\text{Na}^+\text{-H}^+$ exchange mediated by NHE3 appears to function primarily in HCO_3^- absorption by the TAL.¹⁷⁵ There is thus a considerable upregulation of both apical $\text{Na}^+\text{-H}^+$ exchange and NHE3 protein in the TAL of acidotic animals, paired with an induction of AE2, a basolateral $\text{Cl}^-\text{-HCO}_3^-$ exchanger.^{176,177} NHE3 in the TAL is also upregulated by increased flow. However, this is not via shear stress, as demonstrated in the PT, but by the production of endogenous O_2^- and activation of PKC, a potential pathway for flow-stimulated bicarbonate reabsorption.¹⁷⁸

Apical K^+ Channels

Microperfused TALs develop a lumen-positive PD during perfusion with $\text{Na}^+\text{-Cl}^-$.^{179,180} This lumen-positive PD plays a critical role in the physiology of the TAL, driving the paracellular transport of Na^+ , Ca^{2+} , and Mg^{2+} (Fig. 6.14). Originally attributed to electrogenic Cl^- transport, the lumen-positive transepithelial PD in the TAL is generated by the combination of apical K^+ channels and basolateral Cl^- channels.^{14,167,180} The conductivity of the apical membrane of TAL cells is predominantly, if not exclusively, K^+ selective. Luminal recycling of K^+ via $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransport and apical K^+ channels, along with basolateral depolarization due to Cl^- exit through Cl^- channels, results in the lumen-positive transepithelial PD.^{14,167}

Several lines of evidence have indicated that apical K^+ channels are required for transepithelial $\text{Na}^+\text{-Cl}^-$ transport by the TAL.^{14,167} First, the removal of K^+ from luminal perfusate results in a marked decrease in $\text{Na}^+\text{-Cl}^-$ reabsorption by the TAL, as measured electrophysiologically; the residual $\text{Na}^+\text{-Cl}^-$ transport in the absence of luminal K^+ is sustained by K^+ movement into the luminal fluid apical K^+ channels, because the combination of K^+ removal and a luminal K^+ channel inhibitor (barium) almost abolishes the equivalent short circuit current.¹⁴ Apical K^+ channels are thus required for continued functioning of NKCC2, the apical $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter; the low luminal concentration of K^+ in this nephron segment would otherwise become limiting for transepithelial $\text{Na}^+\text{-Cl}^-$ transport.

Second, the net transport of K^+ across perfused TAL is less than 10% that of Na^+ and Cl^- ; about 90% of the K^+ transported by NKCC2 is recycled across the apical membrane via K^+ channels, resulting in minimal net K^+ absorption by the TAL.^{14,181}

Third, the intracellular K^+ activity of perfused TAL cells is about 15 to 20 mV above equilibrium due to furosemide-sensitive entry of K^+ via NKCC2.¹⁸² Given an estimated apical K^+ conductivity of about 12 mS/cm^2 , this intracellular K^+ activity yields a calculated K^+ current of about $200 \text{ }\mu\text{A/cm}^2$, which corresponds quantitatively to the uptake of K^+ by the apical $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter.¹⁶⁷

Fourth, the observation that Bartter syndrome can be caused by mutations in renal outer medullary potassium (ROMK, encoded by *KCNJ1*) provides genetic proof for the importance of K^+ channels in $\text{Na}^+\text{-Cl}^-$ absorption by the TAL (see later).¹⁸³ Finally, a novel ROMK inhibitor functions as a potent diuretic *in vivo*, primarily due to inhibition of TAL $\text{Na}^+\text{-Cl}^-$ transport.¹⁸⁴

Three types of apical K^+ channels have been identified in the TAL, a channel with a conductance of 30 picosiemens (30 pS), a channel with a conductance of 70 pS, and a high-conductance, calcium-activated maxi- K^+ channel (Fig. 6.14).^{185–187} The higher P_o and greater density of the 30-pS and 70-pS channels versus the maxi- K^+ channel suggest that these are the primary routes for K^+ recycling across the apical membrane; the 70-pS channel in turn appears to mediate about 80% of the apical K^+ conductance of TAL cells.¹⁸⁸ The low-conductance, 30-pS channel shares several electrophysiologic and regulatory characteristics with ROMK, the cardinal inward-rectifying K^+ channel that was initially cloned from renal outer medulla.¹⁴ In humans, three isoforms of ROMK (ROMK1, 2, and 3) are generated by alternative splicing of the *KCNJ1* gene; ROMK3 has not been detected in rat or mouse.¹⁸⁹ ROMK2 has the shortest amino terminus, ROMK1 has an additional 16 residues, and ROMK3 an additional 22 residues (compared with ROMK2). ROMK1 mRNA is expressed in the mid and late distal tubule and the CCD, and in the outer medullary collecting duct (OMCD), but not along the TAL (see later). ROMK2 mRNA is expressed from the medullary TAL through the CCD, but is absent from the OMCD.¹⁸⁹ ROMK3 is expressed from the medullary TAL through the DCT. ROMK protein has been identified at the apical membrane of medullary TAL, cortical TAL, and macula densa.¹⁹⁰ Furthermore, the 30-pS channel is also absent from the apical membrane of mice with homozygous deletion of the gene encoding ROMK.¹⁹¹ Notably, not all cells in the TAL are labeled with ROMK antibody, suggesting that ROMK

might be absent in the so-called HBC cells with HBC and low apical K^+ conductance (also see earlier discussion).^{140,142} HBC cells are thought to correspond to the smooth-surfaced morphologic subtype of TAL cells (S cells)¹⁴⁰; however, distribution of ROMK protein by immunoelectron microscopy has not as yet been reported.

ROMK clearly plays a critical role in Na^+ - Cl^- absorption by the TAL, given that loss-of-function mutations in this gene are associated with Bartter syndrome.¹⁸³ The role of ROMK in Bartter syndrome was initially discordant with the data, suggesting that the 70-pS K^+ channel is the dominant conductance at the apical membrane of TAL cells; heterologous expression of the ROMK protein in *Xenopus* oocytes had yielded a channel with a conductance of about 30 pS, suggesting that the 70-pS channel was distinct from ROMK.^{14,188} This paradox has been resolved by the observation that the 70-pS channel is absent from the TAL of ROMK knockout mice, indicating that ROMK proteins form a subunit of the 70-pS channel.¹⁹² A recent study specifically examined the effects of ROMK1 disruption in mice, and consistent with its absence along the TAL, discussed earlier, these mice did not display a Bartter syndrome phenotype.¹⁹³

ROMK activity in the TAL is clearly modulated by association with other proteins, such that coassociation with other subunits to generate the 70-pS channel is perfectly compatible with the known physiology of this protein. ROMK thus associates with scaffolding proteins NHERF-1 and NHERF-2 (see the “[Regulation of Proximal Tubular \$Na^+\$ - \$Cl^-\$ Transport: Neurohumoral Influences](#)” section) via the carboxyl-terminal PDZ-binding motif of ROMK; NHERF-2 is coexpressed with ROMK in the TAL.¹⁹⁴ The association of ROMK with NHERFs serves to bring ROMK into closer proximity to the cystic fibrosis transmembrane regulator protein (CFTR).¹⁹⁴ This ROMK–CFTR interaction is, in turn, required for the native ATP and glibenclamide sensitivity of apical K^+ channels in the TAL.¹⁹⁵

Paracellular Transport

TALs perfused with Na^+ - Cl^- develop a lumen-positive, transepithelial PD generated by the combination of apical K^+ secretion and basolateral Cl^- efflux.^{14,167,179,180,182} This lumen-positive PD plays a critical role in the paracellular reabsorption of Na^+ , Ca^{2+} , and Mg^{2+} by the TAL (Fig. 6.14). In the transepithelial transport of Na^+ , the stoichiometry of NKCC2 ($1Na^+:1K^+:2Cl^-$) is such that other mechanisms are necessary to balance the exit of Cl^- at the basolateral membrane; consistent with this requirement, data from mouse TAL have indicated that about 50% of transepithelial Na^+ transport occurs via the paracellular pathway.^{3,196} For example, the ratio of net Cl^- transepithelial absorption to net Na^+ absorption through the paracellular pathway is 2.4 ± 0.3 in micropperfused mouse medullary TAL segments, and the expected ratio of 50% of Na^+ transport occurs via the paracellular pathway. In the absence of vasopressin, apical Na^+ - Cl^- cotransport is not K^+ dependent (see the “[Regulation of \$Na^+\$ - \$Cl^-\$ Transport by the Thick Ascending Limb](#)” section), reducing the lumen-positive PD; switching to K^+ -dependent Na^+ - K^+ - $2Cl^-$ cotransport in the presence of vasopressin results in a doubling of Na^+ - Cl^- reabsorption, without an effect on oxygen consumption.^{3,196} Therefore, the combination of a cation-permeable paracellular pathway and an “active transport,” lumen-positive PD,¹⁶⁷ generated indirectly by the basolateral Na^+ - K^+ -ATPase,

results in a doubling of active Na^+ - Cl^- transport for a given level of oxygen consumption.^{3,197}

Unlike in the PT, the voltage-positive PD in the TAL is generated almost entirely by transcellular transport, rather than by diffusion across the lateral tight junction.¹⁵ Mouse TAL segments primarily express claudin-14, -16, -19, and -10.^{14,198–200} In vasopressin-stimulated mouse TAL segments, with a lumen-positive PD of 10 mV, the maximal increase in Na^+ - Cl^- in the lateral interspace is about 10 mmol/L.¹⁹⁶ Tight junctions in the TAL are cation selective, with P_{Na}/P_{Cl} ratios of 2 to 5.^{167,196} Notably, however, P_{Na}/P_{Cl} ratios can be highly variable in individual tubules, ranging from 2 to 5 in a single study of perfused mouse TAL.¹⁹⁶ Recently, it has been suggested that the claudin profile of cells along the TAL represents a mosaic, with some cell interfaces expressing the Na^+ -selective claudin-10, whereas others express claudin-3, -16, or -19. It was suggested that this mosaic pattern reveals the existence of spatially separated paracellular routes for Na^+ and Ca^{2+}/Mg^{2+} .²⁰¹ Regardless, assuming a net P_{Na}/P_{Cl} ratio of about 3, the maximal dilution potential in the mouse TAL is between 0.7 and 1.1 mV, consistent with a dominant effect of transcellular processes on the lumen-positive PD.¹⁹⁶

The reported transepithelial resistance in the TAL is between 10 and 50 $\Omega \cdot cm^2$; although this resistance is higher than that of the PT, the TAL is not considered a tight epithelium.^{14,167} Notably, however, water permeability of the TAL is extremely low, less than 1% that of the PT.¹⁶⁷ These hybrid characteristics—relatively low resistance and very low water permeability—allow the TAL to generate and sustain Na^+ - Cl^- gradients of up to 120 mmol/L.^{14,167} Not unexpectedly, given its lack of water permeability, the TAL does not express aquaporin water channels; as in the PT, the particular repertoire of claudins expressed in the TAL determines the resistance and ion selectivity of this nephron segment.

Mutations in human claudin-16 (previously called paracellin-1) and claudin-19 are associated with hereditary hypomagnesemia, suggesting that these claudins are particularly critical for the cation selectivity of TAL tight junctions.^{14,199} Heterologous expression of claudin-16 (paracellin-1) in the anion-selective LLC-PK1 cell line markedly increases Na^+ permeability without affecting Cl^- permeability; this generates a significant increase in the P_{Na}/P_{Cl} ratio (Fig. 6.15).²⁰² LLC-PK1 cells expressing claudin-16 also have increased permeability to other monovalent cations. There is, however, only a modest increase in Mg^{2+} permeability, suggesting that claudin-16 does not form an Mg^{2+} -specific pathway in the tight junction; rather, it may serve to increase the overall cation selectivity of the tight junction. Claudin-19 appears in turn to reduce P_{Cl} in LLC-PK1 cells, without having much effect on Mg^{2+} or Na^+ permeability.²⁰³ The claudin-16 and claudin-19 proteins interact in multiple systems, and coexpression of claudin-16 and claudin-19 synergistically increases the P_{Na}/P_{Cl} ratio in LLC-PK1 cells.^{203,204} Knockdown of claudin-16 in transgenic mice increases Na^+ absorption in the downstream collecting duct, with the development of hypovolemic hyponatremia after treatment with amiloride; claudin-19 knockdown mice exhibit an increase in fractional excretion of Na^+ and a doubling in serum aldosterone levels.^{204,205} In summary, therefore, claudin-16 and claudin-19 interact to confer the cation selectivity of tight junctions in the TAL, contributing significantly to the transepithelial absorption of Na^+ in this nephron segment.

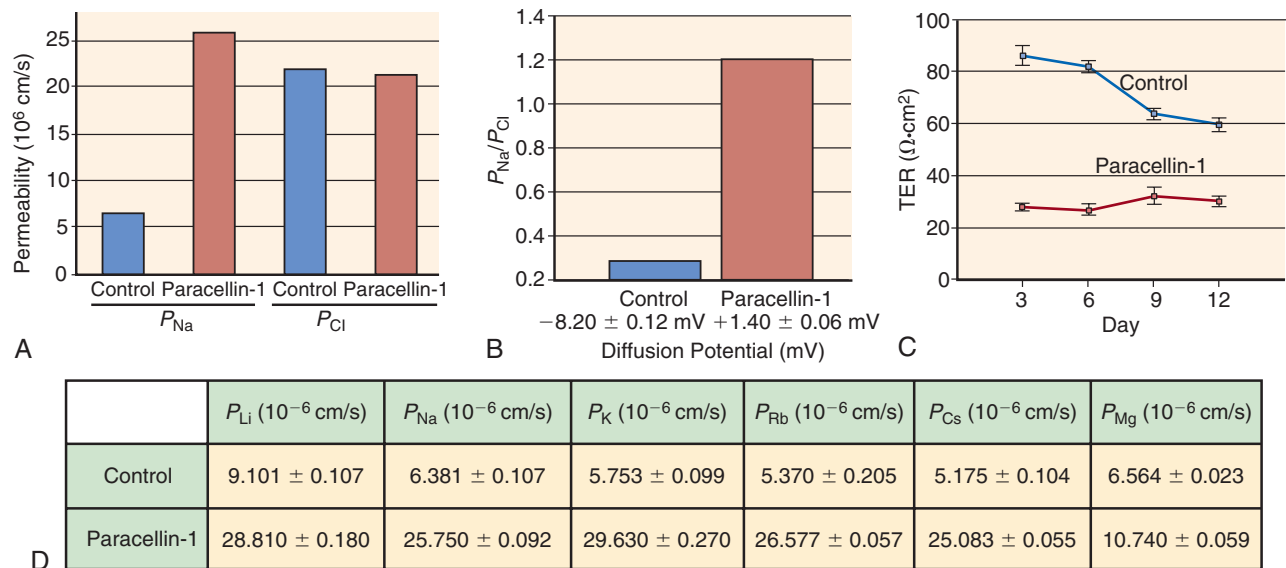


Fig. 6.15 Effect of claudin-16 (formerly called paracellin-1) overexpression in LLC-PK1 cells. (A) Effects of claudin-16 on the permeability of Na^+ and Cl^- in LLC-PK1 cells. (B) Ratio of P_{Na} to P_{Cl} and diffusion potential (bottom) across an LLC-PK1 cell monolayer. (C) Transepithelial resistance across an LLC-PK1 cell monolayer over a period of 12 days in cells expressing claudin-16 and control cells. (D) Summary of the effects of claudin-16 on permeability of various cations in LLC-PK1 cells. (From Hou J, Paul DL, Goodenough DA. Paracellin-1 and the modulation of ion selectivity of tight junctions. *J Cell Sci.* 2005;118:5109–5118.)

Other claudins expressed in the TAL modulate the function of claudin-16–claudin-19 heterodimers or have independent effects. Claudin-14 interacts with claudin-16, disrupting cation selectivity of the paracellular barrier in cells that also coexpress claudin-19.²⁰⁶ Claudin-14 expression in the TAL is calcium dependent via the CaSR, providing a novel axis for calcium-dependent regulation of paracellular calcium transport (see later).^{206–208} Claudin-10 appears to modulate paracellular Na^+ permeability specifically, with impaired paracellular Na^+ transport in claudin-10 knockout mice and a salt-wasting nephropathy in humans bearing compound heterozygous claudin-10 mutations.^{33,209}

Basolateral Mechanisms

The basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$ is the primary exit pathway for Na^+ at the basolateral membrane of TAL cells. The Na^+ gradient generated by $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity is also thought to drive the apical entry of Na^+ , K^+ , and Cl^- via NKCC2, the furosemide-sensitive $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter.¹⁴ Inhibition of $\text{Na}^+\text{-K}^+\text{-ATPase}$ with ouabain thus collapses the lumen-positive PD and abolishes transepithelial $\text{Na}^+\text{-Cl}^-$ transport in the TAL.^{179,180,197} Basolateral exit of Cl^- from TAL cells is primarily but not exclusively electrogenic, mediated by an approximately 10-pS Cl^- channel.^{14,167,210} Reductions in basolateral Cl^- depolarize the basolateral membrane, whereas decreases in intracellular Cl^- induced by luminal furosemide have a hyperpolarizing effect.¹⁴ Intracellular Cl^- activity during transepithelial $\text{Na}^+\text{-Cl}^-$ transport is above its electrochemical equilibrium,¹⁴ with an intracellular negative voltage of -40 to -70 mV that drives basolateral Cl^- exit.^{14,167}

At least two CLC chloride channels, CLC-K1 and CLC-K2 (CLC-NKA and CLC-NKB in humans), are coexpressed in this nephron segment.^{158,163} However, an increasing body of evidence has indicated that the dominant Cl^- channel in the TAL is encoded by CLC-K2. First, CLC-K1 is heavily expressed

at the apical and basolateral membranes of the thin ascending limb, and the phenotype of the corresponding knockout mouse is consistent with primary dysfunction of thin ascending limbs, rather than the TAL (see $\text{Na}^+\text{-Cl}^-$ transport by the thin ascending limb).^{3,137,156,159} Second, loss-of-function mutations in CLC-NKB are associated with Bartter syndrome, providing genetic evidence for a dominant role of this channel in $\text{Na}^+\text{-Cl}^-$ transport in the TAL.¹⁶⁰ More recently, a very common T481S polymorphism in human CLC-NKB was shown to increase channel activity by a factor of 20; preliminary data have indicated an association with hypertension, suggesting that this gain-of-function in CLC-NKB increases $\text{Na}^+\text{-Cl}^-$ transport by the TAL and/or other segments of the distal nephron.^{211–213} Finally, CLC-K2 protein is heavily expressed at the basolateral membrane of the mouse TAL, with additional expression in the DCT, CNT, and α -intercalated cells.²¹⁴ Recently, *Clc-k2* was deleted from mice, leading to salt wasting, resembling Bartter syndrome, and to a loss of the around 10-pS chloride channel in TAL cells.¹⁶² Deletion of this gene also abrogated the response to furosemide, indicating the central role that this channel plays in transepithelial NaCl transport. This approach also permitted conclusions to be drawn regarding the location of CLC-K1 and CLC-K2 in the kidney, even though antibodies cross-react against the two related chloride channels. In the *Clc-k2* knockout mice, the reaction product could be seen to extend beyond the thin ascending limb into the medullary TAL, suggesting that CLC-K1 plays a role there.

A key advance was the characterization of the Barttin subunit of CLC-K channels, which is coexpressed with CLC-K1 and CLC-K2 in several nephron segments, including TAL.^{158,163} Unlike rat CLC-K1, the rat CLC-K2, human CLC-NKA, and human CLC-NKB paralogs are not functional in the absence of Barttin coexpression.^{163,164} CLC-NKB coexpressed with Barttin is highly selective for Cl^- , with a permeability series

of $\text{Cl}^- \gg \text{Br}^- = \text{NO}_3^- > \text{I}^-$.^{14,158,163} CLC-NKB/Barttin channels are activated by increases in extracellular Ca^{2+} and are pH sensitive, with activation at an alkaline extracellular pH and marked inhibition at an acidic pH.¹⁶³ CLC-NKA/Barttin channels have similar pH and calcium sensitivities, but exhibit higher permeability to Br^- .¹⁶³ Strikingly, despite the considerable homology between the CLC-NKA/NKB proteins, these channels also differ considerably in pharmacologic sensitivity to various Cl^- channel blockers, potential lead compounds for the development of paralog-specific inhibitors.²¹⁵

Correlation between functional characteristics of CLC-K proteins with native Cl^- channels in TAL has been problematic, but the recent knockout work discussed earlier has begun to provide clarity. Wide variation in single-channel conductance has been reported for basolateral Cl^- channels in the TAL.²¹⁶ This is perhaps due to the use of collagenase and other conditions for the preparation of tubule fragments and/or basolateral vesicles, manipulations that potentially affect channel characteristics.²¹⁶ Notably, single-channel conductance has not been reported for CLC-NKB/Barttin channels because of the difficulty in expressing the channel in heterologous systems; this complicates the comparison of CLC-NKB/Barttin with native Cl^- channels. Single-channel conductance has, however, been reported for the V166E mutant of rat CLC-K1, which alters gating of the channel without expected effects on single-channel amplitude – coexpression with Barttin increases the single-channel conductance of V166E CLC-K1 from about 7 to 20 pS.¹⁶⁴ Therefore, part of the reported variability in native single-channel conductance may reflect heterogeneity in the interaction between CLC-NKB and/or CLC-NKA with Barttin. Regardless, a study using whole-cell recording techniques has suggested that CLC-K2 (CLC-NKB in humans) is the dominant Cl^- channel in TAL and other segments of the distal nephron.²¹⁶ Like CLC-NKB/Barttin, this native channel is highly Cl^- selective, with considerably weaker conductance for Br^- and I^- ; CLC-NKA/Barttin channels exhibit higher permeability to Br^- .^{14,158,163,216} TAL cells from wild type mice exhibited a dominant basolateral chloride conductance of 8 pS, which was entirely absent in *Clc-k2* knockout mice. Coupled with the strong Bartter syndrome phenotype, these results support the key role of this chloride channel in driving transepithelial NaCl transport along the TAL.

Electroneutral $\text{K}^+\text{-Cl}^-$ cotransport has also been implicated in transepithelial $\text{Na}^+\text{-Cl}^-$ transport in the TAL (Fig. 6.14), functioning in K^+ -dependent Cl^- exit at the basolateral membrane.¹⁴ The KCC KCC4 is thus expressed at the basolateral membrane of medullary and cortical TAL, in addition to the macula densa.^{217,218} There is also functional evidence for $\text{K}^+\text{-Cl}^-$ cotransport at the basolateral membrane of this section of the nephron. First, TAL cells contain a Cl^- -dependent NH_4^+ transport mechanism that is sensitive to 1.5-mmol/L furosemide and 10-mmol/L barium (Ba^{2+}).²¹⁹ NH_4^+ ions have the same ionic radius as K^+ and are transported by KCC4 and other KCCs; KCC4 is also sensitive to Ba^{2+} and millimolar furosemide, consistent with the pharmacology of $\text{NH}_4^+\text{-Cl}^-$ cotransport in the TAL.^{219–221} Second, to account for the effects on the transmembrane PD of basolateral Ba^{2+} and/or increased K^+ , it has been suggested that the basolateral membrane of TAL contains a Ba^{2+} -sensitive $\text{K}^+\text{-Cl}^-$ transporter; this is also consistent with the known expression of Ba^{2+} -sensitive KCC4 at the basolateral membrane.^{14,217,218,221} Third,

increases in basolateral K^+ cause Cl^- -dependent cell swelling in *Amphiuma* early distal tubule, an analog of the mammalian TAL; in *Amphiuma* LBC cells with low basolateral conductance, analogous to mammalian LBC cells (see the “ $\text{Na}^+\text{-Cl}^-$ Transport by the Thick Ascending Limb: Apical $\text{Na}^+\text{-Cl}^-$ Transport” section), this cell swelling was not accompanied by changes in basolateral membrane voltage or resistance, consistent with $\text{K}^+\text{-Cl}^-$ transport.^{140,142,222}

There is thus considerable evidence for basolateral $\text{K}^+\text{-Cl}^-$ cotransport in the TAL, mediated by KCC4.^{217,218} However, direct confirmation of a role for basolateral $\text{K}^+\text{-Cl}^-$ cotransport in transepithelial $\text{Na}^+\text{-Cl}^-$ transport is lacking. Indeed, KCC4-deficient mice do not have a prominent defect in function of the TAL, but exhibit instead a renal tubular acidosis.²¹⁷ The renal tubular acidosis in these mice has been attributed to defects in acid extrusion by $\text{H}^+\text{-ATPase}$ in α -intercalated cells; however, this phenotype is conceivably the result of a reduction in medullary NH_4^+ reabsorption by the TAL due to the loss of basolateral NH_4^+ exit mediated by KCC4.^{217,220,223}

Finally, there is also evidence for the existence of Ba^{2+} -sensitive K^+ channel activity at the basolateral membrane of the TAL, providing an alternative exit pathway for K^+ to that mediated by KCC4.^{224–226} Such channel activity may help stabilize the basolateral membrane potential above the equilibrium potential for Cl^- , thus maintaining a continuous driving force for Cl^- exit via CLC-NKB/Barttin Cl^- channels.²²⁶ Patch-clamp experiments have identified two types of K^+ channels in the basolateral membrane of the TAL: a 40-pS inwardly rectifying K^+ channel and an Na^+ - and Cl^- -activated, 80- to 150-pS K channel (Kca4.1 or slo2.2).^{225,227} The 40-pS K^+ channel was absent in the TAL of *Kcnj10* knockout mice, suggesting that the 40-pS K^+ channel is a KIR4.1 and KIR4.5 heterotetramer.²²⁷ Although KIR4.1 is also detected in human TAL, loss-of-function *KCNJ10* mutations do not show the phenotype of Bartter syndrome, suggesting that the disruption of KIR4.1 has no significant effect on transport function in the TAL.²²⁸ This may reflect secondary activation of alternative K^+ channels along TAL, as *Kcnj10* knockout mice demonstrate vasopressin-induced stimulation of the 80- to 150-pS K channel. Basolateral K^+ channels may also attenuate the increases in intracellular K^+ that are generated by the basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$, thus maintaining transepithelial $\text{Na}^+\text{-Cl}^-$ transport.^{224–226}

REGULATION OF $\text{Na}^+\text{-Cl}^-$ TRANSPORT BY THE THICK ASCENDING LIMB

Activating Influences

Transepithelial $\text{Na}^+\text{-Cl}^-$ transport by the TAL is regulated by a complex blend of competing neurohumoral influences, which are required to maintain the urinary concentrating capacity, and modulate salt balance. In particular, increases in intracellular cAMP tonically stimulate ion transport in the TAL; the stimulatory hormones and mediators that increase cAMP in this nephron segment include vasopressin, PTH, glucagon, calcitonin, and β -adrenergic activation (Fig. 6.9). These overlapping cAMP-dependent stimuli are thought to result in maximal baseline stimulation of transepithelial $\text{Na}^+\text{-Cl}^-$ transport.⁹⁴ For example, characterization of the in vivo effect of these hormones requires the prior simultaneous suppression or absence of circulating vasopressin, PTH, calcitonin, and glucagon.⁹⁴ This baseline activation is, in turn,

modulated by a number of negative influences, most prominently prostaglandin E₂ (PGE₂) and extracellular Ca²⁺ (Fig. 6.9). Other hormones and autocrine factors working through cGMP-dependent signaling, including NO, have potent negative effects on Na⁺-Cl⁻ transport within the TAL.²³⁰ By contrast, Ang II has a stimulatory effect on Na⁺-Cl⁻ transport within the TAL.²³¹

Vasopressin is perhaps the most extensively studied positive modulator of transepithelial Na⁺-Cl⁻ transport in the TAL. The TAL, with the exception of macula densa cells,²³² expresses type 2 vasopressin receptors (V₂Rs) at both the mRNA and protein levels, and microdissected TALs respond to the hormone with an increase in intracellular cAMP.²³³ Vasopressin activates apical Na⁺-K⁺-2Cl⁻ cotransport within minutes in perfused mouse TAL segments and also exerts a longer-term influence on NKCC2 expression and function. The acute activation of apical Na⁺-K⁺-2Cl⁻ cotransport is achieved at least in part by the stimulated exocytosis of NKCC2 proteins, from subapical vesicles to the plasma membrane.²³⁴ This trafficking-dependent activation is abrogated by treatment of perfused tubules with tetanus toxin, which cleaves the vesicle-associated membrane proteins VAMP-2 and VAMP-3.²³⁴ As V₂Rs are prominently expressed along the collecting duct, as well as along the TAL and DCT, it has been difficult to separate the roles of vasopressin signaling in the two sites. This limitation was recently overcome by introducing a dominant-negative V₂R mutant into TAL cells of rat. The rats demonstrated polyuria and defective urinary concentration, as well as hypercalciuria, reminiscent of mild Bartter syndrome.²³² The absence of V₂R expression by macula densa cells was suggested to maintain TGF independent of vasopressin signaling.

Activation of NKCC2 by vasopressin is also associated with the phosphorylation of a cluster of amino-terminal threonines in the transporter protein; treatment of rats with the V₂ agonist desmopressin (DDAVP) induces phosphorylation of these residues in vivo, as measured with a phosphospecific antibody.²³⁴ These threonine residues are substrates for SPAK (STE20/SPS1-related proline/alanine-rich kinase) and OSR1 (oxidative stress-responsive kinase 1), first identified by Gagnon and colleagues as key regulatory kinases for NKCC1 and other cation-chloride cotransporters.²³⁵ SPAK and OSR1 are in turn activated by upstream WNK (with no lysine [K] kinases), such that SPAK or OSR1 requires coexpression with WNK4 to activate NKCC1 fully, at least in in vitro systems.²³⁵ By contrast, two reports of *Wnk4* knockout mice suggested that WNK4 does not regulate NKCC2 in vivo.^{236,237} Regardless, expression of WNK3 in *Xenopus* oocytes results in activatory phosphorylation of the amino-terminal threonines in NKCC2 that are phosphorylated in TAL cells after treatment with DDAVP.^{234,238}

The amino-terminal phosphorylation of NKCC2 by SPAK and/or OSR1 kinases appears to be important for activity of the transporter in the native TAL. The amino-terminus of NKCC2 contains a predicted binding site for SPAK,²³⁹ proximal to the sites of regulatory phosphorylation; the analogous binding site is required for activation of the NKCC1 cotransporter.²⁴⁰ SPAK also requires the sorting receptor SORLA (sorting protein-related receptor with A-type repeats) for proper trafficking within TAL cells, such that targeted deletion of *Sorla* results in a marked reduction in amino-terminal NKCC2 phosphorylation.²⁴¹ The role of the upstream WNK kinases is illustrated by the phenotype of a “knock-in” mouse

strain, in which the knocked-in mutant SPAK cannot be activated by upstream WNK kinases; these mice have a marked reduction in amino-terminal phosphorylation of both NKCC2 and the thiazide-sensitive NCC, with associated salt-sensitive hypotension.²⁴² The upstream WNK kinases appear to regulate SPAK and NKCC2 in a chloride-dependent fashion, phosphorylating and activating SPAK and the transporter in response to a reduction in intracellular chloride concentration (see the “Integrated Na⁺-Cl⁻ and K⁺ Transport in the Distal Nephron” section).²⁴³

Of the two kinases, SPAK and OSR1, OSR1 is perhaps more critical for NKCC2 function in the TAL, because pan-renal epithelial-specific deletion of *Osr1* leads to decreased amino-terminal phosphorylation of NKCC2, and a Bartter syndrome-like phenotype.²⁴⁴ Several groups reported an increase in NKCC2 amino-terminal phosphorylation and an increased response to furosemide in *Spak* knockout mice, suggesting overcompensation by OSR1.^{245–247} By contrast, it was reported that baseline Na⁺ absorption by isolated perfused TAL segments is profoundly impaired in SPAK-deficient mice, though this may reflect an absence of activating factors present in vivo.²⁴⁸ Truncated species of SPAK protein have also been detected in kidney due to generation of alternative mRNA species that lack the amino-terminal kinase domain and to proteolytic degradation; both forms of SPAK function as dominant-negative inhibitors of the full-length kinase, abrogating the usual stimulatory effect on coexpressed NKCC2 or NKCC1.^{247,249} A lack of these species in SPAK knockout mice may also contribute to the increased amino-terminal NKCC2 phosphorylation observed, due to the removal of a dominant-negative effect on OSR1. Further complexity arises from the influence of the adaptor protein calcium-binding protein 39 (CAB39, also called mouse protein-25, MO25²⁵⁰), which can both increase SPAK/OSR1-driven phosphorylation of NKCC2 and also activate SPAK/OSR1 directly, without the need for upstream phosphorylation by WNK kinases, by promoting dimerization of the kinases.^{250,251} WNK4 is also capable of direct interaction with CAB39, promoting activation of NKCC2 in the absence of SPAK or OSR1 expression.²⁵² In support of SPAK/OSR1-independent activation of NKCC2, in mice in which both SPAK and OSR1 were disrupted in the kidney, significant amino-terminal phosphorylation of NKCC2 was still detected. However, although this could reflect a direct effect of WNK4 on NKCC2, it could also result from phosphorylation by an as-yet unidentified kinase. Therefore, there appear to be at least three potential pathways for NKCC2 activation—a WNK4-dependent SPAK/OSR1 pathway, a WNK4-independent SPAK/OSR1 pathway, and a SPAK/OSR1-independent WNK4 pathway.²⁵² It should be noted, however, that amino-terminal phosphorylation has a smaller effect on NKCC2 activity than it does on NKCC1 and NCC.²³⁸ Thus changes in NKCC2 phosphorylation may not always reflect changes in NKCC2 activity.²⁵³

Vasopressin has also been shown to alter the stoichiometry of furosemide-sensitive apical Cl⁻ transport in the TAL, from a K⁺-independent Na⁺-Cl⁻ mode to the classic Na⁺-K⁺-2Cl⁻ cotransport stoichiometry.³ In the absence of vasopressin, ²²Na⁺ uptake by mouse medullary TAL cells is not dependent on the presence of extracellular K⁺, whereas the addition of the hormone induces a switch to K⁺-dependent ²²Na⁺ uptake. Underscoring the metabolic advantages of paracellular Na⁺ transport, which is critically dependent on the apical entry

of K^+ via $Na^+K^+-2Cl^-$ cotransport (see previously), vasopressin accomplishes a doubling of transepithelial Na^+Cl^- transport without affecting $^{22}Na^+$ uptake (an indicator of transcellular Na^+Cl^- transport); this doubling in transepithelial absorption occurs without an increase in O_2 consumption, highlighting the energy efficiency of ion transport by the TAL.³ The mechanism of this shift in the apparent stoichiometry of NKCC2 is not completely clear. However, splice variants of mouse NKCC2 with a novel, shorter carboxyl-terminus have been found to confer sensitivity to cAMP when coexpressed with full-length NKCC2.²⁵⁴ Notably, these shorter splice variants appear to encode furosemide-sensitive, K^+ -independent NCCs when expressed alone in *Xenopus* oocytes.²⁵⁵ The in vivo relevance of these phenomena is not clear, however, nor is it known whether similar splice variants exist in species other than mouse.

In addition to its acute effects on NKCC2, the apical $Na^+K^+-2Cl^-$ cotransporter, vasopressin increases transepithelial Na^+Cl^- transport by activating apical K^+ channels and basolateral Cl^- channels in the TAL.^{94,233} Details have yet to emerge of the regulation of the basolateral $ClC-K2/Barttin$ Cl^- channel complex by vasopressin, cAMP, and related pathways. However, the apical K^+ channel ROMK is directly phosphorylated by PKA on three serine residues (S25, S200, and S294 in the ROMK2 isoform). Phosphorylation of at least two of these three serines is required for detectable K^+ channel activity in *Xenopus* oocytes; mutation of all three serines to alanine abolishes phosphorylation and transport activity, and all three serines are required for full channel activity.²⁵⁶ These three phospho acceptor sites have distinct effects on ROMK activity and expression.²⁵⁷ Phosphorylation of the amino-terminal S25 residue appears to regulate trafficking of the channel to the cell membrane, without effects on channel gating; this serine is also a substrate for the SGK1 kinase, which activates the channel via an increase in expression at the membrane.²⁵⁷ By contrast, phosphorylation of the two carboxyl-terminal serines modulates open channel probability via effects on pH-dependent gating and on activation by the binding of phosphatidylinositol 4,5-bisphosphate (PIP2) to the carboxyl-terminal domain of the channel.^{258,259}

Vasopressin also has considerable long-term effects on transepithelial Na^+Cl^- transport by the TAL. Sustained increases in circulating vasopressin result in marked hypertrophy of medullary TAL cells, accompanied by a doubling in baseline active Na^+Cl^- transport.²³³ Water restriction or treatment with DDAVP also results in an increase in abundance of the NKCC2 protein in rat TAL cells. Consistent with a direct effect of vasopressin-dependent signaling, expression of NKCC2 is reduced in mice with a heterozygous deletion of the G_s stimulatory G protein, through which the V_2R activates cAMP generation.²³³ Increases in cAMP are thought to induce transcription of the *Slc12a1* gene that encodes NKCC2 directly, given the presence of a cAMP-response element in the 5' promoter.^{233,234} Abrogation of the tonic negative effect of PGE_2 on cAMP generation with indomethacin also results in a considerable increase in abundance of the NKCC2 protein.²³³ Finally, in addition to these effects on NKCC2 expression, water restriction or DDAVP treatment increases abundance of the ROMK protein at the apical membrane of TAL cells.²⁶⁰

Recently, a role for β_3 -adrenergic receptors in activation of NKCC2 was proposed, with administration of the selective

agonist BRL37344 increasing NKCC2 phosphorylation ex vivo in wild type but not β_3 -adrenergic receptor knockout mice, which also displayed a mild Na^+ - and K^+ -wasting phenotype.²⁶¹

Inhibitory Influences

The stimulation of transepithelial Na^+Cl^- transport by cAMP-generating hormones (e.g., vasopressin, PTH) is modulated by a number of negative neurohumoral influences (Fig. 6.9).⁹⁴ In particular, extracellular Ca^{2+} and PGE_2 exert dramatic inhibitory effects on ion transport by this and other segments of the distal nephron through a plethora of synergistic mechanisms. Extracellular Ca^{2+} and PGE_2 both activate the G_i inhibitory G protein in TAL cells, opposing the stimulatory, G_s -dependent effects of vasopressin on intracellular levels of cAMP.^{262,263} Extracellular Ca^{2+} exerts its effect through the CaSR, which is heavily expressed at the basolateral membrane of TAL cells; PGE_2 primarily signals through EP_3 prostaglandin receptors.^{94,263,264} The increases in intracellular Ca^{2+} due to the activation of the CaSR and other receptors directly inhibit cAMP generation by a Ca^{2+} -inhibitable adenylate cyclase that is expressed in the TAL, accompanied by an increase in phosphodiesterase-dependent degradation of cAMP (Fig. 6.16).^{263,265} These negative stimuli likely inhibit baseline transport in the TAL; abrogation of the negative effect of PGE_2 with indomethacin results in a considerable increase in abundance of the NKCC2 protein, whereas targeted deletion of the CaSR in mouse TAL activates NKCC2 via increased amino-terminal phosphorylation.^{189,209}

Activation of the CaSR and other receptors in the TAL also results in the downstream generation of arachidonic acid metabolites, with potent negative effects on Na^+Cl^- transport (Fig. 6.16). Extracellular Ca^{2+} thus activates PLA_2 in TAL cells, leading to the liberation of arachidonic acid. This arachidonic acid is in turn metabolized by cytochrome P450 ω -hydroxylase to 20-HETE or by COX-2 to PGE_2 ; cytochrome P450 ω -hydroxylation generally predominates in response to activation of the CaSR in TAL.²⁶³ 20-HETE has very potent negative effects on apical $Na^+K^+-2Cl^-$ cotransport, apical K^+ channels, and the basolateral $Na^+K^+-ATPase$.^{94,263} PLA_2 -dependent generation of 20-HETE also underlies in part the negative effect of bradykinin and Ang II on Na^+Cl^- transport.^{94,263} Activation of the CaSR also induces tumor necrosis factor- α expression in the TAL, which activates COX-2 and thus generation of PGE_2 (Fig. 6.16); this PGE_2 in turn results in additional inhibition of Na^+Cl^- transport.²⁶³

The relative importance of the CaSR in the regulation of Na^+Cl^- transport by the TAL is dramatically illustrated by the phenotype of a handful of patients with gain-of-function mutations in this receptor. In addition to suppressed PTH and hypocalcemia, the usual phenotype caused by gain-of-function mutations in the CaSR (autosomal dominant hypoparathyroidism), these patients manifest a hypokalemic alkalosis, polyuria, and increases in circulating renin and aldosterone.^{266,267} Therefore, the persistent inhibition of Na^+Cl^- transport in the TAL by these overactive mutants of the CaSR causes a rare subtype of Bartter syndrome, type V, in the genetic classification of this disease.²⁶³

Activation of the CaSR also affects claudin expression in TAL cells via downregulation of microRNAs, leading to PTH-independent hypercalciuria (see Chapter 7).^{206–208,268}

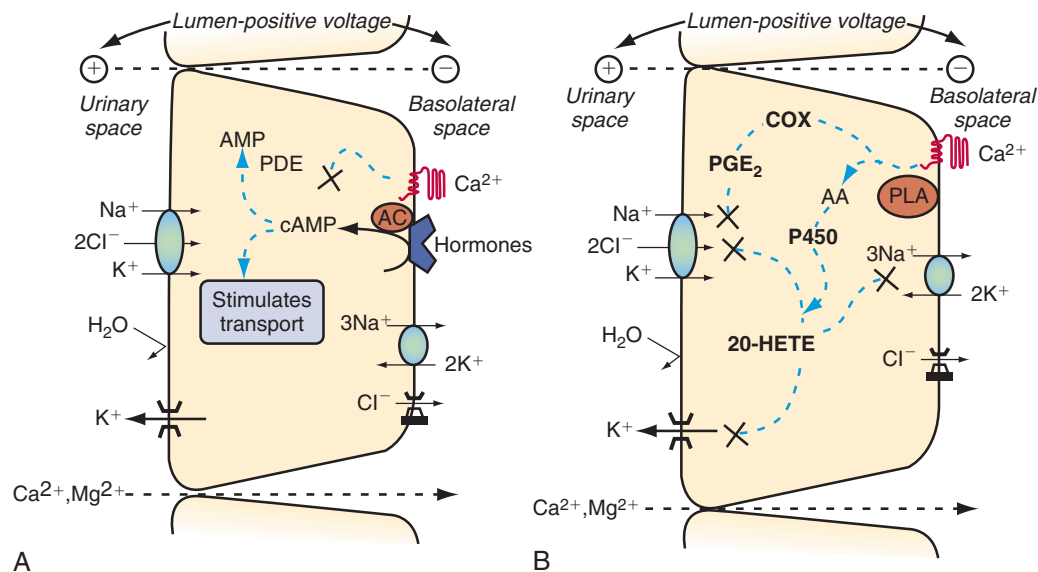


Fig. 6.16 Inhibitory effects of the calcium-sensing receptor (CaSR) on transepithelial Na^+-Cl^- transport in the thick ascending limb. (A) Activation of the basolateral CaSR inhibits the generation of cAMP in response to vasopressin and other hormones (see text for details). (B) Stimulation of phospholipase A_2 by the CaSR leads to liberation of arachidonic acid, which is in turn metabolized by cytochrome P450 ω -hydroxylase to 20-HETE (20-hydroxyeicosatetraenoic acid), or by cyclooxygenase-2 (COX-2) to prostaglandin E_2 (PGE_2). 20-HETE is a potent natriuretic factor, inhibiting apical $\text{Na}^+-\text{K}^+-2\text{Cl}^-$ cotransport, apical K^+ channels, and the basolateral $\text{Na}^+-\text{K}^+-\text{ATPase}$. Activation of the CaSR also induces tumor necrosis factor- α (TNF- α) expression in the TAL, which activates COX-2 and thus generation of PGE_2 , leading to additional inhibition of Na^+-Cl^- transport. (From Hebert SC. Calcium and salinity sensing by the thick ascending limb: a journey from mammals to fish and back again. *Kidney Int Suppl.* 2004;91:S28–S33.)

Uromodulin

TAL cells are unique in expressing the membrane-bound, glycosylphosphatidylinositol (GPI)-anchored protein, uromodulin (Tamm-Horsfall glycoprotein), which is not expressed by macula densa cells or the downstream DCT. Uromodulin is released by proteolytic cleavage at the apical membrane and is secreted as the most abundant protein in normal human urine (20 to 100 mg/day).²⁶⁹ Uromodulin has a host of emerging roles in the physiology and biology of the TAL. A high-salt diet increases uromodulin expression, suggesting a role in ion transport.²⁶⁹ In this regard, uromodulin facilitates membrane trafficking and function of the NKCC2 protein, with similar effects on apical ROMK protein.^{270,271}

Autosomal dominant mutations in the *UMOD* gene encoding uromodulin are associated with medullary cystic disease type 2 and familial juvenile hyperuricemic nephropathy. Now referred to as uromodulin-associated kidney disease, this syndrome includes progressive tubulointerstitial damage and chronic kidney disease (CKD), variably penetrant hyperuricemia and gout, and variably penetrant renal cysts that are typically confined to the corticomedullary junction.²⁶⁹ The causative mutations tend to affect conserved cysteine residues within the amino-terminal half of the protein, leading to protein misfolding and retention within the endoplasmic reticulum.^{269,272} More common genetic variants in the *UMOD* promoter have recently been linked in genomewide association studies with a risk of CKD and hypertension.²⁶⁹ These susceptibility variants have a high frequency (≈ 0.8) and confer an approximately 20% higher risk for CKD and a 15% risk for hypertension.²⁷³ These polymorphisms are associated with more abundant renal uromodulin transcript and higher urinary uromodulin excretion due to activating effects on the *UMOD* promoter.^{273,274} Overexpression of uromodulin in

transgenic mice leads to distal tubular injury, with segmental dilation and an increased tubular cast area relative to wild type mice; similar lesions are increased in frequency in older adults homozygous for susceptibility variants in *UMOD* when compared with those homozygous for protective variants.²⁷³

Uromodulin-transgenic mice also manifest salt-sensitive hypertension due to activation of the SPAK kinase and activating amino-terminal phosphorylation of NKCC2. Human hypertensive individuals homozygous for susceptibility variants in *UMOD* appear to have an analogous phenotype, with exaggerated natriuresis in response to furosemide compared with those who are homozygous for protective variants.²⁷³ These findings are compatible with the stimulatory effects of uromodulin on NKCC2 and ROMK—that is, a net gain of function in TAL transport.^{270,271} Uromodulin excretion also appears to parallel transport activity of the TAL, with common polymorphisms in the *KCNJ1* gene encoding ROMK and two genes involved in regulating SPAK and OSR1 kinase activity (*SORL1* and *CAB39*).²⁷⁴

ANATOMY OF THE DISTAL NEPHRON

The distal nephron that extends beyond the TAL is the final arbiter of urinary Na^+-Cl^- excretion and a critical target for natriuretic and antinatriuretic stimuli. The understanding of the cellular organization and molecular phenotype of the distal nephron continues to evolve and merits a brief review in this context. The DCT begins at a variable distance after the macula densa, with an abrupt transition between NKCC2-positive cortical TAL cells and DCT cells that express the thiazide-sensitive NCC. Considerable progress has been made in the phenotypic classification of cell types in the DCT and adjacent nephron segments, based on the expression of an

expanding list of transport proteins and other markers.²⁷⁵ This analysis has revealed considerable differences in the organization of the DCT, CNT, and CCD in rodent, rabbit, and human kidneys. In general, rabbit kidneys are unique in the axial demarcation of DCT, CNT, and CCD segments, at both a molecular and morphologic level; the organization of the DCT to CCD is considerably more complex in other species, with boundaries that are much less absolute.²⁷⁵ Notably, however, the overall repertoire of transport proteins expressed does not vary among these species; what differs is the specific cellular and molecular organization of this segment of the nephron.

The early DCT (DCT1) of mouse kidney expresses NCC and a specific marker, parvalbumin, which also distinguishes the DCT1 from the adjacent cortical TAL (Fig. 6.17).²⁷⁶ Targeted deletion of parvalbumin in mice reveals that this intracellular Ca^{2+} -binding protein is required for full activity of NCC in the DCT.²⁷⁷ Cells of the late DCT (DCT2) in mice coexpress NCC with proteins involved in transcellular Ca^{2+} transport, including the apical calcium channel, TRPV5 (previously ECAC1), the cytosolic calcium-binding protein calbindin $\text{D}_{28\text{K}}$, and the basolateral Na^+ - Ca^{2+} exchanger NCX1.²⁷⁶ NCC is coexpressed with ENaC in the late DCT2 of mouse, where the two proteins physically and may functionally interact,²⁷⁸ with robust expression of ENaC continuing in the downstream CNT and CCD.²⁷⁶ By contrast, rabbit kidney does not have a DCT1 or DCT2 and exhibits abrupt transitions between NCC- and ENaC-positive DCT and CNT segments, respectively.²⁷⁵ Human kidneys that have been studied thus far exhibit expression of calbindin $\text{D}_{28\text{K}}$ all along the DCT and CNT, extending into the CCD; however, the intensity of expression varies at these sites. Approximately 30% of cells in the distal convolution of human kidney express NCC, with 70% expressing ENaC (CNT cells); ENaC and NCC overlap in expression at the end of the human DCT segment. Finally, cells of the early CNT of human kidneys express ENaC in the absence of aquaporin-2, the apical vasopressin-sensitive water channel.²⁷⁵

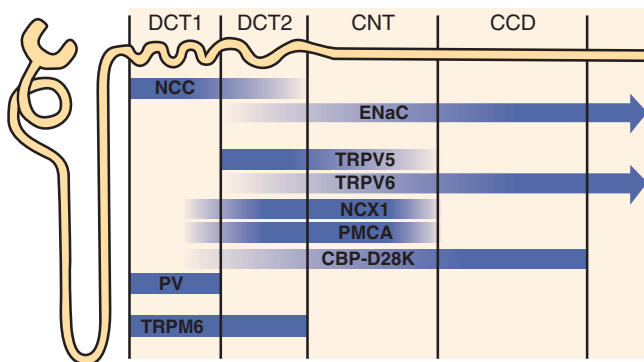


Fig. 6.17 Schematic representation of the segmentation of the mouse distal nephron and distribution and abundance of Na^+ , Ca^{2+} , and Mg^{2+} -transporting proteins. CBP-D28K, calbindin-D28K; CCD, cortical collecting duct; CNT, connecting tubule; DCT1 and DCT 2, distal convoluted tubules 1 and 2, respectively; ENaC, epithelial Na^+ channel; NCC, thiazide-sensitive Na^+ - Cl^- cotransporter; NCX1, Na^+ - Ca^{2+} exchanger; PMCA, plasma membrane Ca^{2+} -ATPase; PV, parvalbumin; TRPM6, apical Mg^{2+} entry channel; TRPV5 and TRPV6, apical Ca^{2+} entry channels.^{276,614,615}

Although primarily contiguous with the DCT, CNT cells share several traits with principal cells of the CCD, including apical expression of ENaC and ROMK, the K^+ secretory channel; the capacity for Na^+ - Cl^- reabsorption and K^+ secretion in this nephron segment is as much as 10 times higher than that of the CCD.²⁷⁹ Intercalated cells are the minority cell type in the distal nephron, emerging within the DCT and CNT and extending into the early inner medullary collecting duct (IMCD).²⁸⁰ Three subtypes of intercalated cells have been defined, based on differences in the subcellular distribution of the H^+ -ATPase and the presence or absence of the basolateral AE1 Cl^- - HCO_3^- exchanger. Type A intercalated cells extrude protons via an apical H^+ -ATPase in series with basolateral AE1; type B intercalated cells secrete HCO_3^- and OH^- via an apical anion exchanger (SLC26A4 or pendrin) in series with basolateral H^+ -ATPase.²⁸⁰ In rodents, the most prevalent subtype of intercalated cells in the CNT is the non-A, non-B intercalated cell, which possesses an apical Cl^- - HCO_3^- exchanger (SLC26A4 or pendrin) along with apical H^+ -ATPase.²⁸⁰ Although intercalated cells play a dominant role in acid-base homeostasis, Cl^- transport by type B intercalated cells performs an increasingly appreciated role in distal nephron Na^+ - Cl^- transport (see the “Connecting Tubules and the Cortical Collecting Duct: Cl^- Transport” section).

The OMCD encompasses two separate subsegments corresponding to the outer and inner stripes of the outer medulla, OMCD_o and OMCD_i, respectively. OMCD_o and OMCD_i contain principal cells with apical amiloride-sensitive Na^+ channels (ENaCs); however, the primary role of this nephron segment is renal acidification, with a particular dominance of type A intercalated cells in OMCD_i.^{6,281} The OMCD may also play a role in K^+ reabsorption via the activity of apical H^+ - K^+ -ATPase pumps,²⁸²⁻²⁸⁴ although deletion of this transport protein does not alter the ability to conserve K^+ substantially.²⁸⁵

Finally, the IMCD begins at the boundary between the outer and inner medulla and extends to the tip of the papilla. The IMCD is arbitrarily separated into three equal zones, denoted IMCD1, IMCD2, and IMCD3; at the functional level, an early IMCD (IMCD_i) and a terminal portion (IMCD_t) can be appreciated.⁶ The IMCD plays a particularly prominent role in vasopressin-sensitive water and urea transport.⁶ The early IMCD contains principal cells and intercalated cells; all three subsegments (IMCD1–3) express apical ENaC protein, albeit considerably weaker expression than in the CNT and CCD.²⁸⁶ The roles of the IMCD and OMCD in Na^+ - Cl^- homeostasis have been more elusive than those of the CNT and CCD; however, to the extent that ENaC is expressed in the IMCD and OMCD, homologous mechanisms are expected to function in Na^+ - Cl^- reabsorption by the CNT, CCD, OMCD, and IMCD segments.

DISTAL CONVOLUTED TUBULE

Mechanisms of Na^+ - Cl^- Transport in the Distal Convoluted Tubule

Earlier micropuncture studies that did not distinguish between early and late DCT indicated that this nephron segment reabsorbs about 10% of filtered Na^+ - Cl^- .^{5,287} The apical absorption of Na^+ and Cl^- by the DCT is mutually dependent; ion substitution does not affect transepithelial voltage, suggesting electroneutral transport.²⁸⁸ The absorption of Na^+ by perfused

DCT segments is also inhibited by chlorothiazide, localized proof that this nephron segment is the target for thiazide diuretics.²⁸⁹ Similar thiazide-sensitive $\text{Na}^+\text{-Cl}^-$ cotransport exists in the urinary bladder of winter flounder, the species in which the thiazide-sensitive NCC was first identified by expression cloning.²⁹⁰ Functional characterization of rat NCC indicates very high affinities for both Na^+ and Cl^- (Michaelis–Menten constants of 7.6 ± 1.6 and 6.3 ± 1.1 mmol/L, respectively); equally high affinities had previously been obtained by Velázquez and associates in perfused rat DCT.^{288,291} The measured Hill coefficients of rat NCC are about 1 for each ion, consistent with electroneutral cotransport.²⁹¹

NCC expression is the defining characteristic of the DCT (Fig. 6.18).²⁷⁵ There is also evidence for expression of this transporter in osteoblasts, peripheral blood mononuclear cells, and intestinal epithelium; however, the functional significance of expression reported for these sites remains unclear.^{168,292} The human *SLC12A3* gene encodes three isoforms (NCC1, NCC2, and NCC3), but only NCC3 has been studied extensively, because NCC1 and NCC2 are not expressed in rats or mice.²⁹³ These isoforms may undergo differential regulation. For example, NCC1 and NCC2 contain a region in their carboxyl-termini that is absent from NCC3 and contain a serine (S811), which undergoes phosphorylation and contributes to cotransporter activity.²⁹⁴ Loss-of-function mutations in the *SLC12A3* gene encoding human NCC cause Gitelman syndrome, familial hypokalemic alkalosis with hypomagnesemia, and hypocalciuria (see Chapter 44). Mice with homozygous deletion of the *Slc12a3* gene encoding NCC exhibit marked morphologic defects in the early DCT, with a reduction in the absolute number of DCT cells and changes

in ultrastructural appearance.^{295,296} Similarly, thiazide treatment promotes marked apoptosis of the proximal part of DCT, suggesting that thiazide-sensitive $\text{Na}^+\text{-Cl}^-$ cotransport plays an important role in modulating growth and regression of this nephron segment.²⁹⁷

Coexpression of NCC and ENaC occurs in the “late DCT” (DCT2) and CNT segments of many species, either in the same cells or in adjacent cells in the same tubule.²⁷⁵ Notably, ENaC is the primary Na^+ transport pathway of CNT and CCD cells, rather than DCT. There is, however, evidence for other Na^+ and Cl^- entry pathways in DCT cells. In particular, the $\text{Na}^+\text{-H}^+$ exchanger NHE2 (SLC9A2) is coexpressed with NCC at the apical membrane of rat DCT cells.²⁹⁸ As in the PT, perfusion of the DCT with formate and oxalate stimulates DIDS-sensitive $\text{Na}^+\text{-Cl}^-$ transport that is distinct from the thiazide-sensitive transport mediated by NCC.⁵⁰ Therefore a parallel arrangement of $\text{Na}^+\text{-H}^+$ exchange and Cl^- anion exchangers may play an important role in electroneutral $\text{Na}^+\text{-Cl}^-$ absorption by the DCT (Fig. 6.18). The $\text{Na}^+\text{-H}^+$ antiporter NHA2 (SLC9B2) has been localized to the DCT, and the anion exchanger SLC26A6 may be expressed in DCT cells; NHE2, NHA2, and SLC26A6 are thus candidate mechanisms for this alternative pathway of DCT $\text{Na}^+\text{-Cl}^-$ absorption.^{298–300}

At the basolateral membrane, as in other nephron segments, Na^+ exits via $\text{Na}^+\text{-K}^+\text{-ATPase}$; bearing in mind the considerable caveats in morphologic identification of the DCT, this nephron segment appears to have the highest $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity of the entire nephron (Fig. 6.3).^{17,275} Basolateral membranes of DCT cells in both rabbit and mouse express the KCC KCC4, a potential exit pathway for

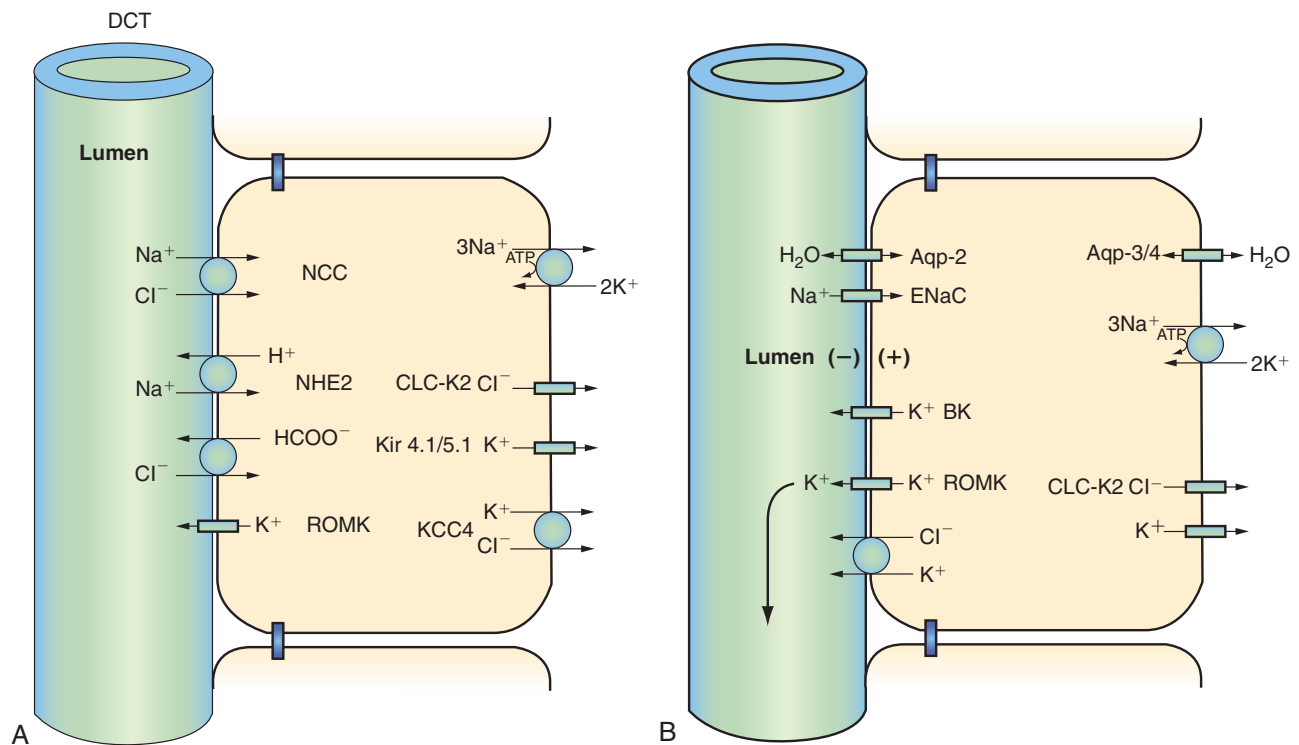


Fig. 6.18 Transport pathways for $\text{Na}^+\text{-Cl}^-$ and K^+ in distal convoluted tubule (DCT) cells (A) and principal cells of the connecting tubule (CNT) and cortical collecting duct (CCD) (B). *Aqp-2*, 3/4, Aquaporin-2, aquaporin-3/4; *ENaC*, epithelial Na^+ channel; *KCC4*, $\text{K}^+\text{-Cl}^-$ cotransporter-4; *NCC*, thiazide-sensitive $\text{Na}^+\text{-Cl}^-$ cotransporter; *NHE-2*, $\text{Na}^+\text{-H}^+$ exchanger-2; *ROMK*, renal outer medullary K^+ channel.

Cl^- .^{218,301} However, several lines of evidence have indicated that Cl^- primarily exits DCT cells via basolateral Cl^- channels. First, the basolateral membrane of rabbit DCT contains Cl^- channel activity, with functional characteristics similar to those of CLC-K2 .^{216,302} Second, CLC-K2 protein is expressed at the basolateral membrane of DCT and CNT cells. Although mRNA for CLC-K1 can also be detected by RT-PCR of microdissected DCT segments,^{158,214} recent data show that deletion of CLC-K2 from mice leads to a loss of response to furosemide, and a markedly blunted one to thiazide, implicating this channel in chloride reabsorption along both the TAL and the DCT.¹⁶² Finally, loss-of-function mutations in CLC-NKB , the human ortholog of CLC-K2 , cause classic Bartter syndrome; this Bartter subtype has a phenotype that is typically intermediate between Bartter syndrome type I, and Gitelman syndrome, consistent with loss of function of DCT segments.^{160,162,303}

K^+ channels at the basolateral membrane of DCT cells play a critical role in the function of this nephron segment. Cell-attached patches in basolateral membranes of microdissected DCTs detect an inward rectifying K^+ channel, with characteristics similar to those of heteromeric KIR4.1/KIR5.1 and KIR4.2/KIR5.1 channels.^{228,304–306} Basolateral membranes of the DCT express immunoreactive KIR4.1 and KIR5.1 protein, and DCT cells express KIR4.2 mRNA.^{228,304–306} Patients with loss-of-function mutations in the *KCNJ10* gene that encodes KIR4.1 develop a syndrome encompassing epilepsy, ataxia, sensorineural deafness, and tubulopathy (EAST or SeSAME syndrome).^{228,307} The associated tubulopathy includes hypokalemia, metabolic alkalosis, hypocalciuria, and hypomagnesemia.^{228,307} Mice in which *Kcnj10* has been deleted following development demonstrate hypokalemic alkalosis with hypocalciuria, and suppression of NCC abundance,³⁰⁸ indicating a key role of this channel in supporting trans-epithelial NaCl transport.^{228,308,309} While KIR4.1 activity is detected in the TAL, KIR4.1 disruption in mice has no significant effect on TAL membrane potential or NKCC2 expression, so the physiological relevance of this is unclear.³¹⁰ By contrast, *Kcnj10* disruption markedly depolarizes the basolateral cell membrane of DCT cells, indicating that these K^+ channels play a key role in setting the membrane potential in the DCT.^{308,311} In addition to sensing the membrane potential, the KIR4.1/KIR5.1 channels at the basolateral membrane of DCT cells are hypothesized to function in basolateral K^+ recycling, maintaining adequate Na^+/K^+ -ATPase activity for Na^+/Cl^- absorption and other aspects of DCT function. Notably, the CaSR coassociates with KIR4.1 and KIR4.2 proteins and inhibits their activity, providing a mechanism for the dynamic modulation of Na^+/Cl^- , calcium, and magnesium transport by the DCT.³⁰⁵

Regulation of Na^+/Cl^- Transport in the Distal Convoluted Tubule

Our understanding of factors that regulate solute transport by the DCT has advanced rapidly, during the past 5 years. Studies performed in the 1990s showed that dietary NaCl deprivation activates thiazide-sensitive NaCl transport along the DCT.⁵ When distal salt delivery was increased further, by administering loop diuretics continuously and administering saline, as drinking solution, additional increases in transport capacity were observed, together with considerable hypertrophy of DCT cells. One contributor to this phenotype is

Ang II. DCT cells express AT_1 receptors, and Ang II activates NCC, through a mechanism that requires the kinase WNK4 (see discussion later).^{237,312–316}

Aldosterone has also been suggested as a factor that modulates NCC, which would make the DCT part of the aldosterone-sensitive distal nephron. Aldosterone, given to adrenalectomized rats, activated NCC.³¹⁷ Dietary salt restriction and exogenous mineralocorticoids were also shown to increase the abundance of NCC, and phosphorylated NCC.³¹⁸ DCT cells, at least in rats, were shown to express 11 β hydroxysteroid dehydrogenase type 2 ($11\beta\text{-HSD2}$),³¹⁹ at least at low levels, and its deletion from mice resulted in a phenotype that includes hypertrophy of DCT cells, and increases in the abundance of phosphorylated NCC.³²⁰ Aldosterone has been reported to enhance NCC activity acutely in cultured DCT cells.³²¹ Roy and colleagues identified two alternatively spliced exons in WNK1 that contain PY motifs, which bind the E3 ubiquitin ligase NEDD4-2 . These motifs were suggested to mediate NCC activation by aldosterone.³²²

It was only the recognition that dietary potassium intake is a powerful NCC and DCT regulatory factor that has modified the view of aldosterone's role. Vallon and colleagues documented that dietary potassium deprivation increased the abundance of phosphorylated NCC in mice.³²³ Two groups subsequently showed that low potassium intake can increase NCC abundance and activity even in the setting of high salt intake, and another showed that high potassium intake can suppress NCC, even when dietary salt intake is low.³²⁴ Thus it appears that the effects of potassium predominate over those of sodium. These effects of high potassium intake raised the possibility that aldosterone may not be playing a dominant role in NCC regulation, as high potassium diets are associated with high aldosterone secretion and low NCC activity.³²⁴ Increasingly, it has become apparent that DCT cells, and NCC, are exquisitely sensitive to plasma $[\text{K}^+]$. These effects occur very rapidly, as a short-term gavage with high potassium solution leads to rapid dephosphorylation of the NCC.³²⁵ Similarly, raising the plasma potassium concentration, either by potassium infusion,³²⁶ by administering the sodium channel blocking drug amiloride,³²⁷ or by deleting Na^+ channels,³²⁸ strikingly reduces the abundance of phosphorylated NCC. These effects on the abundance of phosphorylated NCC are functionally relevant, as several groups have documented that directional changes in pNCC abundance, in the setting of potassium challenge, are associated with concordant changes in thiazide-sensitive NaCl excretion.³²⁹ The mechanisms involved in the potassium effect will be discussed later, with the discussion of WNK kinases.

The recognition that potassium plays a dominant role in regulating NCC and the DCT suggested that some effects observed during aldosterone infusion might be secondary to induced potassium imbalance. Mice with nephron-specific disruption of the mineralocorticoid receptor (MR) did exhibit low NCC activity, as would be expected if aldosterone stimulates NCC directly, but NCC could be stimulated to normal levels by dietary K^+ restriction, proving that it was effects of MR deletion on potassium balance that were responsible for changes in NCC.^{329,330} A second group showed that in mice with deletion of MR in approximately 20% of renal tubule cells, which allowed side-by-side comparison of cells in the same segment, no differences in NCC abundance and phosphorylation were observed between DCT cells that did

and did not express mineralocorticoid receptors; dietary Na^+ restriction upregulated NCC to similar extent in both types of cell.³³¹ By contrast, along the collecting duct ENaC expression and apical membrane localization were not detected on control diet nor in response to dietary Na^+ restriction in knockout cells. Finally, studies in adult mice with inducible disruption of the MR target α -ENaC showed that a high Na^+ /low K^+ diet, which normalized plasma $[\text{K}^+]$, also normalized the reduced abundances of total and phosphorylated NCC seen on a normal diet.³²⁸ In summary, the preponderance of evidence suggests that the effects of plasma $[\text{K}^+]$ are dominant, and that aldosterone plays only a modifying role in regulating NCC; the effects of potassium on NCC and aldosterone on ENaC appear to comprise a potassium switch in the kidney (see later).

A central role in the regulation of DCT Na^+ and Cl^- transport is played by WNK1 and WNK4, key regulatory kinases in the distal nephron that were initially identified as two of the causative genes for familial hyperkalemic hypertension (FHHt; also known as pseudohypoaldosteronism type II or Gordon syndrome). FHHt is in every respect the mirror image of Gitelman syndrome, encompassing hypertension, hyperkalemia, hyperchloremic metabolic acidosis, suppressed plasma renin activity and aldosterone, and hypercalciuria.³³²

Furthermore, FHHt behaves like a gain-of-function in NCC and/or the DCT in that treatment with thiazides typically results in resolution of the entire syndrome³³²; however, simple transgenic overexpression of NCC in DCT cells does not replicate the phenotype in mice, indicating specific effects of the mutant WNK1 and WNK4 alleles.^{332,333}

WNK kinases have pleiotropic effects, and initial experiments involving WNK expression in heterologous systems have often produced results that now seem contradictory. Yet, a consensus view of the predominant effects of WNK kinases in regulating NCC has begun to emerge. According to a highly simplified scheme (Fig. 6.19), WNK kinases activate the downstream kinases SPAK and/or OSR1 by binding to them along their conserved carboxyl-terminal domains, and directly phosphorylating them. When so activated, and enhanced by interactions with MO25 (Cab39),²⁵⁰ these secondary kinases bind to and activate NCC by enhancing phosphorylation of key residues in the NCC amino terminal cytoplasmic domain.³³⁴

Mutations in both WNK1 and WNK4 can cause FHHt. Intronic mutations in WNK1 enhance the expression of a full-length kinase-active form of WNK1 along the DCT, where it is normally expressed only at low levels³³⁵; WNK4-point mutations cluster around an acid-rich conserved region of

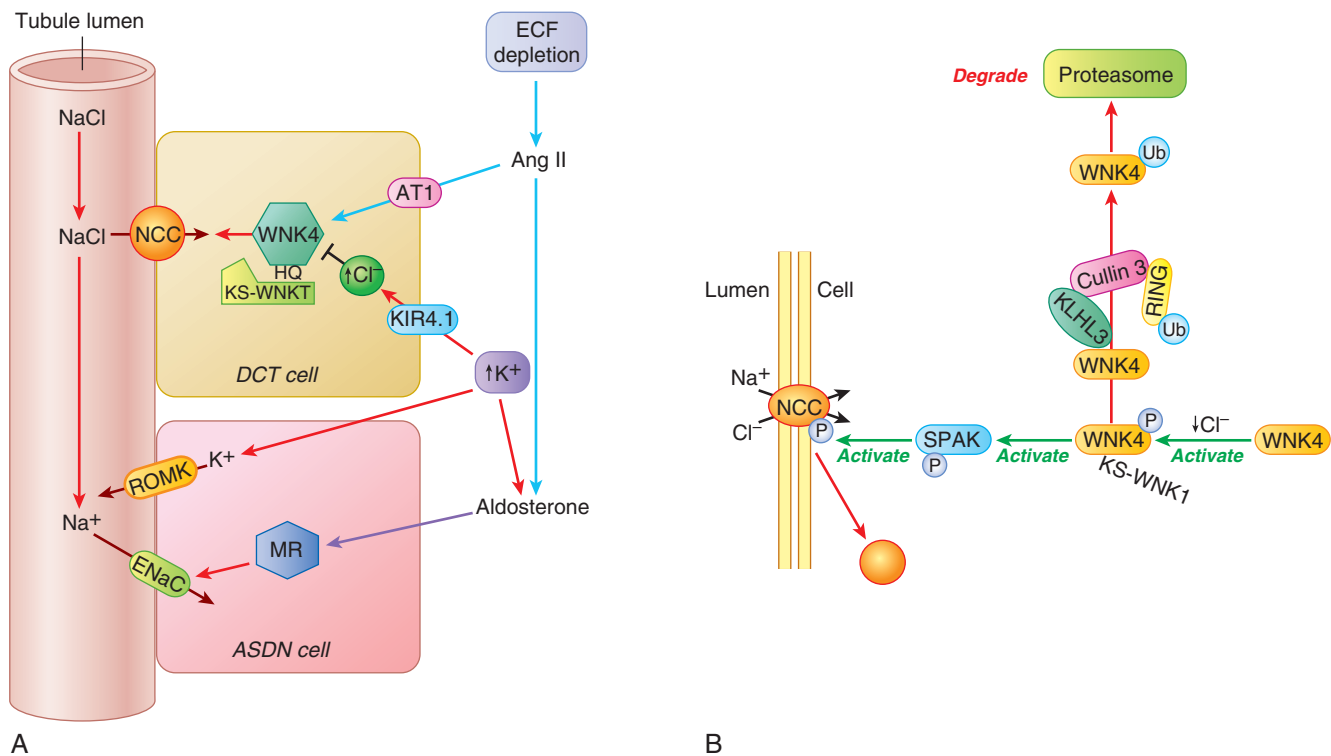


Fig. 6.19 Regulation of distal transport by WNK kinases. (A) Comparison of effects of extracellular fluid (ECF) volume depletion and hyperkalemia ($\uparrow[\text{K}^+]$) on transport along the distal nephron. In the lumen, Na^+ delivery to the aldosterone-sensitive distal nephron (ASDN) is determined by entry into the distal convoluted tubule (DCT) and reabsorption by the thiazide-sensitive NaCl cotransporter (NCC). WNK4, likely together with WNK1-S [also known as kidney-specific (KS)-WNK1], activates NCC.⁶¹⁶ This process is stimulated by angiotensin II (*Ang II*), which also enhances aldosterone secretion by the adrenal gland to stimulate the epithelial Na^+ channel (ENaC), via the mineralocorticoid receptor (MR). Sodium reabsorption along both the DCT and ASDN corrects ECF volume depletion. During hyperkalemia, NCC is turned off via increased intracellular chloride (modulated by K^+ flux through KIR4.1 channels), and Na^+ is reabsorbed primarily by ENaC, favoring K^+ secretion. (B) Detailed scheme of WNK kinase regulation. NCC is phosphorylated and activated by STE20/SPS1-related proline/alanine-rich kinase (SPAK), which is phosphorylated and activated by WNK4, which is activated by low intracellular chloride concentration. WNKs can interact with a cullin-ring ligase, via the adaptor Kelch-Like 3 (KLHL3), leading to WNK ubiquitination and subsequent degradation by the proteasome. When dephosphorylated, NCC can be removed from the apical plasma membrane.

the protein, or near the carboxyl-terminal domain.³³⁶ These mutations disrupt WNK binding to adaptor proteins that are essential for WNK degradation (see later).^{337–339} Thus both WNK mutant forms of FHHt result in increased abundance of WNKs along the DCT.

The WNK1 and WNK4 proteins are coexpressed within the distal nephron in DCT and CCD cells, where they both localize to the cytoplasm (diffusely, and also in punctate structures) and apical membrane.^{340,341} Although both WNK4 and WNK1 can stimulate NCC activity in expression systems, only deletion of WNK4 has been shown to reduce NCC activity in mice.^{236,237} In mice in which an FHHt-like phenotype has been generated by mutation of the protein KLHL3 (see later), increased WNK1 abundance does not compensate for deletion of WNK4, suggesting a key role for WNK4.³³ Generation of mice with WNK1 deletion specifically in the kidney may shed more light on its contribution to NCC regulation in vivo.

WNKs, especially WNK4, appear to be sensitive to inhibition by chloride,^{342,343} so that when intracellular chloride concentration is low, the stimulatory effect is maximal; when intracellular chloride concentration is higher, there is less stimulation.³⁴⁴ For WNK1, chloride binds to the catalytic site of the kinase and inhibits autophosphorylation and activation of the kinase.³⁴² This chloride binding has a major role in the potassium-sensing function of DCT cells. Reduction in potassium intake and/or hypokalemia lead to reduced basolateral K⁺ concentration in the DCT; the subsequent hyperpolarization is dependent on basolateral KIR4.1-containing K⁺ channels.^{308,311,327} Hyperpolarization has been proposed to lead to chloride exit via basolateral CLC-K2 chloride channels and a reduction in intracellular chloride; the reduction in intracellular chloride activates the SPAK and OSR1-WNK cascade, resulting in phosphorylation of NCC and activation of the transporter.³²⁷ This model helps explain the activating effect of potassium depletion on NCC and the inhibitory effect of potassium loading on NCC, and they go a long way to explain the critical role of the DCT and NCC in potassium homeostasis.³²⁷

To develop in vivo models relevant to FHHt, two groups developed mice expressing WNK4 carrying a mutation that causes the disease. Lalioti and coworkers generated BAC-transgenic mice that is an FHHt mutant of WNK4 (TgWnk4^{PHAI}), bearing a Q562E mutation associated with the disease,³⁴⁵ and Yang and colleagues developed a knock-in mouse expressing D568. Both models were hypertensive,^{345a} with biochemical phenotypes similar to that of FHHt (i.e., hyperkalemia, acidosis, and hypercalciuria). TgWnk4^{PHAI} mice also exhibit marked hyperplasia of the DCT. Of particular significance, the DCT hyperplasia of TgWnk4^{PHAI} mice was completely suppressed on an NCC-deficient background, generated by mating TgWnk4^{PHAI} mice with NCC knockout mice.^{295,296} Therefore the DCT is the primary target for FHHt-associated mutations in WNK4. In addition, as suggested by prior studies, changes in Na⁺-Cl⁻ entry via NCC can evidently modulate hyperplasia or regression of the DCT.^{275,295,296,345} Vidal-Petiot and colleagues have also generated mice that lack the orthologous intron of WNK1 involved in patients with FHHt, recapitulating the phenotype.³³⁵ Treatment with the calcineurin inhibitor tacrolimus has a similar effect as in FHHt.³⁴⁸

WNK proteins are regulated by Cullin 3 (CUL3) and Kelch-like 3 (KLHL3), components of an E3 ubiquitin ligase

complex that targets the WNKs for degradation.^{339,349,350} Mutations in the *CUL3* and *KLHL3* genes also cause FHHt and account for the majority of cases. Disease-associated mutations in KLHL3 abrogate binding to WNK4 and vice versa.³³⁹ In turn, disease-associated mutations in CUL3 may deplete levels of KLHL3, preventing WNK degradation.³⁵¹ Physiologically, phosphorylation of KLHL3 by PKC, downstream of Ang II, also abrogates the interaction between KLHL3 and WNK4, leading to NCC activation.³⁵² More information about the mechanisms of FHHt, and the roles of regulatory proteins can be found in Chapters 17 and 44.

The various mechanistic models for the regulation of NCC by upstream WNK1, WNK4, and the SPAK-OSR1 kinases have recently been reviewed; interactions between WNK4 and both WNK3 and SGK1 also contribute to the complexity, as do CUL3 and KLHL3.^{339,349–356} Competing divergent mechanisms can be reconciled by the likelihood that the physiologic context determines whether WNK4 will have an activating or inhibitory effect on NCC. For example, the activation of NCC by the Ang II receptor type 1 appears to require the downstream activation of SPAK by WNK4.^{315,357} Changes in circulating and local levels of Ang II, aldosterone, vasopressin, and K⁺ are thus expected to have different and often opposing effects on the activity of NCC in the DCT (see also Fig. 6.19 and the “Integrated Na⁺-Cl⁻ and K⁺ Transport in the Distal Nephron” section).^{315,353,357–361}

WNK kinases exhibit effects in expression systems that may seem anomalous, but which likely reflect their pleiotropic properties. In *Xenopus* oocytes, for example, WNK4 can inhibit NCC, through its carboxyl-terminal domain, which binds to protein phosphatase 1.³⁶² It now appears that this effect results from WNK4 inhibiting endogenously expressed WNK kinases in the setting of high intracellular chloride concentrations.³⁶³ WNK4 coexpression with NCC reduces transporter expression at the membrane of both *Xenopus* oocytes and mammalian cells, suggesting a prominent effect on membrane trafficking.^{353,357} The WNK4 kinase activates lysosomal degradation of the transporter protein, rather than inducing dynamin- and clathrin-dependent endocytosis.^{364,365} This occurs through effects of WNK4 on the interaction of NCC with the lysosomal targeting receptor sortilin and AP-3 adaptor complex.^{364,365} Dynamin-dependent endocytosis of NCC is induced by ERK1/2 phosphorylation via activation of H-Ras, Raf, and MEK1/2, resulting in ubiquitination of NCC and endocytosis of the transporter, suggesting a significant role in the down-regulation of NCC by PTH.^{366–368} Ubiquitination of NCC is catalyzed by the ubiquitin-ligase NEDD4-2, causing down-regulation of NCC.³²¹ NCC is highly ubiquitinated at multiple specific sites, but it is unclear whether ubiquitination at all of these sites involves NEDD4-2.³⁶⁹ Ubiquitination at these different sites has different effects on NCC, either modulating its endocytosis or degradation. The full implications of these effects observed in cell culture systems regarding in vivo behavior awaits additional study.

CONNECTING TUBULES AND THE CORTICAL COLLECTING DUCT

Apical Na⁺ Transport

The apical membrane of CNT cells and principal cells contain prominent Na⁺ and K⁺ conductances, without a measurable apical conductance for Cl⁻.^{216,279,370,371} The entry

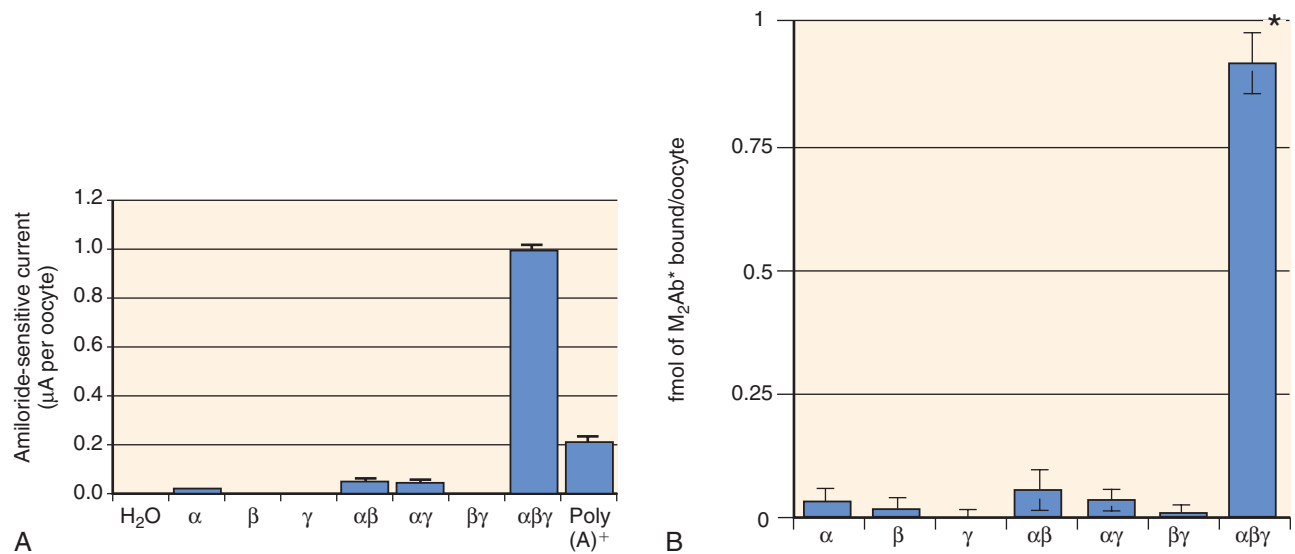


Fig. 6.20 Maximal expression of the amiloride-sensitive epithelial Na⁺ channel (ENaC) at the plasma membrane requires the coexpression of all three subunits (α-, β-, and γ-ENaC). (A) Amiloride-sensitive current in *Xenopus* oocytes expressing the individual subunits and various combinations thereof. (B) Surface expression is markedly enhanced in *Xenopus* oocytes that coexpress all three subunits. The individual complementary DNAs (cDNAs) were engineered with an external epitope tag; expression of the channel proteins at the cell surface is measured by binding of a monoclonal antibody (M₂Ab*) to the tag. Poly A⁺, Polyadenylated messenger RNA (mRNA). (A from Canessa CM, Schild L, Buell G, et al. Amiloride-sensitive epithelial Na⁺ channel is made of three homologous subunits. *Nature*. 1994;367:463–467; B from Firsov, D, Schild L, Gautschi I, et al. Cell surface expression of the epithelial Na channel and a mutant causing Liddle syndrome: a quantitative approach. *Proc Natl Acad Sci U S A*. 1996;93:15370–15375.)

of Na⁺ occurs via the highly selective epithelial Na⁺ channel (ENaC), which is sensitive to micromolar concentrations of amiloride (Fig. 6.20).³⁷² This selective absorption of positive charge generates a lumen-negative PD, the magnitude of which varies considerably as a function of mineralocorticoid status and other factors. This lumen-negative PD serves to drive the following critical processes: (1) K⁺ secretion via apical K⁺ channels; (2) paracellular Cl[−] transport through the adjacent tight junctions; and/or (3) electrogenic H⁺ secretion via adjacent type A intercalated cells.³⁷³

ENaC is a heteromeric channel complex formed by the assembly of separate, homologous subunits, denoted α-, β-, and γ-ENaC.¹⁴ These channel subunits share a common structure, with intracellular amino- and carboxyl-terminal domains, two transmembrane segments, and a large glycosylated extracellular loop.¹⁴ *Xenopus* oocytes expressing α-ENaC alone have detectable Na⁺ channel activity (Fig. 6.20), which facilitated the initial identification of this subunit by expression cloning; functional complementation of this modest activity was then used to clone the other two subunits by expression cloning.¹⁴ Full channel activity requires the coexpression of all three subunits, which causes a dramatic increase in expression of the channel complex at the plasma membrane (Fig. 6.20).³⁷⁴ The subunit stoichiometry has been a source of considerable controversy, with some reports favoring a tetramer with ratios of two α-ENaC proteins to one each of β- and γ-ENaC (2α:1β:1γ) and others favoring a higher-order assembly with a stoichiometry of 3α:3β:3γ.³⁷⁵ Regardless, the single-channel characteristics of heterologously expressed ENaC are essentially identical to those of the amiloride-sensitive channel detectable at the apical membrane of CCD cells.^{14,372}

ENaC plays a critical role in renal Na⁺-Cl[−] reabsorption and maintenance of the extracellular fluid volume. In particular, recessive loss-of-function mutations in the three subunits of ENaC are a cause of pseudohypoaldosteronism type I.^{14,376} Patients with this syndrome typically present with severe neonatal salt wasting, hypotension, acidosis, and hyperkalemia; this dramatic phenotype underscores the critical roles of ENaC activity in renal Na⁺-Cl[−] reabsorption, K⁺ secretion, and H⁺ secretion. Gain-of-function mutations in all three ENaC subunits have been reported to cause Liddle syndrome, an autosomal dominant hypertensive syndrome accompanied by suppressed aldosterone and variable hypokalemia.³⁷⁷ The majority of ENaC mutations associated with Liddle syndrome disrupt interactions between a PPxY motif in the carboxyl-terminus of channel subunits with the NEDD4-2 ubiquitin-ligase leading to increased surface expression of the channel. Mutations in the extracellular loops of α-ENaC and γ-ENaC have also been identified,^{378,379} with the α-ENaC mutation increasing intrinsic activity of the channel.³⁷⁹

The ENaC protein is detectable at the apical membrane of CNT cells and principal cells in the CCD, OMCD, and IMCD.^{281,286} Notably, however, several lines of evidence have supported the hypothesis that the CNT makes the dominant contribution to amiloride-sensitive Na⁺ reabsorption by the distal nephron:

1. Amiloride-sensitive Na⁺ currents in the CNT are twofold to fourfold higher than in the CCD; the maximal capacity of the CNT for Na⁺ reabsorption is estimated to be about 10 times higher than that of the CCD.²⁷⁹
2. Targeted deletion of α-ENaC in the collecting duct abolishes amiloride-sensitive currents in CCD principal

- cells, but does not affect Na^+ or K^+ homeostasis; the residual ENaC expression in the late DCT and CNT of these knockout mice easily compensates for the loss of the channel in CCD cells.³⁸⁰
3. Na^+ - K^+ -ATPase activity in the CCD is considerably less than that of the DCT (see also Fig. 6.4); this speaks to a greater capability for transepithelial Na^+ - Cl^- absorption by the DCT and CNT.¹⁷
 4. The apical recruitment of ENaC subunits in response to dietary Na^+ restriction begins in the CNT, with progressive recruitment of subunits in the downstream CCD at lower levels of dietary Na^+ ; although the CNT plays a dominant role in ENaC-mediated sodium transport, it does so primarily in an aldosterone-independent mechanism, with aldosterone-mediated sodium transport in the CCD involved in the finely tuned regulation of sodium transport.^{381,382}
 5. Patch-clamp analysis of knock-in mice homozygous for a Liddle syndrome ENaC mutant showed that the primary site of increased Na^+ reabsorption is the DCT2/CNT rather than the CNT/CCD.³⁸³

Cl^- Transport

There are two major pathways for Cl^- absorption in the CNT and CCD – paracellular transport across the tight junction and transcellular transport across type B intercalated cells (Fig. 6.21).^{280,384} The CNT and CCD are “tight” epithelia, with comparatively low paracellular permeability that is not selective for Cl^- over Na^+ ; however, voltage-driven, paracellular Cl^- transport in the CCD may play a considerable role in transepithelial Na^+ - Cl^- absorption.³⁸⁵ The CNT, DCT, and collecting duct coexpress claudin-3, -4, and -8; claudin-8 in particular may function as a paracellular cation barrier that prevents backleak of Na^+ , K^+ , and H^+ in this segment of the nephron.^{14,386} Several lines of evidence have indicated that claudin-4 and claudin-8 interact to form a paracellular pathway for Cl^- in the collecting duct, thus mediating transcellular Cl^- absorption via the paracellular pathway.³⁸⁷ CCD-specific knockout of claudin-4 in mice leads to NaCl wasting and hypotension,³⁸⁸ while CCD-specific disruption of claudin-8 causes hypotension, hypokalemia, and metabolic alkalosis, which resembles Gitelman syndrome.³⁸⁹ Regulated changes in paracellular permeability may also contribute to Cl^- absorption by the CNT and CCD. In particular, wild type WNK4 appears to increase paracellular Cl^- permeability in transfected MDCK II cell lines; a WNK4 FHHt mutant has a much larger effect, with no effect seen in cells expressing kinase-dead WNK4 constructs.³⁹⁰ Yamauchi and colleagues have also reported that FHHt-associated WNK4 increases paracellular permeability, due perhaps to an associated hyperphosphorylation of claudin proteins.³⁹¹ The claudin-4-mediated chloride conductance is also negatively regulated by cleavage in its second extracellular loop by channel-activating protease 1 (cap1).³⁸⁸ Similar to WNK4-mediated degradation by CUL3/KLHL3 (see the “Regulation of Na^+ - Cl^- Transport in the Distal Convoluted Tubule” section), claudin-8 was shown to be a target of KLHL3-mediated degradation, with FHHt-associated KLHL3 displaying impaired interaction.³⁸⁹

Transcellular Cl^- absorption across intercalated cells is thought to play a quantitatively greater role in the CNT and CCD than that of paracellular transport.³⁸⁴ In the simplest

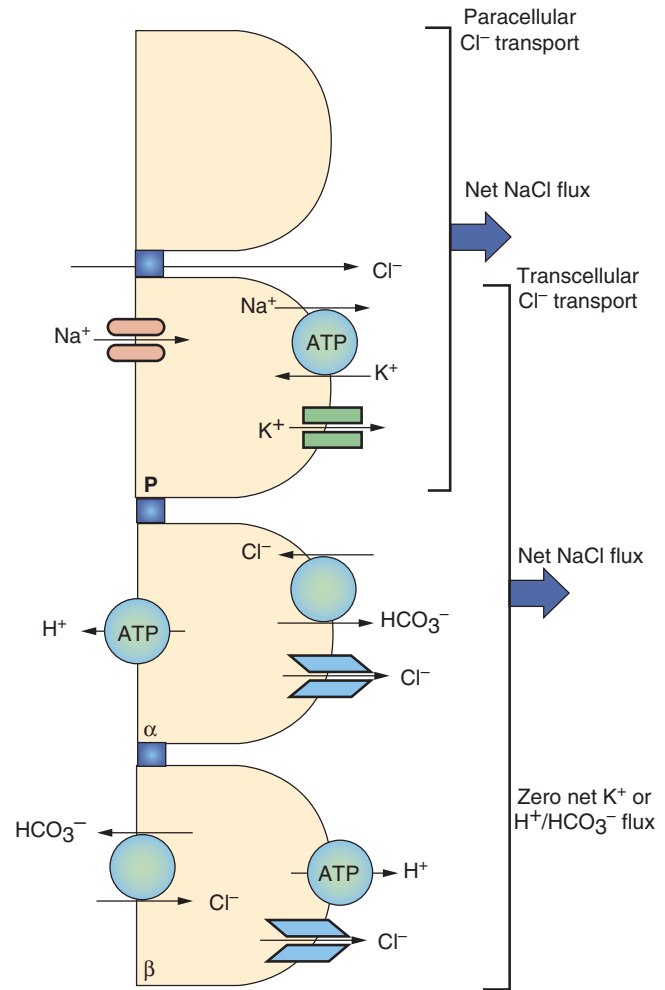


Fig. 6.21 Transepithelial Cl^- transport by principal and intercalated cells. The lumen-negative potential difference (PD) generated by principal cells drives paracellular Cl^- absorption. Alternatively, transepithelial transport occurs in type B intercalated cells via apical Cl^- - HCO_3^- exchange (SLC26A4/pendrin) and basolateral Cl^- exit via CLC-K2 . (Modified from Moe OW, Baum M, Berry CA, Rector Jr FC. Renal transport of glucose, amino acids, sodium, chloride, and water. In: Brenner BM, ed. *Brenner and Rector's the Kidney*. Philadelphia: WB Saunders; 2004:413–452.)

scheme, this process requires the concerted function of type A and type B intercalated cells, achieving net electrogenic Cl^- absorption without affecting HCO_3^- or H^+ excretion (see also Fig. 6.21).³⁸⁴ Chloride thus enters type B intercalated cells via apical Cl^- - HCO_3^- exchange, followed by exit from the cell via basolateral Cl^- channels. Recycling of Cl^- at the basolateral membrane of adjacent type A intercalated cells also results in HCO_3^- absorption and extrusion of H^+ at the apical membrane. The net effect of apical Cl^- - HCO_3^- exchange in type B intercalated cells, leading to apical secretion of HCO_3^- , and recycling of Cl^- at the basolateral membrane of type A intercalated cells, leading to apical secretion of H^+ , is electrogenic Cl^- absorption across type B intercalated cells (Fig. 6.21).

At the basolateral membrane, intercalated cells have a very robust Cl^- conductance, with transport characteristics similar to those of CLC-K2 /Barttin.^{216,392} CLC-K2 protein is

also detected at the basolateral membrane of type A intercalated cells, and CLC-K2 activity has been observed in type B cells.^{214,393} The basolateral $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter NKCC1 in adjacent type A intercalated cells also plays an evident role in transepithelial Cl^- absorption by the CCD.³⁹⁴ At the apical membrane, the SLC26A4 exchanger (also known as pendrin) has been conclusively identified as the elusive $\text{Cl}^-/\text{HCO}_3^-$ exchanger of type B and non-A, non-B intercalated cells; this exchanger functions as the apical entry site during transepithelial Cl^- transport by the distal nephron.²⁸⁰ Human *SLC26A4* is mutated in Pendred syndrome, which encompasses sensorineural hearing loss and goiter; these patients do not have an appreciable renal phenotype.²⁸⁰ However, *Slc26a4* knockout mice are sensitive to restriction of dietary $\text{Na}^+\text{-Cl}^-$, developing hypotension during severe restriction.³⁹⁵ *Slc26a4* knockout mice are also resistant to mineralocorticoid-induced hypertension.³⁹⁶ Pendrin has indirect effects on ENaC abundance and activity, apparently by modulating luminal ATP and HCO_3^- concentrations; pendrin and ENaC are also both coactivated by Ang II^{397–400} and pendrin expression is also induced by aldosterone.⁴⁰¹ The overexpression of pendrin in intercalated cells thus causes hypertension in transgenic mice, with an increase in ENaC activity and activity of electroneutral $\text{Na}^+\text{-Cl}^-$ absorption (see later).⁴⁰² Conversely, disruption of pendrin in mice decreased ENaC-mediated Na^+ absorption by reducing channel open probability and channel density at the apical membrane.⁴⁰³ Finally, dietary Cl^- restriction with provision of $\text{Na}^+\text{-HCO}_3^-$ results in Cl^- wasting in *Slc26a4* knockout mice and increased apical expression of SLC26A4 protein in the type B intercalated cells of normal littermate controls.⁴⁰⁴ Several groups have reported that SLC26A4 expression is exquisitely responsive to changes in distal chloride delivery.⁴⁰⁵ Therefore SLC26A4 plays a critical role in distal nephron Cl^- absorption, underlining the particular importance of transcellular Cl^- transport in this process. Of broader relevance, these studies have served to underline the important role for Cl^- homeostasis in the maintenance of extracellular volume and pathogenesis of hypertension.⁴⁰⁵

Electroneutral $\text{Na}^+\text{-Cl}^-$ Cotransport

Thiazide-sensitive $\text{Na}^+\text{-Cl}^-$ cotransport is considered the exclusive provenance of the DCT, which expresses the canonical thiazide-sensitive transporter NCC (see the “Mechanisms of $\text{Na}^+\text{-Cl}^-$ Transport in the Distal Convulated Tubule” section). However, Tomita and coworkers demonstrated many years ago that approximately 50% of $\text{Na}^+\text{-Cl}^-$ transport in rat CCD is electroneutral, amiloride resistant, and thiazide sensitive.^{406,407} Thiazide-sensitive electroneutral $\text{Na}^+\text{-Cl}^-$ transport has also been demonstrated in mouse CCD.⁴⁰⁸ This transport activity is preserved in CCDs from mice with genetic disruption of NCC and ENaC, indicating independence from the dominant apical Na^+ transport pathways in the distal nephron. This thiazide-sensitive, electroneutral $\text{Na}^+\text{-Cl}^-$ transport appears to be mediated by the parallel activity of the Na^+ -driven SLC4A8 $\text{Cl}^-/\text{HCO}_3^-$ exchanger and SLC26A4 $\text{Cl}^-/\text{HCO}_3^-$ exchanger (pendrin; see earlier discussion).⁴⁰⁸ *Slc4a8* knockout mice display only a mild perturbation of $\text{Na}^+\text{-Cl}^-$ and water balance due to compensation by NCC; combined SLC4A8 and NCC disruption caused intravascular volume contraction and hypokalemia.⁴⁰⁹ Notably, however, heterologously expressed recombinant SLC4A8 and SLC26A4 are resistant

and partially sensitive to thiazide, respectively, such that the in vivo pharmacology of this electroneutral $\text{Na}^+\text{-Cl}^-$ absorption is not completely explained. Immunolocalization of Slc4a8 within the CCD has been problematic; hence, it is unknown whether SLC4A8 and SLC26A4 are coexpressed in type B intercalated cells. Regardless, the combined activity of SLC4A8 and SLC26A4 appears to play a major role in $\text{Na}^+\text{-Cl}^-$ transport within the CCD, with significant implications for $\text{Na}^+\text{-Cl}^-$ and K^+ homeostasis (see also the “Integrated $\text{Na}^+\text{-Cl}^-$ and K^+ Transport in the Distal Nephron” section).

The apical entry of Na^+ via SLC4A8 requires a basolateral exit of Na^+ in type B intercalated cells, evidently mediated by the basolateral $\text{Na}^+\text{-HCO}_3^-$ transporter SLC4A9.² Another puzzle was the energetics of transcellular $\text{Na}^+\text{-Cl}^-$ transport in intercalated cells, which possess minimal, if any, detectable $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity. A series of elegant experiments have revealed that electroneutral $\text{Na}^+\text{-Cl}^-$ transport in type B intercalated cells is energized by and thus dependent on the activity of the basolateral $\text{H}^+\text{-ATPase}$.² Type B intercalated cells are therefore unique among mammalian renal epithelial cells in that transcellular ion transport is driven by $\text{H}^+\text{-ATPase}$ rather than $\text{Na}^+\text{-K}^+\text{-ATPase}$ activity.

REGULATION OF $\text{Na}^+\text{-Cl}^-$ TRANSPORT IN THE CONNECTING TUBULE AND CORTICAL COLLECTING DUCT

Aldosterone

The DCT, CNT, and collecting ducts collectively constitute the aldosterone-sensitive distal nephron, expressing the mineralocorticoid receptor and $11\beta\text{-HSD2}$ enzyme that protects against illicit activation by glucocorticoids.²⁷⁵ Aldosterone plays a dominant positive role in the regulation of distal nephron $\text{Na}^+\text{-Cl}^-$ transport, with a plethora of mechanisms and transcriptional targets.⁴¹⁰ For example, aldosterone increases the expression of the $\text{Na}^+\text{-K}^+\text{-ATPase}$ α_1 - and β_1 -subunits in the CCD, in addition to inducing SLC26A4, the apical $\text{Cl}^-/\text{HCO}_3^-$ exchanger of intercalated cells.^{396,411} Aldosterone may also affect paracellular permeability of the distal nephron via posttranscriptional modification of claudins and other components of the tight junction.⁴¹² However, particularly impressive progress has been made in the understanding of the downstream effects of aldosterone on synthesis, trafficking, and membrane-associated activity of ENaC subunits. A detailed discussion of aldosterone's actions may be found in Chapter 12; here we summarize the major findings of relevance to $\text{Na}^+\text{-Cl}^-$ transport.

Aldosterone increases abundance of $\alpha\text{-ENaC}$ via a glucocorticoid response element in the promoter of the *SCNN1A* gene that encodes this subunit.⁴¹³ Aldosterone also relieves a tonic inhibition of the *SCNN1A* gene by a complex that includes the Dot1a (disruptor of telomere silencing splicing variant *a*) and AF9 and AF17 transcription factors.⁴¹⁴ An aldosterone-dependent reduction in promoter methylation is also involved.⁴¹⁵ This transcriptional activation results in an increased abundance of $\alpha\text{-ENaC}$ protein in response to exogenous aldosterone or dietary $\text{Na}^+\text{-Cl}^-$ restriction (Fig. 6.22); the response to $\text{Na}^+\text{-Cl}^-$ restriction is blunted by spironolactone, indicating involvement of the mineralocorticoid receptor.^{416–418} At baseline, $\alpha\text{-ENaC}$ transcripts in the kidney are less abundant than those encoding β - and $\gamma\text{-ENaC}$ ⁴¹⁹ (Fig. 6.22). All three subunits are required for efficient processing

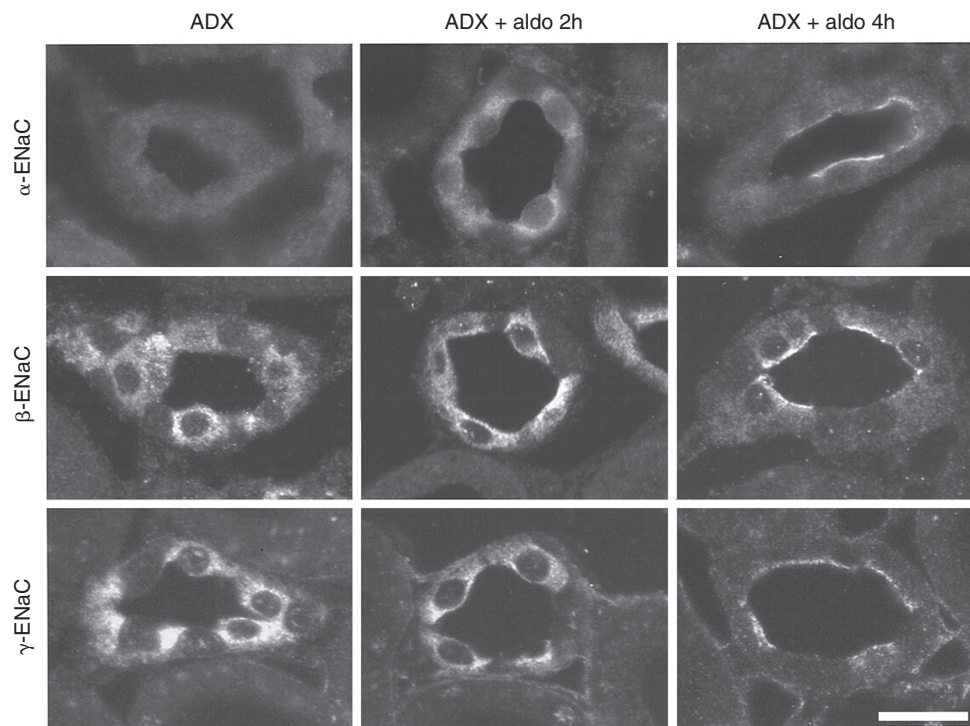


Fig. 6.22 Immunofluorescence images of connecting tubule profiles in kidneys from adrenalectomized rats (ADX) and from ADX rats 2 and 4 hours after aldosterone (aldo) injection. Antibodies against the α , β , and γ subunits of epithelial Na^+ channel (ENaC) reveal absent expression of the former in ADX rats, with progressive induction by aldosterone. All three subunits traffic to the apical membrane in response to aldosterone. This coincides with rapid aldosterone induction of the SGK kinase in the same cells; SGK is known to increase the expression of ENaC at the plasma membrane (see text for details). Bar $\equiv$ 15 μm . (From Loffing J, Zecevic M, Féraille E, et al. Aldosterone induces rapid apical translocation of ENaC in early portion of renal collecting system: possible role of SGK. *Am J Physiol Renal Physiol*. 2001;280:F675–F682.)

of heteromeric channels in the endoplasmic reticulum and trafficking to the plasma membrane (Fig. 6.20), such that the induction of α -ENaC is thought to relieve a major bottleneck in the processing and trafficking of active ENaC complexes.⁴¹⁹

Aldosterone also plays an indirect role in the regulated trafficking of ENaC subunits to the plasma membrane via the regulation of accessory proteins that interact with preexisting ENaC subunits. Aldosterone rapidly induces expression of a serine-threonine kinase denoted SGK1 (serum and glucocorticoid-induced kinase-1); coexpression of SGK1 with ENaC subunits in *Xenopus* oocytes results in a dramatic activation of the channel due to increased expression at the plasma membrane.^{418,420,421} Notably, an analogous redistribution of ENaC subunits occurs in the CNT and early CCD, from a largely cytoplasmic location during dietary $\text{Na}^+\text{-Cl}^-$ excess to a purely apical distribution after aldosterone or $\text{Na}^+\text{-Cl}^-$ restriction (Fig. 6.22).^{381,416,418} Furthermore, there is a temporal correlation between the appearance of induced SGK1 protein in the CNT and the redistribution of ENaC protein to the plasma membrane.⁴¹⁸

SGK1 modulates membrane expression of ENaC by interfering with regulated endocytosis of its channel subunits. Specifically, the kinase interferes with interactions between ENaC subunits and the ubiquitin ligase NEDD4-2.⁴¹⁹ PPxY domains in the C termini of all three ENaC subunits bind to WW domains of NEDD4-2⁴²²; these PPxY domains are deleted, truncated, or mutated in patients with Liddle syndrome, leading to a gain of function in channel activity.^{374,377} Coexpres-

sion of NEDD4-2 with the wild type ENaC channel results in a marked inhibition of channel activity due to retrieval from the cell membrane, whereas channels bearing Liddle syndrome mutations are resistant; NEDD4-2 is thought to ubiquitinate ENaC subunits, resulting in the removal of channel subunits from the cell membrane and degradation in lysosomes and the proteasome.⁴¹⁹ A PPxY domain in SGK1 also binds to NEDD4-2, which is a phosphorylation substrate for the kinase; phosphorylation of NEDD4-2 by SGK1 abrogates its inhibitory effect on ENaC subunits.^{423,424} Aldosterone also stimulates NEDD4-2 phosphorylation in vivo.⁴²⁵ NEDD4-2 phosphorylation in turn results in ubiquitin-mediated degradation of SGK1, suggesting that there is considerable feedback regulation in this system.⁴²⁶ Aldosterone also reduces NEDD4-2 protein expression in cultured CCD cells, and acetylation of ENaC antagonizes ENaC ubiquitination, suggesting additional levels of in vivo regulation.^{427,428}

The induction of SGK1 by aldosterone thus appears to stimulate the redistribution of ENaC subunits from the cytoplasm to the apical membrane of CNT and CCD cells. This phenomenon involves SGK1-dependent phosphorylation of the NEDD4-2 ubiquitin ligase, which is coexpressed with ENaC and SGK1 in the distal nephron.⁴²⁷ Of note, there is considerable axial heterogeneity in the recruitment and redistribution of ENaC to the plasma membrane, which begins in the CNT and only extends into the CCD and OMCD in $\text{Na}^+\text{-Cl}^-$ -restricted or aldosterone-treated animals.^{275,418} The underlying causes of this progressive axial recruitment are not as yet clear.²⁷⁵ However, NEDD4-2 expression is inversely

related to the apical distribution of ENaC, with low expression in the CNT and increased expression levels in the CCD. In all likelihood, the relative balance among SGK1, ENaC, and NEDD4-2 figures prominently in the recruitment of the channel subunits to the apical membrane.⁴²⁷

NEDD4-2 and ENaC are part of a larger regulatory complex that includes the signaling protein Raf-1, stimulatory aldosterone-induced chaperone GILZ1 (glucocorticoid-induced leucine zipper-1), and scaffolding protein CNK3.^{429,430} The mTORC2 (mammalian target of rapamycin complex 2) kinase complex is another component, catalyzing upstream activation of SGK1 and thus inducing activation of ENaC.^{431,432}

Although many studies have supported a central role for SGK1 in mediating the effects of aldosterone on ENaC, a recent study found that despite altered apical trafficking of ENaC in SGK1 knockout mice, ENaC activity is normal even after aldosterone administration. This suggests that other aldosterone-induced proteins play a role in ENaC activation via mineralocorticoid receptors.^{433,434} For example, another aldosterone-induced protein, Ankyrin G, a cytoskeletal protein involved in vesicular trafficking, increases ENaC activity in cultured CCD cells by promoting its plasma membrane insertion from recycling endosomes.⁴³⁵

Finally, aldosterone indirectly activates ENaC channels through the induction of channel-activating proteases, which increase open channel probability by cleavage of the extracellular domains of α - and γ -ENaC. Western blotting of renal tissue from rats subjected to $\text{Na}^+\text{-Cl}^-$ restriction or treatment with aldosterone has revealed α - and γ -ENaC subunits of lower molecular mass than those detected in control animals, indicating that aldosterone induces proteolytic cleavage.^{417,436} Proteases that have been implicated in the processing of ENaC include furin, elastase, and three membrane-associated proteases denoted CAP1–3 (channel activating proteases-1, -2, and -3).^{437–439} Filtered proteases such as plasmin may also contribute to ENaC activation in nephrotic syndrome.⁴³⁹ CAP1 was initially identified from *Xenopus* A6 cells as an ENaC-activating protease; the mammalian ortholog is an aldosterone-induced protein in principal cells.^{440,441} Urinary excretion of CAP1, also known as prostaticin, is increased in hyperaldosteronism, with a reduction after adrenalectomy.⁴⁴¹ CAP1 is tethered to the plasma membrane by a GPI linkage, whereas CAP2 and CAP3 are transmembrane proteases.^{438,440} All three of these proteases activate ENaC by increasing the P_o of the channel, without increasing expression at the cell surface.⁴³⁸ However, analysis of CAP2 knockout mice indicates that it does not play a role in ENaC regulation and sodium balance in vivo.⁴⁴² Proteolytic cleavage of ENaC appears to activate the channel by removing the self-inhibitory effect of external Na^+ ; in the case of furin-mediated proteolysis of α -ENaC, this appears to involve the removal of an inhibitory domain from within the extracellular loop.^{438,443} Extracellular Na^+ appears to interact with a specific acidic cleft in the extracellular loop of α -ENaC, causing inhibition of the channel.⁴⁴⁴ The structures of the extracellular domains of ENaC and related channels resemble an outstretched hand holding a ball, with defined subdomains termed the “wrist,” “finger,” “thumb,” “palm,” “ β -ball,” and “knuckle”; functionally relevant proteolytic events target the finger domains of ENaC subunits.⁴³⁹ Unprocessed channels at the plasma membrane are thought to function as a reserve pool, capable of rapid activation by membrane-associated luminal proteases.⁴³⁷

Vasopressin and Other Factors

Although not typically considered an antinatriuretic hormone, vasopressin has well-characterized stimulatory effects on $\text{Na}^+\text{-Cl}^-$ transport by the CCD.^{94,445} Vasopressin directly activates ENaC in murine CCD, increasing the open probability (P_o) of the channel.⁴⁴⁶ In perfused rat CCD segments, vasopressin and aldosterone can have synergistic effects on Na^+ transport, with a combined effect that exceeds that of the individual hormones.⁴⁴⁵ In addition, water and Na^+ restriction synergistically increase the P_o of ENaC in murine CCDs.⁴⁴⁶ Prostaglandins inhibit this effect of vasopressin, particularly in the rabbit CCD; this inhibition occurs at least in part through reductions in vasopressin-generated cAMP.^{94,445} There are, however, considerable species-dependent differences in the interactions between vasopressin and negative modulators of $\text{Na}^+\text{-Cl}^-$ transport in the CCD, which include prostaglandins, bradykinin, endothelin, and α_2 -adrenergic tone.^{94,445} Regardless, cAMP causes a rapid increase in the Na^+ conductance of apical membranes in the CCD; this effect appears to be due to increases in the surface expression of ENaC subunits at the plasma membrane⁴⁴⁷ in addition to effects on open channel probability.^{446,448} Notably, cAMP inhibits retrieval of ENaC subunits from the plasma membrane via PKA-dependent phosphorylation of the phosphoacceptor sites in NEDD4-2 that are targeted by SGK1; therefore both aldosterone and vasopressin converge on NEDD4-2 in the regulation of ENaC activity in the distal nephron.⁴⁴⁹ Analogous to the effect on trafficking of aquaporin-2 in principal cells, cAMP also seems to stimulate exocytosis of ENaC subunits to the plasma membrane.⁴⁴⁸ Finally, similar to the long-term effects of vasopressin on aquaporin-2 expression and NKCC2 expression, chronic treatment with DDAVP results in an increase in abundance of the β - and γ -ENaC subunits.^{233,450}

The activation of ENaC by vasopressin appears to have additional direct effects on water homeostasis. Hypernatremic mice treated with the ENaC inhibitor benzamil thus exhibit further increases in tonicity due to a reduction in urinary osmolality.⁴⁵¹ In adrenalectomized mice, which lack circulating aldosterone, vasopressin maintains ENaC activity in the distal nephron.⁴⁵² This vasopressin-dependent activation of ENaC may, by extension, play a role in generating hyponatremia in the setting of primary adrenal failure. Systemic generation of circulating Ang II induces aldosterone release by the adrenal gland, with downstream activation of ENaC. However, Ang II also directly activates amiloride-sensitive Na^+ transport in perfused CCDs; blockade by losartan or candesartan suggests that this activation is mediated by Ang II receptors type 1.⁴⁵³ Of particular significance, the effect of luminal Ang II (10^{-9} M) was greater than that of bath Ang II, suggesting that intratubular Ang II may regulate ENaC in the distal nephron. Ang II also activates chloride absorption across intercalated cells via a pendrin (SLC26A4) and an $\text{H}^+\text{-ATPase}$ -dependent mechanism.⁴⁵⁴ Stimulation of ENaC is seen when tubules are perfused with Ang I; this effect is blocked by ACE inhibition with captopril, suggesting that intraluminal conversion of Ang I to Ang II can occur in the CCD.⁴⁵⁵ Notably, CNT cells express considerable amounts of immunoreactive renin versus the vanishingly low expression of renin mRNA in the PT.⁴⁵⁶ Angiotensinogen secreted into the tubule by PT cells may thus be converted to Ang II in the CNT via locally generated renin and ACE and/or related proteases.⁴⁵⁶

Luminal perfusion with ATP or uridine triphosphate (UTP) inhibits amiloride-sensitive Na^+ transport and reduces ENaC P_o in the CCD via activation of luminal P2Y_2 purinergic receptors.^{457,458} Targeted deletion of the murine P2Y_2 receptor results in salt-resistant hypertension due in part to an upregulation of NKCC2 activity in the TAL; resting ENaC activity is also increased, but suppressed aldosterone and downregulation of the α -subunit of ENaC blunts the role of amiloride-sensitive transport.^{458,459} Clamping mineralocorticoid activity at higher levels, via the administration of exogenous mineralocorticoid, reveals that P2Y_2 receptor activation may be a major mechanism for the modulation of ENaC P_o in response to changes in dietary $\text{Na}^+\text{-Cl}^-$.³²³ Increased dietary $\text{Na}^+\text{-Cl}^-$ thus leads to increased urinary ATP and UTP excretion in mice⁴⁵⁸; endogenous ATP from principal cells inhibits ENaC , and ENaC activity is not responsive to increased dietary $\text{Na}^+\text{-Cl}^-$ in P2Y_2 receptor knockout mice.^{323,458} In addition, the activation of apical ionotropic purinergic receptors, likely P2X_4 and/or $\text{P2X}_4/\text{P2X}_6$, can inhibit or activate ENaC , depending on luminal Na^+ concentration; these receptors may also participate in fine-tuning ENaC activity in response to dietary $\text{Na}^+\text{-Cl}^-$.⁴⁶⁰

As in other segments of the nephron, $\text{Na}^+\text{-Cl}^-$ transport by the CNT and CCD is modulated by metabolites of arachidonic acid generated by cytochrome P450 monooxygenases. In particular, arachidonic acid inhibits ENaC channel activity in the rat CCD via generation of the epoxygenase product 11,12-EET by the CYP2C23 enzyme expressed in principal cells.⁴⁶¹ Targeted deletion of the murine *Cyp4a10* gene, encoding, another P450 monooxygenase, results in salt-sensitive hypertension; urinary excretion of 11,12-EET is reduced in these knockout mice, with a blunted effect of arachidonic acid on ENaC channel activity in the CCD.⁴⁶² These mice also became normotensive after treatment with amiloride, indicative of *in vivo* activation of ENaC . It appears that deletion of *Cyp4a10* reduces activity of the murine ortholog of rat CYP2C23 (*Cyp2c44* in mouse) and/or related epoxygenases via reduced generation of a ligand for $\text{PPAR}\alpha$ (peroxisome proliferator-activated receptor α) that induces epoxygenase activity.⁴⁶² The mechanism(s) whereby 11,12-EET inhibits ENaC are unknown as yet. However, renal 11,12-EET production is known to be salt-sensitive, suggesting that generation of this mediator may serve to reduce ENaC activity during high dietary $\text{Na}^+\text{-Cl}^-$ intake.⁴⁶¹

Finally, activation of $\text{PPAR}\gamma$ by thiazolidinediones (TZDs) results in amiloride-sensitive hypertension, suggesting *in vivo* activation of ENaC .^{463,464} TZDs (e.g., rosiglitazone, pioglitazone, troglitazone) are insulin-sensitizing drugs used for the treatment of type II diabetes. Treatment with these agents is frequently associated with fluid retention, suggesting an effect on renal $\text{Na}^+\text{-Cl}^-$ transport. Given robust expression of $\text{PPAR}\gamma$ in the collecting duct, activation of ENaC was an attractive hypothesis for this TZD-associated edema syndrome.^{463,464} This appears to be the case, in that selective deletion of the murine $\text{PPAR}\gamma$ gene in principal cells abrogates the increase in amiloride-sensitive transport seen in response to TZDs.^{463,464} Conversely, mice with disruption of α - ENaC specifically along CNT/CCD display blunted increases in total body water and extracellular fluid volume in response to rosiglitazone administration, providing direct evidence that the effects of $\text{PPAR}\gamma$ are mediated through ENaC .⁴⁶⁵ TZDs appear to induce transcription of the *Scn1g* gene encoding γ - ENaC in addition

to inducing SGK1 ; targeted deletion of SGK1 in knockout mice attenuates but does not abolish TZD-associated edema.^{463,466,467} Notably, however, other studies have failed to detect an effect of TZDs on ENaC activity, which may instead activate a nonspecific cation channel within the IMCD.^{468,469} Regardless, the beneficial effect of spironolactone in type II diabetics with TZD-associated volume expansion is consistent with *in vivo* activation of $\text{Na}^+\text{-Cl}^-$ absorption in the aldosterone-responsive distal nephron.⁴⁷⁰ In addition, the risk of peripheral edema is increased considerably in patients treated with both TZDs and insulin therapy. Notably, insulin appears to activate ENaC via SGK1 -dependent mechanisms; $\text{PPAR}\gamma$ is required for the full activating effect of insulin on ENaC , such that this clinical observation may reflect synergistic activation of ENaC by insulin and TZDs.^{468,471,472}

POTASSIUM TRANSPORT

Maintenance of K^+ balance is important for a multitude of physiologic processes. Changes in intracellular K^+ affect cell volume regulation, regulation of intracellular pH, enzymatic function, protein synthesis, DNA synthesis, and apoptosis.¹⁴ Changes in the ratio of intracellular to extracellular K^+ affect the resting membrane potential, leading to depolarization in hyperkalemia and hyperpolarization in hypokalemia. Thus, disorders of extracellular K^+ have a dominant effect on excitable tissues, chiefly heart and muscle. In addition, a growing body of evidence has implicated hypokalemia and/or reduced dietary K^+ in the pathobiology of hypertension, heart failure, and stroke; these and other clinical consequences of K^+ disorders are reviewed in Chapter 17.

Potassium is predominantly an intracellular cation, with only 2% of total body K^+ residing in the extracellular fluid. Extracellular K^+ is maintained within a very narrow range by three primary mechanisms. First, the distribution of K^+ between the intracellular and extracellular space is determined by the activity of a number of transport pathways—namely, $\text{Na}^+\text{-K}^+\text{-ATPase}$, the $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter NKCC1 , the four KCCs , and a plethora of K^+ channels. In particular, skeletal muscle contains as much as 75% of total body potassium and exerts considerable influence on extracellular K^+ . Short-term and long-term regulation of muscle $\text{Na}^+\text{-K}^+\text{-ATPase}$ play a dominant role in determining the distribution of K^+ between the intracellular and extracellular spaces; the various hormones and physiologic conditions that affect the uptake of K^+ by skeletal muscle are reviewed in Chapter 17. Second, the colon has the ability to absorb and secrete K^+ , with considerable mechanistic and regulatory similarities to renal K^+ secretion. K^+ secretion in the distal colon is increased after dietary loading and in end-stage renal disease.^{14,473,474} However, the colon has a relatively limited capacity for K^+ excretion, such that changes in renal K^+ excretion play the dominant role in responding to changes in K^+ intake. In particular, regulated K^+ secretion by the CNT and CCD plays a critical role in the response to hyperkalemia and K^+ loading; increases in the reabsorption of K^+ by intercalated cells of the CCD and OMCD function in the response to hypokalemia or K^+ deprivation.

This section reviews the mechanisms and regulation of transepithelial K^+ transport along the nephron. As in other sections of this chapter, the emphasis is on particularly recent

developments in the molecular physiology of renal K^+ transport. Of note, transport pathways for K^+ play important roles in renal Na^+-Cl^- transport, particularly within the TAL. Furthermore, Na^+ absorption via ENaC in the aldosterone-sensitive distal nephron generates a lumen-negative PD that drives distal K^+ excretion. These pathways are primarily discussed in the section on renal Na^+-Cl^- transport; related issues relevant to K^+ homeostasis per se will be specifically addressed in this section.

PROXIMAL TUBULE

The PT reabsorbs some 50% to 70% of filtered K^+ (Fig. 6.23). PTs generate minimal transepithelial K^+ gradients, and fractional reabsorption of K^+ is similar to that of Na^+ .²⁸² K^+ absorption follows that of fluid, Na^+ , and other solutes, such that this nephron segment does not play a direct role in regulated renal excretion.^{475,476} Notably, however, changes in Na^+-Cl^- reabsorption by the PT have considerable effects on distal tubular flow and distal tubular Na^+ delivery, with attendant effects on the excretory capacity for K^+ (see the section “Distal Nephron, K^+ secretion” below).

The mechanisms involved in transepithelial K^+ transport by the PT are not completely clear, although active transport does not appear to play a major role.^{476,477} Luminal barium has modest effects on transepithelial K^+ transport by the PT, suggesting a component of transcellular transport via barium-sensitive K^+ channels.⁴⁷⁸ However, the bulk of K^+ transport is thought to occur via the paracellular pathway, driven by the lumen-positive PD in the mid to late PT (Fig. 6.2).^{478,479} The total K^+ permeability of the PT is thus rather high, apparently due to features of the paracellular pathway.^{478,479}

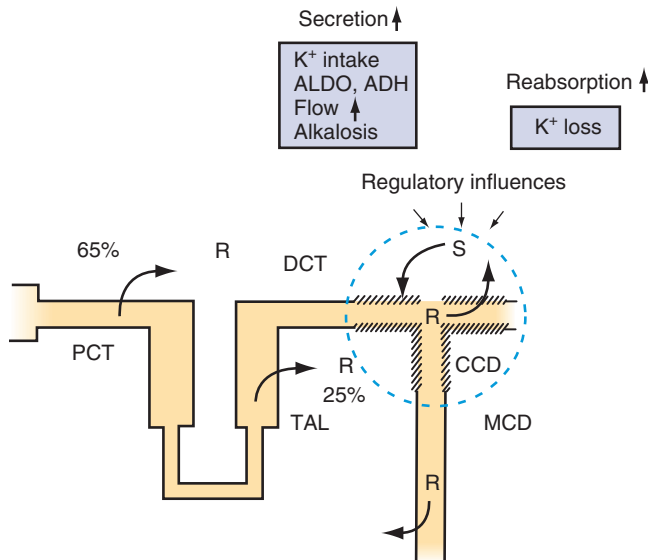


Fig. 6.23 K^+ transport along the nephron. Approximately 90% of filtered K^+ is reabsorbed by the proximal tubule and the loop of Henle. K^+ is secreted along the initial and cortical collecting duct. Net reabsorption occurs in response to K^+ depletion, primarily within the medullary collecting duct. ADH, Antidiuretic hormone; ALDO, aldosterone; CCD, cortical collecting duct; DCT, distal convoluted tubule; MCD, medullary collecting duct; PCT, proximal convoluted tubule; R, reabsorption; S, secretion; TAL, thick ascending limb.

The combination of luminal K^+ concentrations that are about 10% higher than that of plasma, a lumen-positive PD of about 2 mV (Fig. 6.2), and high paracellular permeability leads to considerable paracellular absorption in the PT. This absorption is thought to primarily proceed via convective transport—solvent drag due to frictional interactions between water and K^+ —rather than diffusional transport.⁴⁸⁰ Notably, however, the primary pathway for water movement in the PT is conclusively transcellular via aquaporin-1 and aquaporin-7 water channels in the apical and basolateral membrane.^{19,34,35} Therefore the apparent convective transport of K^+ would have to constitute so-called pseudosolvent drag, with hypothetical uncharacterized interactions between water traversing the transcellular route and diffusion of K^+ along the paracellular pathway.⁴⁸⁰

LOOP OF HENLE

Transport by the loop of Henle plays a critical role in medullary K^+ recycling (Fig. 6.24). Several lines of evidence have indicated that a considerable fraction of K^+ secreted by the CCD is reabsorbed by the medullary collecting ducts and then secreted into the late PT and/or descending thin

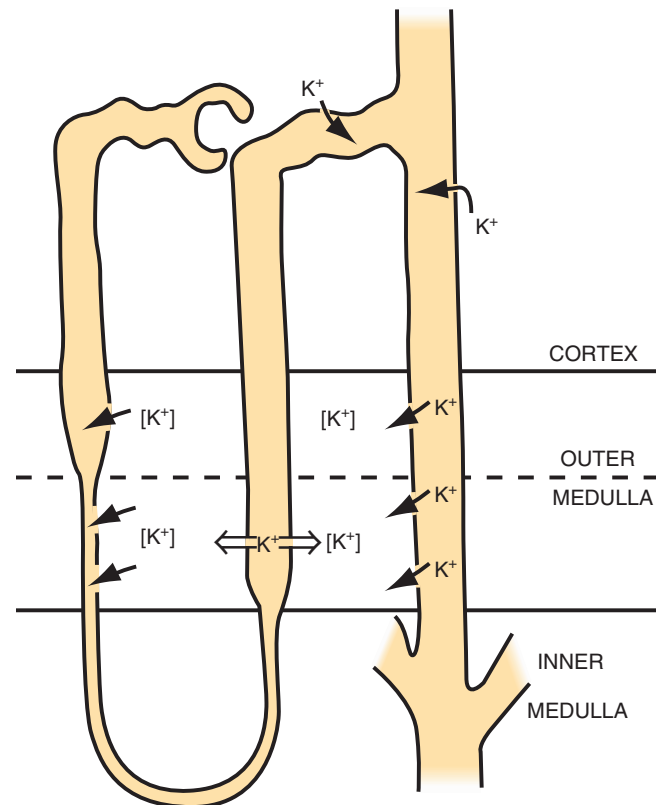


Fig. 6.24 Schematic representation of medullary K^+ recycling. Medullary interstitial K^+ increases considerably after dietary K^+ loading due to the combined effects of secretion in the cortical collecting duct, absorption in the outer medullary collecting duct, thick ascending limb, and inner medullary collecting duct, and secretion in the descending thin limb. See text for details. (From Stokes JB. Consequences of potassium recycling in the renal medulla. Effects of ion transport by the medullary thick ascending limb of the loop of Henle. *J Clin Invest.* 1982;70:219–229.)

limbs of long-looped nephrons.⁴⁸¹ In potassium-loaded rats, there is thus a doubling of luminal K^+ in the terminal thin descending limbs, with a sharp drop after inhibition of CCD K^+ secretion by amiloride.⁴⁸² Enhancement of CCD K^+ secretion by treatment with DDAVP also results in an increase in luminal K^+ in the descending thin limbs.⁴⁸³ This recycling pathway (secretion in CCD, absorption in OMCD and IMCD, secretion in descending thin limb) is associated with a marked increase in medullary interstitial K^+ . Passive transepithelial K^+ absorption by the thin ascending limb and active absorption by the TAL also contribute to this increase in interstitial K^+ (Fig. 6.24).¹⁸¹ Specifically, the absorption of K^+ by the ascending thin limb, TAL, and OMCD exceeds the secretion by the descending thin limbs, thus trapping K^+ in the interstitium.

The physiologic significance of medullary K^+ recycling is not completely clear. However, an increase in interstitial K^+ concentration from 5 to 25 mmol/L dramatically inhibits Cl^- (and to a lesser extent, Na^+) transport by perfused TALs.¹⁸¹ By inhibiting Na^+-Cl^- absorption by the TAL, increases in interstitial K^+ contribute to the well-documented diuretic effects of a high- K^+ diet,⁴⁸⁴ and would increase Na^+ delivery to the CNT and CCD, thus enhancing the lumen-negative PD in these tubules and increasing K^+ secretion.¹⁸¹ Alternatively, the marked increase in medullary interstitial K^+ after dietary K^+ loading serves to limit the difference between luminal and peritubular K^+ in the CCD, thus minimizing passive K^+ loss from the collecting duct.

K^+ is secreted into the descending thin limbs by passive diffusion, driven by the high medullary interstitial K^+ concentration. Descending thin limbs thus have a very high- K^+ permeability, without evidence for active transepithelial K^+ transport.⁴⁸⁵ Transepithelial K^+ transport by ascending thin limbs has not to our knowledge been measured; however, as is the case for Na^+-Cl^- transport (see the “[Na⁺-Cl⁻ Transport by the Thin Ascending Limb](#)” section), the absorption of K^+ by the thin ascending limbs is presumably passive. Active transepithelial K^+ transport across the TAL includes a transcellular component, via apical $Na^+-K^+-2Cl^-$ cotransport mediated by NKCC2, and a paracellular pathway (Fig. 6.14). Luminal K^+ channels play a critical role in generating the lumen-positive PD in the TAL, as summarized earlier (see the “[Na⁺-Cl⁻ Transport by the Thick Ascending Limb: Apical K⁺ Channels](#)” section). Secretion of K^+ through these may also play a role in the response to high dietary K^+ . Patch-clamp analysis of split-open TALs revealed that 70-pS ROMK exhibited a higher open probability (P_o) in mice placed on a low- Na^+ /high- K^+ diet.⁴⁸⁶ This may be dependent on NKCC2 activity, because micropuncture showed that furosemide increased K^+ secretion in the early distal tubule in mice on a normal diet, but decreased it in mice on the low- Na^+ /high- K^+ diet.

DISTAL NEPHRON

K^+ SECRETION

Approximately 90% of filtered K^+ is reabsorbed by the PT and loop of Henle (Fig. 6.23); the fine-tuning of renal K^+ excretion occurs in the remaining distal nephron. The bulk of regulated secretion occurs in principal cells within the CNT and CCD, whereas K^+ reabsorption primarily occurs in the OMCD (see later). A low rate of K^+ secretion is initially detectable in the early DCT, in which NCC-positive cells

express ROMK, the apical K^+ secretory channel.^{190,487} Generally, the CCD is considered the primary site for distal K^+ secretion, partially due to the greater ease with which this segment is perfused and studied. However, as is the case for Na^+-Cl^- absorption (see the “[Connecting Tubules and the Cortical Collecting Duct: Apical Na⁺ Transport](#)” section), the bulk of distal K^+ secretion appears to occur prior to the CCD, within the CNT.^{282,371}

In principal cells, apical Na^+ entry via ENaC generates a lumen-negative PD, which drives passive K^+ exit through apical K^+ channels. Distal K^+ secretion is therefore dependent on delivery of adequate luminal Na^+ to the CNT and CCD, essentially ceasing when luminal Na^+ drops below 8 mmol/L.^{488–490} Dietary Na^+ intake also influences K^+ excretion, such that excretion is enhanced by excess Na^+ intake and reduced by Na^+ restriction.^{488,489} Secreted K^+ enters principal cells via the basolateral $Na^+-K^+-ATPase$, which also generates the gradient that drives apical Na^+ entry via ENaC (Fig. 6.23).

Two major subtypes of apical K^+ channels function in secretion by the CNT and CCD, with or without the DCT; a small-conductance (SK), 30-pS channel and a large-conductance, Ca^{2+} -activated, 150-pS (maxi-K or BK) channel.^{191,371,491} The density and high P_o of the SK channel indicates that this pathway alone is sufficient to mediate the bulk of K^+ secretion in the CCD under baseline conditions—hence, its designation as the secretory K^+ channel.⁴⁹² Notably, SK channel density is considerably higher in the CNT than in the CCD, consistent with the greater capacity for Na^+ absorption and K^+ secretion in the CNT.³⁶ The characteristics of the SK channel are similar to those of the ROMK K^+ channel, and ROMK protein has been localized at the apical membrane of principal cells.^{188,493} SK channel activity is absent from apical membranes of the CCD in homozygous knockout mice with a targeted deletion of the *Kcnj1* gene that encodes ROMK, definitive proof that ROMK is the SK channel.¹⁹¹ The observation that these knockout mice are normokalemic, with an increased excretion of K^+ , illustrates the considerable redundancy in distal K^+ secretory pathways; distal K^+ secretion in these mice is mediated by apical BK channels (see later).^{191,494} However, dietary K^+ loading induced hyperkalemia in ROMK1 knockout mice, consistent with a role in K^+ secretion along the CCD.¹⁹³ Of interest, loss-of-function mutations in human *KCNJ1* genes are associated with Bartter syndrome; ROMK expression is critical for the 30- and 70-pS channels that generate the lumen-positive PD in the TAL (Fig. 6.14).^{191,192} These patients typically have slightly higher serum K^+ levels than those with other genetic forms of Bartter syndrome, and affected patients with severe neonatal hyperkalemia have also been described; this neonatal hyperkalemia is presumably the result of a transient developmental deficit in apical BK channel activity.^{14,183} Note that Bartter syndrome due to *KCNJ1* disruption may specifically reflect a defect in ROMK2 and/or ROMK3 function since mice with disruption of ROMK1, consistent with its absence along the TAL, do not display a Bartter syndrome phenotype.¹⁹³

The apical Ca^{2+} -activated BK channel plays a critical role in flow-dependent K^+ secretion by the CNT and CCD.⁴⁹¹ BK channels have a heteromeric structure, with α -subunits that form the ion channel pore and modulatory β -subunits that affect the biophysical, regulatory, and pharmacologic characteristics of the channel complex.⁴⁹¹ BK α -subunit transcripts

are expressed in multiple nephron segments, and channel protein is detectable at the apical membrane of principal and intercalated cells in the CCD and CNT.⁴⁹¹ The β -subunits are differentially expressed within the distal nephron. Thus β_1 -subunits are restricted to the CNT, with no expression in intercalated cells, whereas β_4 -subunits are detectable at the apical membranes of the TAL, DCT, and intercalated cells.^{491,495} Increased distal flow has a well-established stimulatory effect on K^+ secretion, due in part to enhanced delivery and absorption of Na^+ and to increased removal of secreted K^+ .^{488,489} The pharmacology of flow-dependent K^+ secretion in the CCD is consistent with dominant involvement of BK channels, and flow-dependent K^+ secretion is reduced in mice with targeted deletion of the α_1 - and β_1 -subunits.^{491,496–498} Both mice strains develop hyperaldosteronism that is exacerbated by a high- K^+ diet, leading to hypertension in the α_1 -subunit knockout.⁴⁹⁸ Disruption of the β_2 subunit also leads to hyperaldosteronism, but flow-induced K^+ secretion is normal, suggesting compensation by other isoforms.⁴⁹⁹ Ca^{2+} -dependence of BK activation involves TRPV4.^{500–502} A high- K^+ diet increases TRPV4 expression, and leads to its redistribution to the apical membrane in CCD. The importance of this process in BK activation was demonstrated in TRPV4 knockout mice, which display decreased BK activity in CCD, and hyperkalemia after dietary K^+ loading.

One enigma has been the greater density of BK channels in intercalated cells in both the CCD and CNT.^{503,504} This has suggested a major role for intercalated cells in K^+ secretion; however, the much lower density of Na^+ - K^+ -ATPase activity in intercalated cells has been considered inadequate to support K^+ secretion across the apical membrane.⁵⁰⁵ More recent evidence has revealed a major role for the basolateral Na^+ - K^+ - $2Cl^-$ cotransporter NKCC1 in K^+ secretion mediated by apical BK channels. NKCC1 is expressed almost exclusively at the basolateral membrane of intercalated cells, providing an alternative entry pathway for basolateral K^+ secreted at the apical membrane.^{506,507} This still begs the question of how basolateral Na^+ recycles across the basolateral membrane in the absence of significant Na^+ - K^+ -ATPase activity; one possibility is an alternative basolateral Na^+ pump, the ouabain-insensitive furosemide-sensitive Na^+ -ATPase, a transport activity that has been detected in cell culture models of intercalated cells.⁵⁰⁶ At the apical membrane, BK-mediated K^+ secretion is only partially dependent on luminal Na^+ ; K^+ secretion would eventually hyperpolarize the membrane in the absence of apical Na^+ entry, which is mediated by ENaC in principal cells.⁵⁰⁸ An intriguing possibility is that apical Cl^- channels allow for the parallel secretion of K^+ and Cl^- in intercalated cells.⁵⁰⁹

BK channels also play a critical role in cell volume regulation by intercalated cells, with indirect, flow-mediated influences on distal K^+ secretion. MDCK-C11 cells have an intercalated cell phenotype and express BK α - and β_4 -subunits, as do intercalated cells; shear stress activates BK channels in these cells, leading to loss of K^+ and cell shrinkage.^{495,510} Mice with a targeted deletion of the β_4 -subunit exhibit normal K^+ excretion on a normal diet.⁵⁰⁵ However, when fed a high- K^+ diet, which increases urinary and tubular flow rates and tubular shear stress, the β_4 -knockout mice develop hyperkalemia with a blunted increase in K^+ excretion and urinary flow rates. Intercalated cells from β_4 -knockouts fail to significantly decrease cell volume in response to high- K^+ diet.

Intercalated cells thus function as so-called speed bumps that protrude into the lumen of distal tubules; flow-activated BK channels reduce the cell volume of intercalated cells after K^+ loading, reducing tubular resistance, increasing tubular flow rates, and increasing distal K^+ secretion.⁵⁰⁵

The physiologic rationale for the presence of two apical secretory K^+ channels, ROMK/SK and BK channels, is not completely clear. However, the high density and higher P_o of ROMK/SK channels are perhaps better suited for a role in basal K^+ secretion, with additional recruitment of the higher capacity, flow-activated BK channels when additional K^+ secretion is required.⁴⁹¹ Evolving evidence has also indicated that BK channels function in partially Na^+ -independent K^+ secretion by intercalated cells, with ROMK functioning in ENaC- and Na^+ -dependent K^+ excretion by DCT, CNT, and CCD cells. Regardless, at the whole-organ level, the two K^+ channels can substitute for one another, with BK-dependent K^+ secretion in ROMK knockout mice and an upregulation of ROMK in the distal nephron of α_1 -subunit BK knockouts.^{494,497}

Other K^+ channels reportedly expressed at the luminal membranes of the CNT and CCD include voltage-sensitive channels such as Kv1.3, the calcium-activated, small-conductance SK3 channel, and double-pore K^+ channels, such as TWIK-1 and KCNQ1.^{511–514} KCNQ1 mediates K^+ secretion in the inner ear and is expressed at the apical membrane of principal cells in the CCD, whereas TWIK-1 is expressed at the apical membrane of intercalated cells.^{514,515} The roles of these channels in renal K^+ secretion or absorption are not fully characterized. However, Kv1.3 may play a role in distal K^+ secretion in that luminal margatoxin, a specific blocker of this channel, reduces K^+ secretion in CCDs of rat kidneys from animals on a high- K^+ diet.⁵¹⁶ Other apical K^+ channels in the distal nephron may subserve other physiologic functions. For example, the apical Kv1.1 channel is critically involved in Mg^{2+} transport by the DCT, likely by hyperpolarizing the apical membrane and increasing the driving force for Mg^{2+} influx via TRPM6 (transient receptor potential cation channel 6); missense mutations in *KVL1* are a cause of genetic hypomagnesemia.⁵¹⁷

K^+ channels present at the basolateral membrane of principal cells appear to set the resting potential of the basolateral membrane and function in K^+ secretion and Na^+ absorption at the apical membrane, the latter via K^+ recycling at the basolateral membrane to maintain activity of the Na^+ - K^+ -ATPase. A variety of different K^+ channels have been described in the electrophysiologic characterization of the basolateral membrane of principal cells, which has a number of technical barriers to overcome.⁵¹⁸ However, a single predominant activity has been identified in principal cells from the rat CCD using whole-cell recording techniques under conditions in which ROMK is inhibited (low intracellular pH or presence of the ROMK inhibitor tertiapin-Q).⁵¹⁸ This basolateral current is tetraethylammonium-insensitive, barium-sensitive, and acid-sensitive ($pK_a \approx 6.5$), with a conductance of about 17 pS and weak inward rectification. These properties do not correspond exactly to specific characterized K^+ channels or combinations thereof. However, candidate inward-rectifying K^+ channel subunits that have been localized at the basolateral membrane of the CCD include KIR4.1, KIR5.1, KIR7.1, and KIR2.3.⁵¹⁸ KIR4.1 and KIR5.1 channels generate a predominant 40-pS basolateral K^+ channel in murine

principal cells,⁵¹⁹ with both KIR4.1 and KIR5.1 participating in generating the membrane potential that permits Na⁺ entry through ENaC.^{227,520} Notably, basolateral K⁺ channel activity increases on a high-K⁺ diet, suggesting a role in transepithelial K⁺ secretion.⁵¹⁸ Disruption of KIR5.1 in rats reveals it plays a critical role in mediating collecting duct function, particularly with respect to maintenance of K⁺ homeostasis.⁵²⁰ Activation of KIR4.1/5.1 by insulin and insulin-like growth factor-1 (IGF-1) may also facilitate Na⁺ reabsorption along the CCD by hyperpolarizing the basolateral membrane.⁵²¹

In addition to apical K⁺ channels, considerable evidence has implicated apical K⁺-Cl⁻ cotransport (or functionally equivalent pathways) in distal K⁺ secretion.^{67,488,522,523} Thus, in rat distal tubules, a reduction in luminal Cl⁻ markedly increases K⁺ secretion; the replacement of luminal Cl⁻ with SO₄²⁻ or gluconate has an equivalent stimulatory effect on K⁺ secretion.⁵²⁴ This anion-dependent component of K⁺ secretion is not influenced by luminal Ba²⁺, suggesting that it does not involve apical K⁺ channel activity.⁵²⁴ Perfused surface distal tubules are a mixture of the DCT, connecting segment, and initial collecting duct; however, Cl⁻-coupled K⁺ secretion is detectable in the DCT and in early CNT.⁵²⁵ In addition, similar pathways are detectable in rabbit CCD, where a decrease in luminal Cl⁻ concentration from 112 to 5 mmol/L increases K⁺ secretion by 48%.⁵²⁶ A reduction in basolateral Cl⁻ also decreases K⁺ secretion without an effect on transepithelial voltage or Na⁺ transport, and the direction of K⁺ flux can be reversed by a lumen to bath Cl⁻ gradient, resulting in K⁺ absorption.⁵²⁶ In perfused CCDs from rats treated with mineralocorticoid, vasopressin increases K⁺ secretion; because this increase in K⁺ secretion is resistant to luminal Ba²⁺ (2 mmol/L), vasopressin may stimulate Cl⁻-dependent K⁺ secretion.^{14,527} Pharmacologic study results of perfused tubules are consistent with K⁺-Cl⁻ cotransport mediated by the KCCs; however, of the three renal KCCs, only KCC1 is apically expressed along the nephron.^{67,523} Other functional possibilities for Cl⁻-dependent K⁺ secretion include parallel operation of apical H⁺-K⁺-exchange and Cl⁻-HCO₃⁻ exchange in type B intercalated cells.⁵²²

A provocative study by Frindt and Palmer serves to underline the importance of ENaC-independent K⁺ excretion, whether it is mediated by apical K⁺-Cl⁻ cotransport and/or by other mechanisms (see also the “Integrated Na⁺-Cl⁻ and K⁺ Transport in the Distal Nephron” section).⁵²⁸ Rats were infused with amiloride via osmotic minipumps, generating urinary concentrations considered sufficient to inhibit more than 98% of ENaC activity. Whereas amiloride almost abolished K⁺ excretion in rats on a normal K⁺ intake, acute and long-term high-K⁺ diets led to an increasing fraction of K⁺ excretion that was independent of ENaC activity (≈50% after 7–9 days on a high-K⁺ diet).

K⁺ REABSORPTION BY THE COLLECTING DUCT

In addition to K⁺ secretion, the distal nephron is capable of considerable reabsorption, primarily during restriction of dietary K⁺.^{282–284} This reabsorption is accomplished largely by intercalated cells in the OMCD via the activity of apical H⁺/K⁺-ATPase pumps. Under K⁺-replete conditions, apical H⁺/K⁺-ATPase activity recycles K⁺ with an apical K⁺ channel, without an effect on transepithelial K⁺ absorption. Under K⁺-restricted basolateral conditions, K⁺ absorbed via apical

H⁺/K⁺-ATPase appears to exit intercalated cells via a K⁺ channel, thus achieving the transepithelial transport of K⁺.⁵²⁹

H⁺-K⁺-ATPase holoenzymes are members of the P-type family of ion transport ATPases, which also includes subunits of the basolateral Na⁺-K⁺-ATPase.⁵³⁰ HKα₁ and HKα₂ are also referred to as the gastric and colonic subunits, respectively; humans also have an HKα₄-subunit.^{530,531} A specific HKβ subunit interacts with the HKα subunits to ensure delivery to the cell surface and complete expression of H⁺-K⁺-ATPase activity; HKα₂ and HKα₄ subunits are also capable of interaction with Na⁺-K⁺-ATPase β-subunits.^{14,532} The pharmacology of H⁺-K⁺-ATPase holoenzymes differs considerably, such that the gastric HKα₁ subunit is typically sensitive to the H⁺-K⁺-ATPase inhibitors SCH-28080 and omeprazole and resistant to ouabain; the colonic HKα₂ subunit is usually sensitive to ouabain and resistant to SCH-28080.⁵³² Within the kidney, the HKα₁ subunit is expressed at the apical membrane of at least a subset of type A intercalated cells in the distal nephron.⁵³¹ HKα₂-subunit distribution in the distal nephron is more diffuse, with robust expression at the apical membrane of types A and B intercalated cells and connecting segment cells and lesser expression in principal cells.^{533–535} The human HKα₄ subunit is reportedly expressed in intercalated cells.⁵³¹

HKα₁ and HKα₂ subunits are both constitutively expressed in the distal nephron. However, tubule perfusion of K⁺-replete animals suggests a functional dominance of omeprazole/SCH-28080-sensitive, ouabain-resistant H⁺-K⁺-ATPase activity, consistent with holoenzymes containing the HKα₁ subunit.⁵³⁶ K⁺ deprivation increases the overall activity of H⁺-K⁺-ATPase in the collecting duct, with the emergence of ouabain-sensitive H⁺-K⁺-ATPase activity; this is consistent with a relative dominance of the HKα₂ subunit during K⁺-restricted conditions.¹⁴ K⁺ restriction also induces a dramatic upregulation of the HKα₂-subunit transcript and protein in the outer and inner medulla during K⁺ depletion; HKα₁-subunit expression is unaffected.¹⁴ Mice with targeted deletion of the HKα₂ subunit exhibit lower plasma and muscle K⁺ than wild type littermates when maintained on a K⁺-deficient diet. However, this appears to be due to marked loss of K⁺ in the colon rather than in the kidney, because renal K⁺ excretion is appropriately reduced in the K⁺-depleted knockout mice.²⁸⁵ Presumably the lack of an obvious renal phenotype in HKα₁- or HKα₂-subunit knockout mice reflects the marked redundancy in the expression of HKα subunits in the distal nephron.^{285,537} Indeed, collecting ducts from the HKα₁-subunit knockout mice have significant residual ouabain-resistant and SCH-28080-sensitive H⁺-K⁺-ATPase activities, consistent with the expression of other HKα-subunits that confer characteristics similar to the “gastric” H⁺-K⁺-ATPase.⁵³⁸ However, data from HKα₁- and HKα₂-subunit knockout mice have suggested that compensatory mechanisms in these mice are not accounted for by ATPase-type mechanisms.⁵³⁹

The importance of K⁺ reabsorption mediated by the collecting duct is dramatically illustrated by the phenotype of transgenic mice with generalized overexpression of a gain-of-function mutation in H⁺-K⁺-ATPase, effectively bypassing the redundancy and complexity of this reabsorptive pathway. This transgene expresses a mutant form of the HKβ subunit, in which a tyrosine-to-alanine mutation within the carboxyl-terminal tail abrogates regulated endocytosis from the plasma membrane; these mice have higher plasma K⁺ than their

wild type littermates, with approximately half the fractional excretion of K^+ .⁵⁴⁰

The H^+ , K^+ -ATPase may also serve to recycle K^+ , facilitating downregulation of ENaC by the purinergic signaling system during high dietary Na^+ intake. ENaC activity in $HK\alpha_1$ knockout mice is uncoupled from dietary Na^+ intake, and urinary [ATP] is not increased by a high- Na^+ diet, as in wild type mice.⁵⁴¹

REGULATION OF DISTAL K^+ TRANSPORT

MODULATION OF RENAL OUTER MEDULLARY POTASSIUM CHANNEL ACTIVITY

ROMK and other Kir channels are inward-rectifying—that is, K^+ flows inward more readily than outward (Kir, inward rectifying renal K^+ channel). Even though outward conductance is usually less than inward conductance, K^+ efflux through the ROMK predominates in the CNT and CCD because the membrane potential is more positive than the equilibrium potential for K^+ . Intracellular magnesium (Mg^{2+}) and polyamines play key roles in inward rectification, binding, and blocking the pore of the channel from the cytoplasmic side.^{542–544} A single transmembrane residue, asparagine-171 in ROMK1, controls the affinity and blocking effect of Mg^{2+} and polyamines.^{542,543} Intracellular Mg^{2+} in the TAL, DCT, CNT, and principal cells is thought to have a significant effect on ROMK activity because it inhibits outward ROMK-dependent currents in principal cells.⁵⁴⁵ The blocking affinity of Mg^{2+} is enhanced at lower extracellular K^+ concentrations, which should aid in reducing K^+ secretion during hypokalemia and K^+ deficiency.⁵⁴⁵ A reduction of this intracellular Mg^{2+} block may also explain the hypokalemia associated with hypomagnesemia, wherein distal K^+ secretion is enhanced.^{544,545}

In addition to inward rectification, the endogenous ROMK channels in the TAL and principal cells exhibit a very high channel P_o . The high P_o of ROMK is maintained by the combined effects of binding of PIP2 to the channel protein, direct channel phosphorylation by PKA, ATP binding to the ROMK-CFTR complex, and cytoplasmic pH. PIP2 binding to ROMK is thus required to maintain the channel in an open state, whereas cytoplasmic acidification inhibits the channel.⁵⁴⁶ PKA phosphorylates ROMK protein at one amino-terminal serines and two carboxyl-terminal serines—S25, S200, and S294 in the ROMK2 isoform.²⁵⁶ Phosphorylation of all three sites is required for full channel function. Phosphorylation of the amino-terminal site overrides the effect of a carboxy-terminal endoplasmic reticulum retention signal, thus increasing expression of the channel protein at the cell membrane.⁵⁴⁷ Phosphorylation of S200 and S294 maintains the channel in a high P_o state, in part by modulating the effects of PIP2, ATP, and pH.^{195,258,259}

Because ROMK channels exhibit such a high P_o , physiologic regulation of the channel is primarily achieved by regulated changes in the number of active channels on the plasma membrane. The associated mechanisms are discussed in the context of the adaptation to K^+ loading and hyperkalemia and K^+ deprivation and hypokalemia.

ALDOSTERONE AND K^+ LOADING

Aldosterone has a potent kaliuretic effect, with important interrelationships between circulating K^+ and aldosterone.⁵⁴⁸ Aldosterone release by the adrenal is thus induced by

hyperkalemia and/or a high- K^+ diet, suggesting an important feedback effect of aldosterone on K^+ homeostasis.⁵⁴⁹ Aldosterone also has clinically relevant effects on K^+ homeostasis, with a clear relationship at all levels of serum K^+ between circulating levels of the hormone and the ability to excrete K^+ .

Aldosterone has no effect on the density of apical ROMK channels in the CCD; it does, however, induce a marked increase in the density of apical Na^+ channels in the CNT and CCD.⁵⁵⁰ This hormone activates ENaC via interrelated effects on the synthesis, trafficking, and membrane-associated activity of the subunits encoding the channel (see the “[Regulation of \$Na^+\$ - \$Cl^-\$ Transport in the Connecting Tubule and Cortical Collecting Duct](#)” section). Aldosterone is thus induced by a high- K^+ diet and strongly stimulates apical ENaC activity, which provides the lumen-negative PD that stimulates K^+ secretion by principal cells.

The important relationships between K^+ and aldosterone notwithstanding, it is increasingly clear that much of the adaptation to a high- K^+ intake is aldosterone independent. For example, a high- K^+ diet in adrenalectomized animals increases apical Na^+ reabsorption and K^+ secretion in the CCD.⁵⁵¹ At the tubular level, when basolateral K^+ is increased, there is significant activation of Na^+ - K^+ -ATPase, accompanied by a secondary activation of apical Na^+ and K^+ channels.⁵⁵² Increased dietary K^+ also markedly increases the density of ROMK channels in the CCD, along with a modest increase in Na^+ channel (ENaC) density; this is associated with changes in the subcellular distribution of the ROMK protein, with an increase in apical expression.^{550,553} Notably, this increase in ENaC and ROMK density in the CCD occurs within hours of consuming a high- K^+ diet, with a minimal associated increase in circulating aldosterone (Table 6.1).⁵⁵⁴ By contrast, a week of low Na^+ - Cl^- intake, with almost a 1000-fold increase in aldosterone, has no effect on ROMK channel density, nor for that matter does 2 days of aldosterone infusion, despite the development of hypokalemia (Table 6.1).⁵⁵⁴ Unlike the marked increase seen in the CCD, the density of ROMK channels in the CNT is not increased by high dietary K^+ ,^{371,550,554} but this may reflect difficulties in estimating channel densities in small membrane patches. Measurement of whole cell currents using the ROMK inhibitor tertiapin-Q indicates an upregulation of ROMK activity in the CNT by a high- K^+ diet.⁵⁵⁵

BK channels in the CNT and CCD play an important role in the flow-activated component of distal K^+ excretion; these channels are also activated by dietary K^+ loading.⁴⁹¹ Flow-stimulated K^+ secretion by the CCD of mice and rats is thus enhanced on a high- K^+ diet, with an absence of flow-dependent K^+ secretion in rats on a low- K^+ diet.^{494,556} This is accompanied by the appropriate changes in transcript levels for α and β_{2-4} subunits of the BK channel proteins in microdissected CCDs (β_1 subunits are restricted to the CNT).⁴⁹¹ Trafficking of BK subunits is also affected by dietary K^+ , with a largely intracellular distribution of α subunits in K^+ -restricted rodents and prominent apical expression in K^+ -loaded rodents.^{500,556} Aldosterone does not contribute to the regulation of BK channel activity or expression in response to a high- K^+ diet.⁵⁵⁷

The changes in trafficking and/or activity of the ROMK channel that are induced by dietary K^+ appear in large part to involve tyrosine phosphorylation and dephosphorylation of the ROMK protein (see later). However, a series of reports have linked changes in expression of WNK1 kinase subunits

Table 6.1 Effect of High-K⁺ Diet, Aldosterone, and/or Na⁺-Cl⁻ Restriction on SK Channel Density in the Rat Cortical Collecting Duct

Parameter	K ⁺ Channel Density (μm ²)	Plasma Aldosterone (ng/dL)	Plasma K (mmol/L)
Control	0.41	15	3.68
High-K ⁺ diet, 6 hours	1.51	36	NM
High-K ⁺ diet, 48 hours	2.13	98	4.37
Low-Na ⁺ diet, 7 days	0.48	1260	NM
Aldosterone infusion, 48 hours	0.44	550	2.44
Aldosterone + high-K ⁺ diet	0.32	521	3.80

NM, Not measured.

Modified from Palmer LG, Frindt G. Regulation of apical K channels in rat cortical collecting tubule during changes in dietary K intake. *Am J Physiol.* 1999;277:F805–F812.

in the response to a high-K⁺ diet. WNK1 and WNK4 were initially identified as causative genes for FHHt (see also the “Regulation of Na⁺-Cl⁻ Transport in the Distal Convulated Tubule” section). ROMK expression at the membrane of *Xenopus* oocytes is dramatically reduced by coexpression of WNK4; FHHt-associated mutations dramatically increase this effect, suggesting a direct inhibition of SK channels in FHHt.⁵⁵⁸ This is further supported by a recent patch-clamp study that showed DCT2/CNT isolated from transgenic mice expressing FHHt-causing mutant WNK4 has lower ENaC and ROMK activity, suggesting direct, NCC-independent effects of the mutant.⁵⁵⁹ The aldosterone-induced SGK1 is activated by the upstream kinase mTORC2. Disruption of mTORC2 in renal epithelia in mice causes profound hyperkalemia when they are placed on a high-K⁺ diet. Patch-clamp analysis showed that in CNT/CCD, while ENaC activity was unaltered, Ba²⁺-sensitive K⁺ currents were almost absent. SGK1 phosphorylation was ablated, suggesting that hyperkalemia may result from unchecked WNK4-mediated ROMK endocytosis.⁵⁶⁰

The study of WNK1 is further complicated by the transcriptional complexity of its gene, which has at least three separate promoters and a number of alternative splice forms. In particular, the predominant intrarenal WNK1 isoform is generated by a distal nephron transcriptional site that bypasses the amino-terminal exons that encode the kinase domain, yielding a kinase-deficient short form of the protein (WNK1-S, also known as kidney-specific [KS]-WNK1).⁵⁶¹ Full-length WNK1 (WNK1-L) inhibits ROMK activity by inducing endocytosis of the channel protein; kinase activity and/or the amino-terminal kinase domain of WNK1 appear to be required for this effect, although Cope and colleagues have reported that a kinase-dead mutant of WNK1 is unimpaired.^{562–564} WNK1 and WNK4 induce endocytosis of ROMK via interaction with intersectin, a multimodular endocytic scaffold protein.⁵⁶⁵ Additional binding of ROMK to the clathrin adaptor protein termed “autosomal recessive hypercholesterolemia” (ARH) is required for basal and WNK1-stimulated endocytosis of the channel protein.⁵⁶⁶ Ubiquitination of ROMK protein is also involved in clathrin-dependent endocytosis, requiring interaction between the channel and the U3 ubiquitin ligase POSH (plenty of SH domains).⁵⁶⁷

The shorter WNK1-S isoform, which lacks the kinase domain, appears to inhibit the effect of WNK1-L.^{563,564} The ratio of WNK1-S to WNK1-L transcripts is reduced by K⁺

restriction (greater endocytosis of ROMK) and increased by K⁺ loading (reduced endocytosis of ROMK), suggesting that this ratio between WNK1-S and WNK1-L functions as a type of switch to regulate distal K⁺ secretion.^{563,564,568} The inhibitory effect of WNK1-S tracks to the first 253 amino acids of the protein, encompassing the initial 30 amino acids unique to this isoform and an adjacent autoinhibitory domain.⁵⁶⁹ Transgenic mice that overexpress this inhibitory domain of WNK1-S have lower serum K⁺ concentrations, higher fractional excretion of K⁺, and increased expression of ROMK protein at the apical membrane of CNT and CCD cells—all consistent with an important inhibitory effect of WNK1-S.⁵⁶⁹

The BK channel is also regulated by the WNK kinases. WNK4 thus inhibits BK channel activity and protein expression, whereas FHHt-associated mutations in WNK4 also enhance the inhibitory effect via ubiquitination.^{570–572} A high-K⁺ diet increases WNK1-L expression selectively in intercalated cells,⁵⁷³ and activates BK by reducing ERK1/2 signaling-mediated lysosomal degradation of the channel protein.²²⁹

K⁺ DEPRIVATION

A reduction in dietary K⁺ leads within 24 hours to a dramatic drop in urinary K⁺ excretion.^{568,574} This drop in excretion is due to both an induction of reabsorption by intercalated cells in the OMCD and to a reduction in SK channel activity in principal cells.^{14,283,284} The mechanisms involved in K⁺ reabsorption by intercalated cells are discussed earlier; notably, H⁺/K⁺-ATPase activity in the collecting duct does not appear to be regulated by aldosterone.⁵⁷⁵

Considerable progress has been made in defining the signaling pathways that regulate the activity of the SK channel (ROMK) in response to changes in dietary K⁺. Dietary K⁺ intake modulates trafficking of the ROMK channel protein to the plasma membrane of principal cells, with a marked increase in the relative proportion of intracellular channel protein in K⁺-depleted animals and clearly defined expression at the plasma membrane of CCD cells from animals on a high-K⁺ diet.^{553,576} The membrane insertion and activity of ROMK are modulated by tyrosine phosphorylation of the channel protein, such that phosphorylation of tyrosine residue 337 stimulates endocytosis and dephosphorylation induces exocytosis; this tyrosine phosphorylation appears to play a key role in the regulation of ROMK by dietary K⁺.^{577–579} Whereas the levels of protein tyrosine phosphatase-1D do not vary with K⁺ intake, intrarenal activity of the cytoplasmic

tyrosine kinases c-src and c-yes are inversely related to dietary K^+ intake, with a decrease under high- K^+ conditions and a marked increase after several days of K^+ restriction.^{14,580} Localization studies have indicated coexpression of c-src with ROMK in the TAL and principal cells of the CCD.⁵⁵³ Moreover, inhibition of protein tyrosine phosphatase activity, leading to a dominance of tyrosine phosphorylation, dramatically increases the proportion of intracellular ROMK in the CCD of animals on a high- K^+ diet.⁵⁵³

The neurohumoral factors that induce the K^+ -dependent trafficking and expression of apical ROMK and BK channels have only come into focus rather recently.^{553,556,576} Several studies have implicated the intrarenal generation of superoxide anions in the activation of cytoplasmic tyrosine kinases.^{581–583} Potential candidates for the upstream kaliuretic factor include Ang II and growth factors such as IGF-1.⁵⁸¹ Ang II inhibits ROMK activity in K^+ -restricted rats, but not rats on a normal K^+ diet.⁵⁸⁴ This inhibition involves downstream activation of superoxide production and c-src activity, such that the well-known induction of Ang II by a low- K^+ diet appears to play a major role in reducing distal tubular K^+ secretion.⁵⁸⁵

Reports of transient postprandial kaliuresis in sheep, independent of changes in plasma K^+ or aldosterone, have suggested that an enteric or hepatoportal K^+ sensor controls kaliuresis via a sympathetic reflex; tissue kallikrein (TK) has recently emerged as a candidate mediator for this postprandial kaliuresis (see later).⁵⁸⁶ Regardless of the signaling involved, changes in dietary K^+ absorption have a direct anticipatory effect on K^+ homeostasis in the absence of changes in plasma K^+ . Such a feed-forward control has the theoretical advantage of greater stability because it operates prior to changes in plasma K^+ .⁵⁸⁷ Notably, changes in ROMK phosphorylation status and insulin-sensitive muscle uptake can be seen in K^+ -deficient animals in the absence of a change in plasma K^+ , suggesting that upstream activation of the major mechanisms that serve to reduce K^+ excretion (reduced K^+ secretion in the CNT and CCD, decreased peripheral uptake, and increased K^+ reabsorption in the OMCD) does not require changes in plasma K^+ .⁵⁸⁸ Consistent with this hypothesis, moderate K^+ restriction, without an associated drop in plasma K^+ , is sufficient to induce Ang II-dependent superoxide generation and c-src activation, leading to inhibition of ROMK channel activity.⁵⁸⁵

VASOPRESSIN

Vasopressin has a well-characterized stimulatory effect on K^+ secretion by the distal nephron.^{483,589} From an evolutionary viewpoint, this vasopressin-dependent activation serves to preserve K^+ secretion during dehydration and extracellular volume depletion, when circulating levels of vasopressin are high and tubular delivery of Na^+ and fluid is reduced. The stimulation of basolateral V_2 Rs results in an activation of ENaC, which increases the driving force for K^+ secretion by principal cells; the relevant mechanisms have been discussed earlier in this chapter (see the “Regulation of Na^+ - Cl^- Transport in the Connecting Tubule and Cortical Collecting Duct: Vasopressin and Other Factors” section). In addition, vasopressin activates SK channels directly in the CCD, as does cAMP.^{492,590} The ROMK is directly phosphorylated by PKA on three serine residues (S25, S200, and S294 in the ROMK2 isoform), with phosphorylation of all three sites required

for full activity in *Xenopus* oocytes (see the “Regulation of Na^+ - Cl^- Transport by the Thick Ascending Limb: Activating Influences” section). Finally, the stimulation of luminal V_1 receptors also stimulates K^+ secretion in the CCD, apparently via activation of BK channels.⁵⁹¹

TISSUE KALLIKREIN

The serine protease TK is involved in the generation of kinins, ultimately stimulating the formation of bradykinin.⁵⁹² Within the kidney, TK is synthesized in CNT cells and released into the tubular lumen and peritubular interstitium. Although TK-induced bradykinin has a number of effects on distal tubular physiology, more recent data have revealed a provocative role in postprandial kaliuresis.⁵⁹² Thus oral K^+ - Cl^- loading leads to a spike in urinary K^+ and TK excretion in rats, mice, and humans.⁵⁹² The increase in urinary TK after K^+ loading is not accompanied by changes in urinary aldosterone and can be detected in aldosterone synthase knockout mice.⁵⁹³ Mice deficient in TK demonstrate postprandial hyperkalemia, indicating a role for the protease in postprandial kaliuresis. This transient hyperkalemia is accompanied by a marked increase in K^+ reabsorption by perfused CCDs due to an upregulation of H^+ / K^+ -ATPase activity and an increase in $HK\alpha_2$ -subunit transcript. The addition of luminal but not basolateral TK inhibits the activated CCD H^+ / K^+ -ATPase activity in the TK knockout mice, consistent with direct proteolytic activation. There is also a marked increase in Na^+ reabsorption by perfused CCDs from TK knockout mice, without development of a lumen-negative PD; this is consistent with an increased activity of the electroneutral Na^+ - Cl^- cotransport mediated by the Na^+ -driven SLC4A8 Cl^- - HCO_3^- exchanger and the SLC26A4 Cl^- - HCO_3^- exchanger (see later discussion as well).⁴⁰⁸ This electroneutral transport pathway had previously been shown to be inhibited by bradykinin; hence the activation by TK deletion presumably reflected loss of tonic inhibition by TK-generated bradykinin.⁴⁰⁶ Previous data had indicated that TK mediates proteolytic cleavage of the γ subunit of ENaC, with reduced ENaC activity in TK-deficient mice; net Na^+ balance is thus neutral in these mice.⁵⁹⁴

In summary, TK secretion from CNT cells is induced by oral K^+ - Cl^- loading, causing proteolytic activation of ENaC and thus an increase in ENaC-driven K^+ secretion, bradykinin-dependent inhibition of electroneutral Na^+ - Cl^- cotransport in the CCD.^{406,408,594} There is consequently a further augmentation of electrogenic Na^+ transport (favoring K^+ secretion), and direct luminal inhibition of H^+ / K^+ -ATPase activity and thus a decrease or tonic inhibition of K^+ reabsorption. TK may very well be the postprandial factor that functions in feed-forward control of plasma K^+ .^{586,587}

INTEGRATED Na^+ - Cl^- AND K^+ TRANSPORT IN THE DISTAL NEPHRON

Segmentation of the renal epithelia into distinct sections does not mean that different segments function independently. Increasing evidence from in vivo studies in mice with multiple genes disrupted, or from studies in which knockout mice are subjected to drug treatment or dietary manipulation, shows that effects in one segment exert compensatory effects to maintain homeostasis. For example, *Slc26a4* knockout mice with combined genetic ablation of NCC⁵⁹⁵ or administration of hydrochlorothiazide⁵⁹⁶ display salt wasting and volume

contraction. Similarly, ENaC blockade with amiloride leads to enhanced natriuresis in NCC knockout mice.⁵⁹⁷ In mice with disruption of MR specifically in renal epithelia, while activities of both ENaC and NCC are diminished, the effect on NCC is secondary to the hyperkalemia induced by lower ENaC activity.^{329,330} These mice are hypotensive, but normalization of plasma $[K^+]$ by dietary K^+ restriction stimulates NCC activity resulting in normalization of blood pressure. Grimm and colleagues performed microarray studies on SPAK knockout mice, which display lower NCC activity, and proposed that α -ketoglutarate may serve as a downstream signaling molecule that mediates compensatory responses.⁵⁹⁸ These interactions between segments are likely to be similar to those seen with diuretic treatment and may contribute to diuretic resistance (see Chapter 50). In the following sections, other examples of this intersegment crosstalk are described in more depth.

In the classic model of renal K^+ secretion, the lumen-negative PD generated by Na^+ entry via ENaC induces the exit of K^+ via apical K^+ -selective channels. This general scheme explains much of the known physiology and pathophysiology of renal K^+ secretion, yet has several key consequences that bear emphasis. First, enhanced Na^+Cl^- reabsorption upstream of the CNT and CCD will reduce the delivery of luminal Na^+ to the CNT and CCD, decrease the lumen-negative PD, and thus decrease K^+ secretion; K^+ secretion by the CCD essentially stops when luminal Na^+ drops below 8 mmol/L.^{488–490} In this respect, the increasingly refined phenotypic understanding of FHt, caused by kinase-induced gain of function of the DCT, has served to underscore that variation in NCC-dependent Na^+Cl^- absorption, just upstream of the CNT, has truly profound effects on the ability to excrete dietary K^+ (Fig. 6.19).³⁴⁵ Second, aldosterone is a kaliuretic hormone, induced by hyperkalemia. However, under certain circumstances associated with marked induction of aldosterone, such as dietary sodium restriction, sodium balance is maintained without effects on K^+ homeostasis. This so-called aldosterone paradox—how the kidney independently regulates Na^+Cl^- and K^+ handling by the aldosterone-sensitive distal nephron—is only recently beginning to yield to investigative efforts. The major factors in the integrated control of Na^+Cl^- and K^+ transport appear to include electroneutral thiazide-sensitive Na^+Cl^- transport within the CCD, ENaC-independent K^+ excretion within the distal nephron, and the differential regulation of various signaling pathways by aldosterone, Ang II, and dietary K^+ .^{406–408,528,599,600}

Thiazide-sensitive electroneutral Na^+Cl^- transport within the CCD is evidently mediated by the parallel activity of the Na^+ -driven SLC4A8 $Cl^-HCO_3^-$ exchanger and the SLC26A4 $Cl^-HCO_3^-$ exchanger.⁴⁰⁸ The molecular identity of this transport mechanism has only emerged rather recently, so regulatory influences are not fully characterized.⁴⁰⁸ However, electroneutral Na^+Cl^- transport within the CCD is evidently induced by volume depletion and mineralocorticoid treatment.^{406–408} This mechanism appears to mediate about 50% of Na^+ reabsorption in the CCD under these conditions, all without affecting the luminal PD and thus without direct effect on K^+ excretion. Therefore electroneutral, thiazide-sensitive Na^+Cl^- transport affords the ability to increase the reabsorption of Na^+ within the CCD without affecting K^+ excretion. The converse occurs after several days of accommodation to a high- K^+ diet, which increases the fraction of

ENaC-independent, amiloride-resistant K^+ excretion to about 50%. Again, this presumptively electroneutral, aldosterone-independent pathway for K^+ excretion serves to uncouple distal tubular Na^+ and K^+ excretion.⁵²⁸

WNK-dependent signaling constitutes a major pathway for integrating Na^+Cl^- and K^+ transport within the distal nephron, and is regulated by the differential influence of K^+ intake on circulating Ang II, ROMK activity (i.e., K^+ secretory capacity), ratio of WNK1 isoforms, and activity of NCC in the DCT. Thus Ang II activates NCC via the WNK4-SPAK pathway, reducing delivery of Na^+ to the CNT and limiting K^+ secretion.^{312,315,601} By contrast, Ang II inhibits ROMK activity via several mechanisms, including downstream activation of c-src tyrosine kinases.^{583–585} Whereas K^+ restriction induces renin and circulating Ang II, increases in dietary K^+ are suppressive.^{585,602} A decrease in circulating and local Ang II partially explains why NCC phosphorylation and activity are down-regulated by a high- K^+ diet; teleologically, this serves to increase delivery of Na^+ to the CNT, thus increasing K^+ secretion.³⁶¹

The DCT also clearly functions as a potassium sensor, directly responding to changes in circulating potassium (Fig. 6.19). Evidence that a key role of the DCT is to act as a plasma $[K^+]$ sensor, rather than to maintain plasma $[Na^+]$, is demonstrated by the observation that acute K^+ loading dephosphorylates NCC even in Na^+ -restricted mice.⁶⁰³ Reduction in potassium intake and/or hypokalemia thus lead to reduced basolateral $[K^+]$ in the DCT; the subsequent hyperpolarization of DCT cells is dependent on basolateral KIR4.1-containing K^+ channels.^{308,327} Hyperpolarization leads to chloride exit via basolateral CLC-K2 chloride channels; the resulting decrease in intracellular chloride disinhibits WNK kinases, resulting in phosphorylation of NCC and activation of the transporter by SPAK/OSR1.³²⁷ Ex vivo studies (micro-perfusion and kidney slices) have provided evidence that increased phosphorylation of NCC resulting in response to lower extracellular $[K^+]$ is dependent on increased intracellular chloride.⁶⁰⁴ Along the DCT, WNK4 appears to be the major WNK activating NCC,²³⁶ and WNK4 is also more sensitive to changes in intracellular chloride than WNK1 or WNK3.³⁴³ The essential role of this pathway in mediating responses to dietary K^+ restriction is clearly demonstrated by the development of severe hypokalemia in mice lacking KIR4.1,³¹¹ WNK4,⁶⁰⁵ SPAK/OSR1,⁶⁰⁶ and NCC (D.H. Ellison, unpublished observations; see Fig. 6.19 for summary of model). An alternative mechanism has been proposed in which KLHL3, part of the ubiquitin ligase complex that degrades WNK4, is phosphorylated in response to K^+ depletion. This prevents WNK4 binding to the complex, decreasing WNK4 degradation, which promotes NCC activation.⁶⁰⁷

NCC undergoes rapid dephosphorylation in response to K^+ loading,^{604,608} a process that appears to be largely independent of increased intracellular chloride, which would inhibit WNKs.⁶⁰⁴ This suggests a role for protein phosphatases; one study suggested that PP3 (calcineurin) is responsible for NCC dephosphorylation,⁶⁰⁸ while another found no role for PP1, PP2A, or PP3.⁶⁰⁴ Glucocorticoid-induced leucine zipper protein (GILZ) may also play a role in the response to high dietary K^+ by inhibiting SPAK activity and hence NCC. Compared with wild type mice, *Gilz* knockout mice are more sensitive to NCC inhibition by thiazides and have greater abundance of phosphorylated NCC. *Gilz* knockout

mice display elevated plasma $[K^+]$ at baseline, with dietary Na^+ restriction increasing it further. When plasma $[K^+]$ increases, GILZ-mediated inhibition of NCC may serve to maintain distal Na^+ delivery to ENaC to drive kaliuresis.⁶⁰⁹ Most studies have used diets completely almost deficient in K^+ , or with very high levels, for example, 5% KCl, to examine NCC activation status. Relatively small changes in plasma $[K^+]$ also alter the degree of NCC phosphorylation in vivo, suggesting that the role of NCC to maintain plasma $[K^+]$ is homeostatically important under dietary K^+ levels that do not show extreme deviation from normal.³⁴³

A mouse model in which NCC activity was increased specifically along DCT1, by expressing a constitutively active SPAK mutant only in this segment, suggests that altering NCC activity may affect distal K^+ secretion by inducing remodeling of the CNT, and not just by altering Na^+ delivery to ENaC.⁶¹⁰ These mice displayed hyperkalemia, which was associated with lower volume of the CNT and lower ROMK abundance, which may limit K^+ secretion. Chronic blockade of NCC led to increased CNT volume and ROMK abundance, and normalized plasma K^+ , suggesting that altered Na^+ delivery to the CNT may induce remodeling.

Finally, within principal cells, increases in aldosterone induce the SGK1 kinase, which phosphorylates WNK4 and attenuates the effect of WNK4 on ROMK, while activating ENaC via NEDD4-2-dependent effects.⁶¹¹ However, when dietary K^+ intake is reduced, c-src tyrosine kinase activity increases under the influence of increased Ang II, causing direction inhibition of ROMK activity via tyrosine phosphorylation of the channel.^{577,579,612} The increase in c-src tyrosine kinase activity also abrogates the effect of SGK1 on WNK4.^{600,613} While NEDD4-2 can regulate NCC during K^+ restriction,³²¹ ENaC regulation may be more important. In NEDD4-2 knockout mice, chronic K^+ restriction led to hypokalemia and urinary K^+ wasting that was reversed with the ENaC blocker benzamil.³⁴⁰ Higher phosphorylation of NCC and lower ROMK abundance were observed, but not sufficient to compensate for the effect on ENaC activation. While both insulin and IGF-1 activate ENaC, insulin promotes kaliuresis whereas IGF-1 exerts an antikaliuretic response. Patch-clamp analysis of isolated mouse CCD cells showed that this differential effect on K^+ secretion may be due to differential effects on CLC-K2 (which is expressed in intercalated cells).³⁹² IGF-1 stimulates CLC-K2, which may promote net Na^+ -Cl⁻ reabsorption, thus reducing the driving force for K^+ secretion by the CCD. By contrast, insulin inhibits CLC-K2, which would enhance the generation of the electrogenic drive for K^+ secretion.



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BOARD REVIEW QUESTIONS

1. Despite being electrically “leaky,” the proximal tubule establishes a large transepithelial gradient for which of the following ions,
 - a. Sodium
 - b. Calcium
 - c. Bicarbonate
 - d. Magnesium

Answer: c

Rationale: The proximal tubule actively reabsorbs bicarbonate through the concerted actions of an apical Na-H exchanger and basolateral Na-bicarbonate cotransporter, and this occurs in excess of osmotic water reabsorption so that, by the mid-PT, there is a transepithelial gradient with a higher bicarbonate concentration basolaterally than apically.

2. Glomerulotubular balance is a term that describes the
 - a. Decrease in glomerular filtration rate when distal NaCl delivery is high
 - b. Correspondence between glucose filtration and excretion
 - c. Increased proximal fluid reabsorption when GFR increases
 - d. Correspondence between glomerular size and tubule length

Answer: c

Rationale: Glomerulotubular balance refers to the phenomenon wherein changes in the glomerular filtration rate (GFR) are offset by changes in tubular reabsorption, thus maintaining a constant fractional reabsorption of fluid and $\text{Na}^+\text{-Cl}^-$. Glomerulotubular balance is independent of direct neuronal and systemic hormonal control and is thought to be mediated by the additive effects of luminal and peritubular factors.

3. The thick ascending limb has a transepithelial voltage oriented in the lumen-positive direction primarily because of the combined actions of
 - a. Primary active transport of chloride out of the lumen
 - b. Luminal chloride channels and basolateral potassium channels
 - c. Apical Na-K-Cl cotransport and basolateral Na/K ATPase
 - d. The charge carried by the apical Na-K-Cl cotransporter
 - e. Luminal potassium channels and basolateral chloride channels

Answer: e

Rationale: The lumen-positive transepithelial potential difference in the thick ascending limb of Henle is generated

by the combination of apical K^+ channels and basolateral Cl^- channels. Luminal recycling of K^+ via $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransport and apical K^+ channels, along with basolateral depolarization due to Cl^- exit through Cl^- channels, results in the lumen-positive transepithelial voltage.

4. Which of the following is now known to play a dominant role in modulating the activity of the thiazide-sensitive NaCl cotransporter in the distal convoluted tubule?
 - a. Serum potassium concentration
 - b. Serum sodium concentration
 - c. Serum chloride concentration
 - d. Serum bicarbonate concentration

Answer: a

Rationale: Recent evidence shows that DCT cells, and NCC, are exquisitely sensitive to plasma $[\text{K}^+]$. WNKs, especially WNK4, appear to be sensitive to inhibition by intracellular chloride, and this plays a major role in the potassium-sensing function of DCT cells. Hypokalemia leads to reduced basolateral K^+ concentration in the DCT, and subsequent hyperpolarization that is dependent on basolateral KIR4.1 -containing K^+ channels. This hyperpolarization has been proposed to lead to chloride exit via basolateral CLC-K2 chloride channels and a reduction in intracellular chloride; the reduction in intracellular chloride activates the SPAK and OSR1-WNK cascade, resulting in phosphorylation of NCC and activation of the transporter.

5. Combined genetic deletion of the thiazide-sensitive NaCl cotransporter and pendrin in mice causes which striking phenotype?
 - a. Profound hyperkalemic acidosis
 - b. Hypercalciuria with nephrocalcinosis
 - c. Nephrogenic diabetes insipidus
 - d. Profound salt wasting

Answer: d

Rationale: There is increasing evidence that effects on salt transport in one tubule segment induce compensatory effects in other segments to maintain homeostasis. Whereas single deletion of pendrin or the thiazide-sensitive NaCl cotransporter in mice does not cause salt wasting or excessive diuresis, double knockout mice with combined genetic ablation of pendrin and the thiazide-sensitive NaCl cotransporter display severe salt wasting, increased urine output, volume depletion and renal failure.